SMA 2104: MATHEMATICS FOR SCIENCE

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Course Content

- Surds, indices and logarithms.
- Quadratic equations, functions, and inequalities.
- Remainder theorem and its application to the solution of factorizable polynomial equations.
- Permutations and combinations.
- Sequences and Series: finite, infinite, arithmetic, geometric, and binomial (positive integral index only) including applications to compound interest, approximations, growth, and decay.
- Trigonometry: trigonometric functions including their graphs and inverses in degree and radian measure. Sine and cosine formulae.
- Statistics: collection and representation of data, measures of central tendency and variability by graphical and calculation methods.
- Probability: classical and axiomatic approaches to probability, compound events, conditional probability, tree diagrams, and binomial distribution.

References

1. Mathematics for Science by S. M. Uppal and H. M. Humphreys, 1997, Mc Graw Hill, 2nd edition.
2. Algebra and Trigonometry with Analytical Geometry (12th edition) by Earl Swokowski
3. Pure Mathematics 1 by Backhouse.

Lecture 1

1 Algebra

- Algebra is the part of mathematics dealing with the manipulation of expressions and solutions of equations.
- Since these operations are needed in all branches of mathematics, algebraic skill is a fundamental need for doing mathematics and therefore for working in any discipline that substantially requires mathematics.
- It is the foundation and the core of higher mathematics.
- A strong foundation in Algebra will help individuals become better mathematicians, analyzers, and thinkers.

1.1 Numbers

Numbers can be grouped together and classified into different sets of numbers such as:

1.1.1 Natural numbers ($\mathbb{N}$)

Are those numbers that are used for counting? The set of natural numbers can then be displayed formally as:

$$\mathbb{N} = \{1,2,3,4,5,6,...\}$$

1.1.2 Integers ($\mathbb{Z}$)

Integers consist of positive and negative whole numbers including zero. The set of integers is denoted by $\mathbb{Z}$. Thus,

$$\mathbb{Z} = \{0, \pm1, \pm2, \pm3, \pm4, \pm5, \pm6, ...\}$$

1.1.3 Rational numbers ($\mathbb{Q}$)

A rational number is a number that can be expressed in the simplest form p/q, where p and q are integers and $q \neq 0$. Note that $p/0 = 1$ and $0/0$ is indeterminate. Examples of rational numbers are:

- Integers
- Fractions: $\frac{5}{6}$
- Terminating decimals: 0.25,0.75,1.5
- Recurring/Repeating decimals: $0.5555, 0.333$, etc.

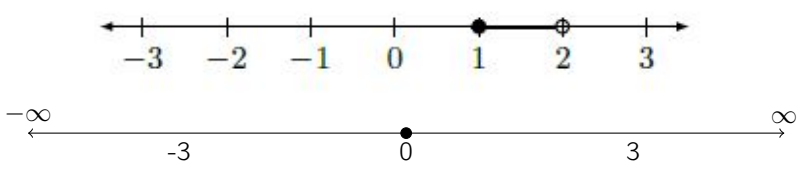
1.1.4 Irrational numbers ($\mathbb{Q}^c$)

Are numbers that cannot be expressed in the simplest form p/q, where p and q are integers and $q \neq 0$. Examples of irrational numbers are:

- Special numbers such as π and e.
- Surds or roots of numbers that do not have finite (exact) answers e.g. $\sqrt{2}, \sqrt{5}, \sqrt[3]{4}$ etc.

1.1.5 Real numbers ($\mathbb{R}$)

A real number is a value that represents a quantity along a continuous line (number line).



All rational numbers and irrational numbers are real numbers. The set of all real numbers is denoted by $\mathbb{R}$.

1.1.6 Complex numbers ($\mathbb{C}$)

Complex number z is a number that can be expressed in the form $z = x + iy$, where x and y are real numbers and i is an imaginary number defined by $i = \sqrt{-1}$. Examples of complex numbers are: $4 + i6, i3, 5 + i\frac{2}{3}$, etc. All real numbers are complex numbers. $\mathbb{C}$ denotes the set of all complex numbers.

1.2 Surds

A surd is an irrational root of a rational number. Examples are: $\sqrt{2}, \sqrt{3}, \sqrt[3]{5}, \sqrt[4]{\frac{1}{3}}, \sqrt[3]{9}$, etc. Note, for instance, that $\sqrt{9}$ is not a surd since $\sqrt{9} = \frac{3}{1}$, which is a rational number. Also, note that $\sqrt[n]{a} = (a)^{\frac{1}{n}}$, where n is an integer different from zero.

Examples:

1. Express $\sqrt[4]{48}$ in the form $p\sqrt[4]{q}$, where p and q are integers and q contains no factors which are exact 4th powers of integers.
   Solution

$$\sqrt[4]{48} = \sqrt[4]{16 \times 3} = \sqrt[4]{16} \times \sqrt[4]{3} = \sqrt[4]{2^4} \times \sqrt[4]{3} = 2\sqrt[4]{3}$$

2. Express $\sqrt[3]{\frac{81}{625}}$ in the form $p\sqrt[3]{q}$, where p and q are rational numbers and q contains no factors which are exact cubes of rational numbers.
   Solution

$$\sqrt[3]{\frac{81}{625}} = \sqrt[3]{\frac{3^4}{5^4}} = \sqrt[3]{\frac{3^3}{5^3} \times \frac{3}{5}} = \sqrt[3]{\frac{3^3}{5^3}} \times \sqrt[3]{\frac{3}{5}} = \frac{3}{5}\sqrt[3]{\frac{3}{5}}$$

3. Express $6\sqrt{5}$ in the form $\sqrt{p}$, where p is an integer.
   Solution

$$6\sqrt{5} = \sqrt{6^2 \times 5} = \sqrt{36 \times 5} = \sqrt{180}$$

4. Express $7\sqrt[4]{3}$ in the form $\sqrt[4]{p}$, where p is an integer.
   Solution

$$7\sqrt[4]{3} = \sqrt[4]{7^4 \times 3} = \sqrt[4]{7203}$$

5. Express $\frac{8}{3}\sqrt[3]{\frac{5}{9}}$ in the form $\sqrt[3]{\frac{p}{q}}$, where p and q are integers.
   Solution

$$\frac{8}{3}\sqrt[3]{\frac{5}{9}} = \sqrt[3]{\left(\frac{8}{3}\right)^3 \times \frac{5}{9}} = \sqrt[3]{\frac{8^3 \times 5}{3^3 \times 9}} = \sqrt[3]{\frac{2560}{243}}$$

1.2.1 Addition and subtraction of surds

Surds can be added or subtracted if they are of the same order and have the same number under the root sign.

Examples:

Simplify the following surds:

1. $6\sqrt{3} + 4\sqrt{3}$
   Solution

$$6\sqrt{3} + 4\sqrt{3} = (6 + 4)\sqrt{3} = 10\sqrt{3}$$

2. $5\sqrt{3}-\sqrt{27}$
   Solution

$$5\sqrt{3}-\sqrt{27}=5\sqrt{3}-\sqrt{3^2\times 3}=5\sqrt{3}-3\sqrt{3}=(5-3)\sqrt{3}=2\sqrt{3}$$

3. $\sqrt[3]{5}+5\sqrt{2}+\frac{5}{2}\sqrt[3]{5}+\frac{3}{7}\sqrt{2}$
   Solution

$$\sqrt[3]{5}+5\sqrt{2}+\frac{5}{2}\sqrt[3]{5}+\frac{3}{7}\sqrt{2}=\left(1+\frac{5}{2}\right)\sqrt[3]{5}+\left(5+\frac{3}{7}\right)\sqrt{2}=\frac{7}{2}\sqrt[3]{5}+\frac{38}{7}\sqrt{2}$$

1.2.2 Multiplication and division of surds

Two or more surds can be multiplied or divided if they are of the same order.
Examples:
Simplify the following surds:

1. $\left(5\sqrt{2}+3\sqrt{3}\right)\left(4\sqrt{2}-5\sqrt{3}\right)$
   Solution

$$\begin{aligned}\left(5\sqrt{2}+3\sqrt{3}\right)\left(4\sqrt{2}-5\sqrt{3}\right) &= 5\sqrt{2}\left(4\sqrt{2}-5\sqrt{3}\right)+3\sqrt{3}\left(4\sqrt{2}-5\sqrt{3}\right)\\ &= 20\sqrt{2}\sqrt{2}-25\sqrt{2}\sqrt{3}+12\sqrt{3}\sqrt{2}-15\sqrt{3}\sqrt{3}\\ &= 20\times 2-25\sqrt{2\times 3}+12\sqrt{2\times 3}-15\times 3\\ &= 40-25\sqrt{6}+12\sqrt{6}-45=-5-13\sqrt{6}\end{aligned}$$

2. $\sqrt[3]{5}\times 5\sqrt[3]{2}$
   Solution

$$\sqrt[3]{5}\times 5\sqrt[3]{2}=5\sqrt[3]{5\times 2}=5\sqrt[3]{10}$$

3. $\frac{\sqrt[3]{18}\times\sqrt{48}}{\sqrt[4]{216}}$
   Solution
   The LCM of 2, 3, and 4 is 12. Expressing the surds as 12th root yields:

$$\begin{aligned}\frac{\sqrt[3]{18}\times\sqrt{48}}{\sqrt[4]{216}} &= \frac{\sqrt[12]{18^4}\times\sqrt[12]{48^6}}{\sqrt[12]{216^3}}=\sqrt[12]{\frac{\cancel{18^1}\times\cancel{18^1}\times\cancel{18^1}\times 18\times\cancel{48^4}\times\cancel{48^4}\times\cancel{48^4}\times 48\times 48\times 48}{\cancel{216^1}\times\cancel{216^1}\times\cancel{216^1}}}\\ &= \sqrt[12]{18\times 4\times 4\times 4\times 48\times 48\times 48}=\sqrt[12]{2^{19}\times 3^5}=\sqrt[12]{2^{12}}\times\sqrt[12]{2^7\times 3^5}\\ &= 2\sqrt[12]{31104}\end{aligned}$$

Exercise:

1. Show that $\sqrt{\frac{85}{12}}\times\sqrt{\frac{63}{136}}=\frac{1}{4}\sqrt{30}$

1.2.3 Rationalizing the denominator

If fractions are involved in surds, it is appropriate to eliminate the surds from the denominator. The aim here is to work with a rational numbers. Rationalizing the denominator means multiplying both the denominator and the numerator by the conjugate of the denominator. Note that the conjugate of $(a+\sqrt{b})$ is $(a-\sqrt{b})$.
Examples:

1. Determine $\frac{1}{\sqrt{7}}$ correct to 6 decimal places without using a calculator, given that $\sqrt{7}=2.6457513$.
   Solution

$$\frac{1}{\sqrt{7}} = \frac{1 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{7}}{7} = \frac{2.6457513}{7} \approx 0.377964$$

2. Rationalize the denominator.

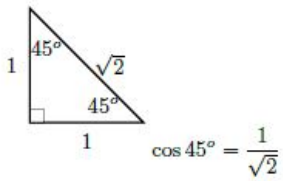
   (a) $\frac{\sqrt{6}+\sqrt{3}}{\sqrt{6}-\sqrt{3}}$
   Solution

$$\frac{\sqrt{6}+\sqrt{3}}{\sqrt{6}-\sqrt{3}} = \frac{\left(\sqrt{6}+\sqrt{3}\right)\left(\sqrt{6}+\sqrt{3}\right)}{\left(\sqrt{6}-\sqrt{3}\right)\left(\sqrt{6}+\sqrt{3}\right)} = \frac{6+2\sqrt{6}\sqrt{3}+3}{6-3} = \frac{9+2\sqrt{18}}{3} = \frac{9+6\sqrt{2}}{3} = 3+2\sqrt{2}$$

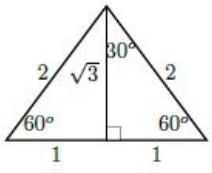
   (b) $\frac{\sqrt{6}}{5\sqrt{2}+\sqrt{3}}$
   Solution

$$\frac{\sqrt{6}}{5\sqrt{2}+\sqrt{3}} = \frac{\sqrt{6}\left(5\sqrt{2}-\sqrt{3}\right)}{\left(5\sqrt{2}+\sqrt{3}\right)\left(5\sqrt{2}-\sqrt{3}\right)} = \frac{5\sqrt{6}\sqrt{2}-\sqrt{6}\sqrt{3}}{50-3} = \frac{5\sqrt{12}-\sqrt{18}}{47} = \frac{10\sqrt{3}-3\sqrt{2}}{47}$$

3. Simplify without using tables or calculator $3\ cos^2 45° cos\ 42° + tan^2 60° sin\ 48° - 9\ cos\ 60°\ cos\ 42°$
   Solution
   From the property of complementary angles, we have $sin\ 48° = cos(90° - 48°) = cos\ 42°$. Also, we use the following special triangles and apply SOHCAHTOA to find the trigonometric ratios of the special angles.



$\cos 45^o = \frac{1}{\sqrt{2}}$



$\cos 60^o = \frac{1}{2}, \tan 60^o = \sqrt{3}$

$$\begin{aligned} 3\cos^2 45^o \cos 42^o + \tan^2 60^o \sin 48^o - 9\cos 60^o \cos 42^o &= 3\left(\frac{1}{\sqrt{2}}\right)^2 \cos 42^o + \left(\sqrt{3}\right)^2 \cos 42^o - 9\left(\frac{1}{2}\right)\cos 42^o \\ &= \left(\frac{3}{2}+3-\frac{9}{2}\right)\cos 42^o = 0 \end{aligned}$$

Exercise:

1. Simplify $\frac{2\sqrt{2}-\sqrt{3}}{5\sqrt{2}-\sqrt{3}}$ by rationalizing the denominator.
2. Simplify $\frac{2\sqrt{5}-3}{3\sqrt{5}-2}$ by rationalizing the denominator.
3. Without using tables or calculator;
   (a) Evaluate $\frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}}$ correct to 6dp, given that $\sqrt{35} = 5.9160798$
   (b) Show that $\frac{sin^2 315°(1-tan^2 210°)}{(1+cos\ 120°)(1+tan^2\ 330°)} = \frac{1}{2}$
4. Simplify $\frac{3}{\sqrt[3]{5}-2}$ by rationalizing the denominator. [Hint: $a^3 - b^3 = (a-b)(a^2+ab+b^2]$
5. Express $\frac{3\sqrt{2}+5\sqrt{6}}{3\sqrt{2}-5\sqrt{6}}$ in the form $a + b\sqrt{c}$, where a, b and c are rational numbers. State the values of a, b and c.
6. Given that $\beta = 8 + 3\sqrt{7}$, express $\beta + \frac{1}{\beta}$ in the form $a + b\sqrt{c}$, where a, b and c are rational numbers.

Lecture 2

2 Indices and Logarithms

2.1 Indices

An index is the number of times a number is multiplied by itself.

I: Laws of indices

There are three laws of indices related to powers of the same base a.

(1) $a^m \times a^n = a^{m+n}$

(2) $a^m \div a^n = a^{m-n}$

(3) $(a^m)^n = a^{m \times n}$

II: Rational indices

Consider $a^{1/n}$, where n is a positive integer. From law (3), we have

$$\left(a^{1/n}\right)^n = a^{(1/n)\times n} = a^1 \quad \Rightarrow \quad \boxed{a^{1/n} = \sqrt[n]{a}}$$

i) Now, $a^{m/n} = \left(a^{1/n}\right)^m = (\sqrt[n]{a})^m$.

ii) Also, $a^{m/n} = (a^m)^{1/n} = \sqrt[n]{a^m}$.

Therefore, $\boxed{\sqrt[n]{a^m} = (\sqrt[n]{a})^m}$. For example,

i) $(16)^{3/4} = \left(16^{1/4}\right)^3 = 2^3 = 8.$

ii) $(16)^{3/4} = \left(16^3\right)^{1/4} = \sqrt[4]{4096} = 8.$

III: Zero and negative indices

i) $a^m \div a^m = \dfrac{a^m}{a^m} = 1.$

ii) Also, $a^m \div a^m = a^{m-m} = a^0$ [from law (2)].

Therefore, $\boxed{a^0 = 1}$. Similarly,

i) $a^0 \div a^n = 1 \div a^n = \dfrac{1}{a^n}.$

ii) Also, $a^0 \div a^n = a^{0-n} = a^{-n}$ [from law (2)].

Therefore, $\boxed{a^{-n} = \dfrac{1}{a^n}}$.

Examples:

1. Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$
   Solution

$$\left(\frac{27}{8}\right)^{-2/3} = \left[\left(\frac{27}{8}\right)^{-1}\right]^{2/3} = \left(\frac{8}{27}\right)^{2/3} = \left(\sqrt[3]{\frac{8}{27}}\right)^{2} = \left(\frac{2}{3}\right)^{2} = \frac{4}{9}$$

2. Simplify $\frac{(1+x)^{\frac{1}{2}} - \frac{1}{2}x(1+x)^{-\frac{1}{2}}}{(1+x)}$
   Solution
   Multiplying both the numerator and denominator by $2(1+x)^{\frac{1}{2}}$, we get

$$\frac{\left[(1+x)^{1/2} - \frac{1}{2}x(1+x)^{-1/2}\right] \cdot 2(1+x)^{1/2}}{(1+x)\cdot 2(1+x)^{1/2}} = \frac{2(1+x)-x}{2(1+x)^{3/2}} = \frac{2+x}{2(1+x)^{3/2}}$$

Exercise:

1. Find the value of the following
   (a) $(0.04)^{-3/2}$. [ans: 125]
   (b) $\dfrac{4^{-3/2}}{8^{-2/3}}$. [ans: 1/2]
2. Simplify:
   (a) $3^{n+1} \times 9^n \div 27^{(2/3)n}$. [ans: 3^{n+1}]
   (b) $9^{-(1/2)n} \times 3^{n+2} \times 81^{-1/4}$. [ans: 3]
   (c) $6^{(1/2)n} \times 12^{n+1} \times 27^{-(1/2)n} \div 32^{(1/2)n}$. [ans: 12]
   (d) $10^{(1/3)n} \times 15^{(1/2)n} \times 6^{(1/6)n} \div 45^{(1/3)n}$. [ans: $10^{n/2}$]
   (e) $\dfrac{x^{p+(1/2)q} \cdot y^{2p-q}}{(xy^2)^p \cdot \sqrt{x^q}}$. [ans: y^{-q}]
3. Simplify:
   (a) $\dfrac{\sqrt{xy} \cdot x^{1/3} \cdot 2y^{1/4}}{(x^{10}y^9)^{1/12}}$. [ans: 2]
   (b) $\dfrac{-\frac{1}{2}x(1-x)^{-1/2} + (1-x)^{1/2}}{x^2}$. [ans: $\dfrac{x-2}{2x^2(1-x)^{1/2}}$]
   (c) $\dfrac{\frac{1}{2}x^{1/2}(1+x)^{-1/2} - \frac{1}{2}x^{-1/2}(1+x)^{1/2}}{x}$. [ans: $\dfrac{-1}{2x^{3/2}(1+x)^{1/2}}$]
   (d) $\dfrac{(1+x)^{1/3} - \frac{1}{3}x(1+x)^{-2/3}}{(1+x)^{2/3}}$. [ans: $\dfrac{3+2x}{3(1+x)^{4/3}}$]
4. Find the values of x which satisfy the following equations.
   (a) $\left(\frac{4}{5}\right)^{5x-4} = \left(\frac{5}{4}\right)^{4x-5}$. [ans: $x = 1$]
   (b) $x^{\sqrt{x}} = (\sqrt{x})^x$. [ans: $x = 0$ or $x = 4$]
   (c) $2^{2x+3} + 2^{x+3} = 1 + 2^x$. [ans: $x = -3$]

Q3. (c) $\dfrac{\frac{1}{2}x^{\frac{1}{2}}(1+x)^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}(1+x)^{\frac{1}{2}}}{x}$

$\Rightarrow \dfrac{\frac{1}{2}\left[x^{\frac{1}{2}}(1+x)^{-\frac{1}{2}} - x^{-\frac{1}{2}}(1+x)^{\frac{1}{2}}\right]}{x}$

$\Rightarrow \frac{1}{2x}\left[x^{\frac{1}{2}}(1+x)^{-\frac{1}{2}} - x^{-\frac{1}{2}}(1+x)^{\frac{1}{2}}\right]$.

$\Rightarrow \frac{1}{2}\left[x^{(\frac{1}{2}-1)}(1+x)^{-\frac{1}{2}} - x^{(-\frac{1}{2}-1)}(1+x)^{\frac{1}{2}}\right]$

$\Rightarrow \frac{1}{2}\left[x^{-\frac{1}{2}}(1+x)^{-\frac{1}{2}} - x^{-\frac{3}{2}}(1+x)^{\frac{1}{2}}\right]$.

Q. $\dfrac{\frac{1}{2}x^{-\frac{1}{2}}(1+x)^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}(1+x)^{\frac{1}{2}}}{x}$

$\Rightarrow \dfrac{\frac{1}{2}x^{-\frac{1}{2}}\left[(1+x)^{-\frac{1}{2}} - (1+x)^{\frac{1}{2}}\right]}{x}$

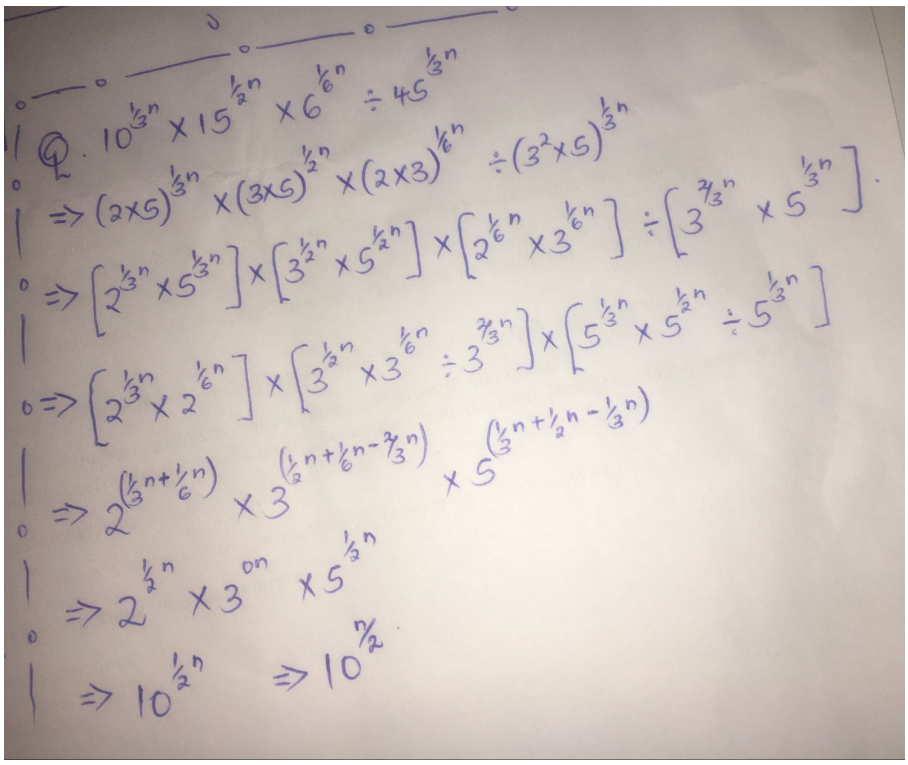
$\Rightarrow \frac{1}{2}x^{(-\frac{1}{2}-1)}\left[\frac{1}{\sqrt{1+x}} - \sqrt{1+x}\right]$

$\Rightarrow \frac{1}{2}x^{-\frac{3}{2}}\left[\frac{1-(1+x)}{\sqrt{1+x}}\right]$

$\Rightarrow \frac{1}{2}x^{-\frac{3}{2}}\left[\frac{-x}{\sqrt{1+x}}\right]$

$\Rightarrow \frac{1}{2}x^{-\frac{1}{2}}\left[\frac{-1}{\sqrt{1+x}}\right]$

$\Rightarrow \dfrac{-1}{2x^{\frac{1}{2}}(1+x)^{\frac{1}{2}}}$.



2.2 Logarithms

Logarithms and indices are closely related. Let a be a positive real number, where $a \neq 1$. If $a^x = y$, then x is the logarithm of y to base a and is written as:

$$x = log_a\ y$$

For example,

1. $10^2 = 100 \implies 2 = log_{10}\ 100$ The index 2 is the logarithm of 100 to base 10
2. $2^3 = 8 \implies 3 = log_2\ 8$ The index 3 is the logarithm of 8 to base 2.
3. $125^{\frac{1}{3}} = 5 \implies \frac{1}{3} = log_{125}\ 5$

→ Note: the base for a logarithm can only be a positive real number. Logarithms to base 10 are called *common logarithms*. In logarithms to base 10, we sometimes suppress the base e.g., $\log_{10} a = \log a$. In advanced mathematics, we have the number $e \approx 2.71828$. Logarithms to base e are called *natural logarithms*/Napierian logarithms. Thus, $\log_e x = \ln x$.

I: Laws of logarithms

(1) $\boxed{\log_a(x \cdot y) = \log_a x + \log_a y}$.

Proof. Let $\log_a x = m$ and $\log_a y = n$. In index form, we have $x = a^m$ and $y = a^n$. Multiplying yields

$$xy = a^m \cdot a^n = a^{m+n} \quad \text{(by 1st law of indices)}$$

In logarithm form, we have

$$\log_a xy = m + n$$

Replacing the values of m and n, we obtain $\log_a(xy) = \log_a x + \log_a y$. □

(2) $\boxed{\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y}$.

Proof. Let $\log_a x = m$ and $\log_a y = n$. In index form, we have $x = a^m$ and $y = a^n$. Dividing x by y yields

$$\frac{x}{y} = \frac{a^m}{a^n} = a^{m-n} \quad \text{(by 2nd law of indices)}$$

In logarithm form, we have

$$\log_a(x/y) = m - n$$

Replacing the values of m and n, we obtain $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$. □

(3) $\boxed{\log_a x^p = p \log_a x}$ for any base $a > 0$.

Proof. Let $\log_a x = m$. In index form, we have $x = a^m$. Raising both sides to power p, we get

$$x^p = (a^m)^p = a^{pm} \quad \text{(by 3rd law of indices)}$$

In logarithm form, we have

$$\log_a x^p = pm$$

Replacing the value of m, we obtain $\log_a x^p = p \log_a x$. □

II: Other laws of logarithms

For any $a > 0$,

(4) $a^0 = 1 \quad \Rightarrow \quad \boxed{\log_a 1 = 0}$.

(5) $a^1 = a \quad \Rightarrow \quad \boxed{\log_a a = 1}$.

Examples:

1. Simplify

(a) $\log_2(8 \times 16)$.

Solution

$$\log_2(8 \times 16) = \log_2 8 + \log_2 16 = \log_2 2^3 + \log_2 2^4 = 3\log_2 2 + 4\log_2 2 = 3 + 4 = 7$$

(b) $\dfrac{\log 125}{\log 25}$.

Solution

$$\frac{\log 125}{\log 25} = \frac{\log 5^3}{\log 5^2} = \frac{3\cancel{\log 5}}{2\cancel{\log 5}} = \frac{3}{2}$$

2. Write $\log\left(\dfrac{a^2b^3}{100\sqrt{c}}\right)$ in terms of $\log a$, $\log b$, and $\log c$.

Solution

$$\begin{aligned}
\log\left(\frac{a^2b^3}{100\sqrt{c}}\right) &= \log(a^2b^3) - \log(100 \times c^{1/2}) \quad \text{(by 2nd law of logarithms)} \\
&= \log a^2 + \log b^3 - \log 10^2 - \log c^{1/2} \quad \text{(by 1st law of logarithms)} \\
&= 2\log a + 3\log b - 2\log_{10} 10 - \frac{1}{2}\log c \quad \text{(by 3rd law of logarithms)} \\
&= 2\log a + 3\log b - 2 - \frac{1}{2}\log c
\end{aligned}$$

3. Solve the equation $\log_5 x + \log_5(2x-3) = 1$.

Solution

$\log_5 x + \log_5(2x-3) = 1 \quad \Rightarrow \quad \log_5 x(2x-3) = 1$. In index form, we have $x(2x-3) = 5$. Expanding and rearranging yields $2x^2 - 3x - 5 = 0$. Factorizing the expression on the left, we get $(x+1)(2x-5) = 0$. Thus, either $(x+1) = 0$ or $(2x-5) = 0$. Hence, $x = -1$ or $x = \dfrac{5}{2}$. But logarithms of negative numbers do not exist. Therefore, $x = \dfrac{5}{2}$.

4. Simplify $\dfrac{\log 75 + \log 9 + \log 5}{\log 5 + \log 45}$ without using a calculator.

Solution

$$\begin{aligned}
\frac{\log 75 + \log 9 + \log 5}{\log 5 + \log 45} &= \frac{\log(75 \times 9 \times 5)}{\log(5 \times 45)} = \frac{\log(3 \times 5^2 \times 3^2 \times 5)}{\log(5 \times 3^2 \times 5)} = \frac{\log(3^3 \times 5^3)}{\log(5^2 \times 3^2)} \\
&= \frac{\log 3^3 + \log 5^3}{\log 3^2 + \log 5^2} = \frac{3\log 3 + 3\log 5}{2\log 3 + 2\log 5} = \frac{3\cancel{(\log 3 + \log 5)}}{2\cancel{(\log 3 + \log 5)}} = \frac{3}{2}
\end{aligned}$$

III: **Changing base of a logarithm**

If $\log_a x = m$, then in index notation we have $x = a^m$. Applying logarithms to base b on both sides, we get $\log_b x = \log_b a^m \quad \Rightarrow \quad \log_b x = m\log_b a \quad \Rightarrow \quad m = \dfrac{\log_b x}{\log_b a}$. Replacing the value of m, we obtain

$$\boxed{\log_a x = \frac{\log_b x}{\log_b a}}$$

Examples:

1. Evaluate $log_7\ 2$

Solution

Changing the base to 10, yields: $\log_7 2 = \dfrac{\log_{10} 2}{\log_{10} 7} = 0.3562 \quad (4\text{dp}).$

2. Show that $log_a x = \frac{1}{log_x a}$
Solution

Changing the base to x, yields: $\log_a x = \dfrac{\log_x x}{\log_x a} = \dfrac{1}{\log_x a}.$

3. Solve the equation $log_3 x - 4log_x 3 + 3 = 0$
Solution

$\log_3 x - 4\log_x 3 + 3 = 0 \quad \Rightarrow \quad \log_3 x - 4\left(\dfrac{1}{\log_3 x}\right) + 3 = 0$. Let $\log_3 x = y$. Thus, we have

$y - 4\left(\dfrac{1}{y}\right) + 3 = 0 \quad \Rightarrow \quad y^2 + 3y - 4 = 0$. Solving using quadratic formula, we get

$$y = \frac{-3 \pm \sqrt{9+16}}{2} = \frac{-3 \pm 5}{2} \quad \Rightarrow \quad y = -4 \text{ or } y = 1$$

Now,

$$\log_3 x = -4 \quad \Rightarrow \quad x = 3^{-4} = \frac{1}{81} \quad \text{and} \quad \log_3 x = 1 \quad \Rightarrow \quad x = 3$$

4. If $x = log_9 5$ and $y = log_3 5$, show that $y = 2x$
Solution

Changing the base to 3, we get

$$x = \log_9 5 = \frac{\log_3 5}{\log_3 9} = \frac{y}{\log_3 3^2} = \frac{y}{2\log_3 3} = \frac{y}{2} \quad \Rightarrow \quad y = 2x \quad \text{as required}$$

5. If $log_{10} 2 = a$, show that $log_8 5 = \frac{1-a}{3a}$
Solution

Changing the base to 10, we get

$$\log_8 5 = \frac{\log_{10} 5}{\log_{10} 8} = \frac{\log_{10}\left(\dfrac{10}{2}\right)}{\log_{10} 2^3} = \frac{\log_{10} 10 - \log_{10} 2}{3\log_{10} 2} = \frac{1-a}{3a}$$

6. Solve for x if $log_3 x + log_9 x^2 = 6$
Solution

Changing the base to 3, we get

$$\log_3 x + \log_9 x^2 = 6 \quad \Rightarrow \quad \log_3 x + \frac{\log_3 x^2}{\log_3 9} = 6 \quad \Rightarrow \quad \log_3 x + \frac{2\log_3 x}{2\log_3 3} = 6$$

$$\Rightarrow \quad \log_3 x + \log_3 x = 6 \quad \Rightarrow \quad 2\log_3 x = 6 \quad \Rightarrow \quad \log_3 x = 3 \quad \Rightarrow \quad x = 3^3 = 27$$

Exercise:

1. (a) Solve for x if $\left(\frac{2}{3}\right)^x = \frac{1}{16}$. [ans: $x =$]

   (b) Solve for x given that $\log_{10} 4 + 2\log_{10} x = 2$. [ans: $x = 5$]

2. Find y in terms of x if $\log\left(\frac{x^2}{y}\right) = 5 - 2\log x$. [ans: $y = \frac{x^4}{10^5}$]

3. Solve for x in the following equations.

   (a) $4^x - 6(2^x) - 16 = 0$. [ans: $x = 3$]

   (b) $\log_2 x = \log_x 16$. [ans: $x = 4$]

   (c) $\log_2 8x^3 - \log_x 8x^3 = 8$. [ans: $x = 8$ or $x = 2^{-1/3}$]

   (d) $\log_3 x - 2\log_x 3 = 1$. [ans: $x = 9$ or $x = \frac{1}{3}$]

   (e) $\log_x 2 + \log_x 3 = 5$. [ans: $x = 6^{1/5}$]

4. Show that $\log_{16} xy = \frac{1}{2}\log_4 x + \frac{1}{2}\log_4 y$. Hence, solve the simultaneous equations:

$$\log_{16} xy = \sqrt{3} \quad (i) \quad \text{and} \quad \frac{\log_4 x}{\log_4 y} = -8 \quad (ii)$$

Lecture 3

3 Quadratic equations, functions and inequalities

3.1 Quadratic equations

A quadratic equation is an equation of the form $ax^2 + bx + c = 0$, where a, b, and c are constants, x is a variable, and $a \neq 0$.

Methods of solving quadratic equations

(a) Factorization method

(b) Completing the square

(c) Quadratic formula

(d) Graphical method

3.1.1 Factorization method

It involves expressing the quadratic expression as a product of two linear functions (polynomials of degree 1) and then solving for the unknown. To factorize a quadratic expression of the form $ax^2 + bx + c$, we determine two numbers whose sum is b and whose product is ac.

Examples:

1. Solve the equation $2x^2 - 5x - 3 = 0$ by factorization method.

   Solution

   We need to find two number whose sum is -5 and product is -6. Thus, the numbers are -6 and 1. So, $2x^2 - 5x - 3 = 0$ becomes

$$2x^2 - 6x + x - 3 = 0 \quad \Rightarrow \quad 2x(x-3) + 1(x-3) = 0 \quad \Rightarrow \quad (2x+1)(x-3) = 0$$

$$\Rightarrow \quad (2x+1) = 0 \quad \text{or} \quad (x-3) = 0 \quad \Rightarrow \quad x = -\frac{1}{2} \quad \text{or} \quad x = 3$$

   Hence, $(2x + 1)$ and $(x - 3)$ are called (linear) factors of the quadratic expression $2x^2 - 5x - 3$ while $x = -\frac{1}{2}$ and $x = 3$ are called roots/zeros/solutions of the quadratic equation $2x^2 - 5x - 3 = 0$.

Exercise:

Solve the following quadratic equations using the factorization method:

(i) $x^2 - 10x + 24 = 0$

(ii) $x^2 + 6x + 9 = 0$

(iii) $4x^2 + 10x - 6 = 0$

(iv) $5x^2 - 17x + 6 = 0$

(v) $25x^2 - 9 = 0$

3.1.2 Completing the square

Consider the expansion of $(x + a)^2$ for a real number a:

$$(x + a)^2 = x^2 + 2ax + a^2$$

For all such expansions, the constant term is the square of half coefficient of x. This forms the basis for the solution of quadratic equations by completing the square method. Thus if $x^2 + bx + c$ is a complete square, then $\boxed{c = \left(\frac{b}{2}\right)^2}$.

Note: This method is appropriate when factorization is not possible.

Examples:

1. Solve the equation $x^2 - 8x + 11 = 0$ by completing the square.
   Solution
   The given equation can be written as $x^2 - 8x = -11 \quad \Rightarrow \quad x^2 - 8x + c = -11 + c$, where $c = \left(\frac{-8}{2}\right)^2 = (-4)^2$. Thus, we have

$$x^2 - 8x + (-4)^2 = -11 + (-4)^2 \quad \Rightarrow \quad (x-4)^2 = 5 \quad \Rightarrow \quad (x-4) = \pm\sqrt{5}$$

$$\Rightarrow \quad x = 4 \pm \sqrt{5}. \quad \text{Either} \quad x = 4 + \sqrt{5} \quad \text{or} \quad x = 4 - \sqrt{5}$$

2. Solve the equation $5x^2 - 6x - 2 = 0$ by completing the square.
   Solution
   We first need to make the coefficient of x^2 be 1. Thus, the given equation can be written as $x^2 - \frac{6}{5}x = \frac{2}{5} \quad \Rightarrow \quad x^2 - \frac{6}{5}x + c = \frac{2}{5} + c$, where $c = \left(\frac{-6}{10}\right)^2 = \left(\frac{-3}{5}\right)^2$. Thus, we have

$$x^2 - \frac{6}{5}x + \left(\frac{-3}{5}\right)^2 = \frac{2}{5} + \left(\frac{-3}{5}\right)^2 \quad \Rightarrow \quad \left(x - \frac{3}{5}\right)^2 = \frac{19}{25} \quad \Rightarrow \quad \left(x - \frac{3}{5}\right) = \pm\sqrt{\frac{19}{25}} = \pm\frac{\sqrt{19}}{5}$$

$$\Rightarrow \quad x = \frac{3}{5} \pm \frac{\sqrt{19}}{5}. \quad \text{Either} \quad x = \frac{3 + \sqrt{19}}{5} \quad \text{or} \quad x = \frac{3 - \sqrt{19}}{5}$$

Exercise:
Solve by completing the square

(i) $2x^2 - 6x - 1 = 0$

(ii) $x^2 + 7x - 3 = 0$

(iii) $5x^2 + 12x + 6 = 0$

(iv) $10 + 3x - 2x^2 = 0$

(v) $2 - 2x - x^2 = 0$

(vi) $-7 + 12x - 3x^2 = 0$

(vii) $2x^2 + 6x + 13 = 0$

3.1.3 Quadratic formula

Solve the quadratic equation $ax^2 + bx + c = 0$ by completing the square method (where a, b, and c are real numbers and $a \neq 0$).
Solution
We first need to make the coefficient of x^2 be 1. Thus, the given equation can be written as $x^2 + \frac{b}{a}x = -\frac{c}{a} \quad \Rightarrow \quad x^2 + \frac{b}{a}x + k = -\frac{c}{a} + k$, where $k = \left(\frac{b}{2a}\right)^2$. Thus, we have

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \quad \Rightarrow \quad \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad \Rightarrow \quad \left(x + \frac{b}{2a}\right) = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\Rightarrow \quad x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \quad \Rightarrow \quad \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}, \quad \text{which is the quadratic formula.}$$

The expression $\boxed{b^2 - 4ac}$ is known as the discriminant. The nature of the roots of a quadratic equation depend on the discriminant.

- If $b^2 - 4ac > 0$, then the roots of the quadratic equation $ax^2 + bx + c = 0$ are real are distinct.
- If $b^2 - 4ac = 0$, then the roots of the quadratic equation $ax^2 + bx + c = 0$ are real are equal.
- If $b^2 - 4ac < 0$, then the roots of the quadratic equation $ax^2 + bx + c = 0$ are complex conjugates.

Examples:

1. Giving your answer correct to 2dp, solve $2x^2 - 6x - 3 = 0$ using the quadratic formula.
   Solution
   Here, $a = 2, b = -6$ and $c = -3$. Substituting these values into the quadratic formula, we get
   $$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{6 \pm \sqrt{36 + 24}}{4} = \frac{6 \pm \sqrt{60}}{4} = \frac{6 \pm 2\sqrt{15}}{4} = \frac{3 \pm \sqrt{15}}{2}$$
   Either $x = \frac{3 + \sqrt{15}}{2} \approx 3.44$ or $x = \frac{3 - \sqrt{15}}{2} \approx -0.44$ (2dp).

2. Find the positive value of k if the equation $x^2 + (2 + k)x + k^2 = 0$ has equal roots.
   Solution
   Here, $a = 1, b = (2 + k)$ and $c = k^2$. For equal roots, we require that the discriminant be zero. That is, $b^2 - 4ac = 0$. Substituting yields $(2 + k)^2 - 4k^2 = 0 \quad \Rightarrow \quad 3k^2 - 4k - 4 = 0$, which is a quadratic equation in the unknown k. Using the quadratic formula, we have
   $$k = \frac{4 \pm \sqrt{16 + 48}}{6} = \frac{4 \pm 8}{6} = \frac{2 \pm 4}{3}. \quad \text{Either} \quad k = 2 \quad \text{or} \quad k = -\frac{2}{3}$$
   Only $k = 2$ is positive.

3.1.4 Graphical method

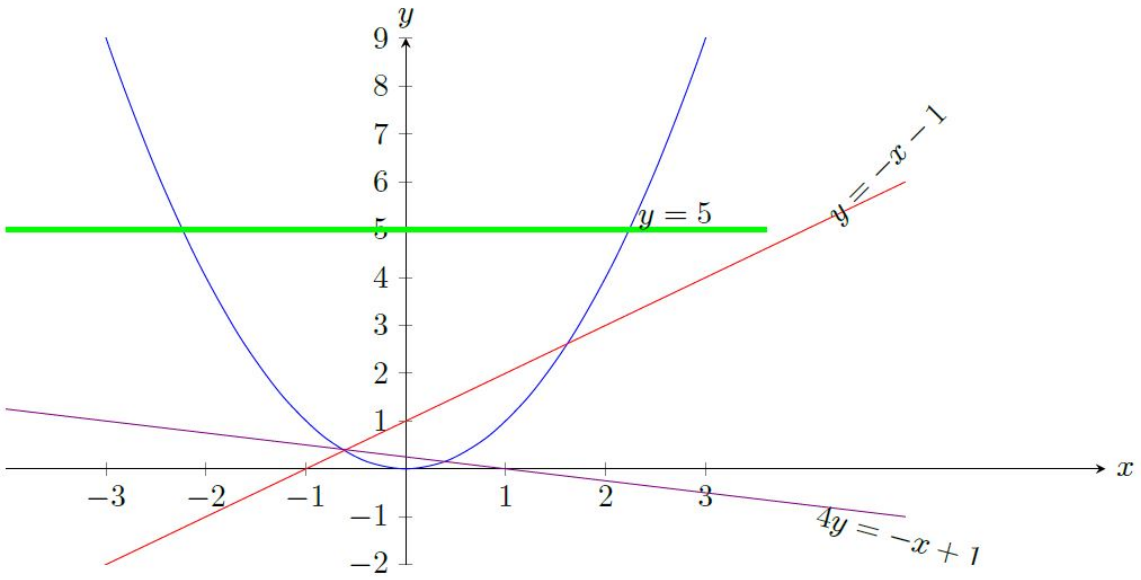
This method has low degree of accuracy since it involves some approximations.

Examples:

1. Draw the graph of $y = x^2$ for $-3 \leq x \leq 3$. Hence, use the graph to:
   (a) find $\sqrt{5}$ correct to 2dp.
   (b) solve the equations
      (i) $x^2 - x - 1 = 0$.
      (ii) $4x^2 + x - 1 = 0$.

Solution

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9



(a) From the line $y = 5$ and the graph of $y = x^2 \quad \Rightarrow \quad x = \sqrt{y} = \pm\sqrt{5} \approx \pm 2.24$ (from the graph).

(b) (i)

$y = x^2$
$0 = x^2 - x - 1$

$y = x + 1$ (plot this line on the same axes then read the x coordinates at the intersection of the graphs of $y = x + 1$ and $y = x^2$, to obtain the required roots). These are: $x = -0.62$ and $x = 1.62$ (2dp).

(ii)

$4y = 4x^2$
$0 = 4x^2 + x - 1$

$4y = -x + 1$ (plot this line on the same axes then read the x coordinates at the intersection of the graphs of $4y = -x + 1$ and $y = x^2$, to obtain the required roots). These are: $x = -0.64$ and $x = 0.39$ (2dp).

Exercise:

Draw the graph of $y = 2x^2 - 12x + 19$ for $1 \leq x \leq 5$. Hence, use the graph to find the roots of the equations.

(a) $x^2 - 6x + 6 = 0$. [hint: $y = 7$, ans: $x \approx 1.3, x \approx 4.7$]

(b) $4x^2 - 25x + 28 = 0$. [hint: $2y = x + 10$, ans: $x \approx 1.45, x \approx 4.80$]

3.1.5 Roots of quadratic equations

Consider the equation $ax^2 + bx + c = 0$. Then on rewriting

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \qquad (a)$$

Now, suppose α and β are the roots of equation (a), then $x = \alpha$ or $x = \beta$. Thus, we have $(x - \alpha) = 0$ or $(x - \beta) = 0 \Rightarrow \quad (x - \alpha)(x - \beta) = 0$. Expanding yields

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \qquad (b)$$

Comparing equations (a) and (b), we obtain

$$\alpha + \beta = -\frac{b}{a} \quad \text{and} \quad \alpha\beta = \frac{c}{a}$$

$\rightarrow$ Note: if we are required to write down an equation whose roots are known, then the required equation is given by

$$x^2 - (\text{sum of the roots})x + (\text{product of the roots}) = 0$$

Examples:

1. Write down the sum and products of the roots of $3x^2 - 2x - 7 = 0$

   Solution

   If the roots are α and β, then

   $$\alpha + \beta = -\left(\frac{-2}{3}\right) = \frac{2}{3} \quad \text{and} \quad \alpha\beta = \frac{c}{a} = -\frac{7}{3}$$

2. Express each of the following in terms of $\alpha + \beta$ and $\alpha\beta$:

   (a) $\alpha^2 + \beta^2$

   (b) $(\alpha - \beta)^2$

   (c) $\alpha^3 + \beta^3$

   Solution

   (a) α^2 and β^2 occur in the expansion of $(\alpha + \beta)^2$. Thus,

   $$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2 \quad \Rightarrow \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

   (b)

   $$\begin{aligned} (\alpha - \beta)^2 &= \alpha^2 - 2\alpha\beta + \beta^2 = \left(\alpha^2 + \beta^2\right) - 2\alpha\beta \\ &= (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta \\ &= (\alpha + \beta)^2 - 4\alpha\beta \end{aligned}$$

   (c) α^3 and β^3 occur in the expansion of $(\alpha + \beta)^3$. Thus,

   $$\begin{aligned} (\alpha + \beta)^3 &= \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta) \\ \Rightarrow \quad \alpha^3 + \beta^3 &= (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \end{aligned}$$

3. The roots of the equation $3x^2 + 4x - 5 = 0$ are α and β. Find the equation with integer coefficients whose roots are:

(a) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

(b) α^2 and β^2.

(c) $(\alpha+1)$ and $(\beta+1)$.

(d) $\alpha^2\beta$ and $\alpha\beta^2$.

(e) $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

(f) $\frac{(1+\alpha)}{\beta}$ and $\frac{(1+\beta)}{\alpha}$.

(g) $(\alpha-1)^2$ and $(\beta-1)^2$.

Solution

From the equation $3x^2+4x-5=0$, we have $(\alpha+\beta)=-\frac{b}{a}=-\frac{4}{3}$ and $\alpha\beta=\frac{c}{a}=-\frac{5}{3}$.

(a) The roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

Sum of the roots: $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{-4/3}{-5/3}=\frac{4}{5}$

Product of the roots: $\frac{1}{\alpha}\cdot\frac{1}{\beta}=\frac{1}{\alpha\beta}=\frac{1}{-5/3}=-\frac{3}{5}$

Equation: $x^2-(\text{sum of the roots})x+(\text{product of the roots})=0$

$\Rightarrow \quad x^2-\frac{4}{5}x-\frac{3}{5}=0 \quad \Rightarrow \quad 5x^2-4x-3=0$

(b) The roots are α^2 and β^2.

Sum of the roots: $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\left(-\frac{4}{3}\right)^2-2\left(-\frac{5}{3}\right)=\frac{16}{9}+\frac{10}{3}=\frac{46}{9}$

Product of the roots: $\alpha^2\cdot\beta^2=(\alpha\beta)^2=\left(-\frac{5}{3}\right)^2=\frac{25}{9}$

Equation: $x^2-(\text{sum of the roots})x+(\text{product of the roots})=0$

$\Rightarrow \quad x^2-\frac{46}{9}x+\frac{25}{9}=0 \quad \Rightarrow \quad 9x^2-46x+25=0$

(c) The roots are $(\alpha+1)$ and $(\beta+1)$.

Sum of the roots: $(\alpha+1)+(\beta+1)=(\alpha+\beta)+2=\left(-\frac{4}{3}\right)+2=\frac{2}{3}$

Product of the roots: $(\alpha+1)(\beta+1)=(\alpha\beta)+(\alpha+\beta)+1=\left(-\frac{5}{3}\right)+\left(-\frac{4}{3}\right)+1=-2$

Equation: $x^2-(\text{sum of the roots})x+(\text{product of the roots})=0$

$\Rightarrow \quad x^2-\frac{2}{3}x-2=0 \quad \Rightarrow \quad 3x^2-2x-6=0$

(d) The roots are $\alpha^2\beta$ and $\alpha\beta^2$.

Sum of the roots: $\alpha^2\beta+\alpha\beta^2=\alpha\beta(\alpha+\beta)=\left(-\frac{5}{3}\right)\left(-\frac{4}{3}\right)=\frac{20}{9}$

Product of the roots: $\alpha^2\beta\cdot\alpha\beta^2=(\alpha\beta)^3=\left(-\frac{5}{3}\right)^3=-\frac{125}{27}$

Equation: $x^2-(\text{sum of the roots})x+(\text{product of the roots})=0$

$\Rightarrow \quad x^2-\frac{20}{9}x-\frac{125}{27}=0 \quad \Rightarrow \quad 27x^2-60x-125=0$

(e) The roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

Sum of the roots: $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{(-4/3)^2-2(-5/3)}{-5/3}=-\frac{46}{15}$

Product of the roots: $\frac{\alpha}{\beta}\cdot\frac{\beta}{\alpha}=1$

Equation: $x^2-(\text{sum of the roots})x+(\text{product of the roots})=0$

$\Rightarrow \quad x^2-\left(-\frac{46}{15}\right)x+1=0 \quad \Rightarrow \quad 15x^2+46x+15=0$

(f) The roots are $\frac{(1+\alpha)}{\beta}$ and $\frac{(1+\beta)}{\alpha}$. [ans: $15x^2+34x+18=0$]

(g) The roots are $(\alpha-1)^2$ and $(\beta-1)^2$. [ans: $9x^2-88x+6=0$]

Exercise:

1. The roots of the equation $x^2+6x+q=0$ are α and $(\alpha-1)$. Find the value of q. [ans: $q=\frac{35}{4}$]
2. Find the equation, with integer coefficients, whose roots are the squares of the roots of the equation $2x^2+5x-6=0$.
3. The roots of the equation $x^2-px+8=0$ are α and $(\alpha+2)$. Find two possible values of p.
4. The roots of the equation $x^2+2px+q=0$ differ by 2. Show that $p^2=1+q$.
5. Prove that if the sum of the squares of the roots of the equation $ax^2+bx+c=0$ is 1, then $b^2=2ac+a^2$.
6. Prove that, if one root of the equation $ax^2+bx+c=0$ is twice the other, then $2b^2=9ac$.
7. Prove that, if the sum of the reciprocals of the roots of the equation $ax^2+bx+c=0$ is 1, then $b+c=0$.

3.2 Quadratic functions

The function

$$f(x) = ax^2 + bx + c, \tag{1}$$

where $a \neq 0$ is called a quadratic function or a polynomial of degree 2 in the variable x. Using the method of completing the square, the function (1) can be written in the form

$$f(x) = a\,(x - p)^2 + q, \tag{2}$$

as follows:

$$\begin{aligned} ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right). \quad \text{Completing the square in the bracket yields.} \\ &= a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}\right] = a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a^2}\right] \\ &= a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a} \\ &= a\,(x - p)^2 + q, \end{aligned}$$

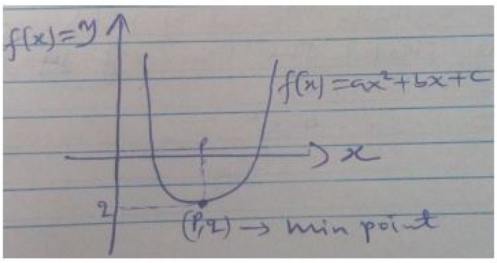
where $p = -\dfrac{b}{2a}$ and $q = \dfrac{4ac - b^2}{4a}$. In this case, q is either the minimum or maximum value of the function depending on the value of a.

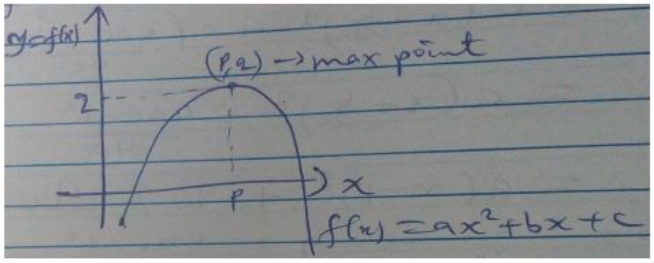
- If $a > 0$ (positive), then q is the minimum value.
- If $a < 0$ (negative), then q is the maximum value.

Graphs of quadratic functions can be used to solve financial problems, science problems, etc. Given the algebraic equation, we can calculate any point on the function including critical values like minimum/maximum, point of symmetry and x and y-intercepts.

Graph of $y = ax^2 + bx + c$

The graph of $y = ax^2 + bx + c$ is a parabola. The parabola opens upwards if $a > 0$ and downwards if $a < 0$. From equation (2), $f(p) = q$. Thus, (p, q) is a minimum point when $a > 0$ and (p, q) is a maximum point when $a < 0$.





- The line of symmetry of the function $f(x)$ is $x = p = -\dfrac{b}{2a}$ (i.e., the value of x for which $f(x)$ is a maximum/minimum, depending on the value of a).
- The x-intercepts are the values of x at the points where $f(x)$ cuts the x-axis. These occur when $f(x) = 0$ i.e., $ax^2 + bx + c = 0$. These values can only be real if $b^2 - 4ac > 0$.
- The y-intercept is the value of y at the point where the curve cuts the y-axis. To obtain this value, we put $x = 0$.

The graph of $y = a(x - p)^2 + q$ is that of $y = x^2$ shifted horizontally p units, stretched by a factor of a, then shifted vertically q units.

Examples:

1. Determine the line of symmetry, minimum or maximum values, x and y-intercepts of $f(x)$. Hence, sketch the curves of $f(x)$.

   (a) $f(x) = 2x^2 + 3x - 5$.

   (b) $f(x) = 2 - 3x - 4x^2$.

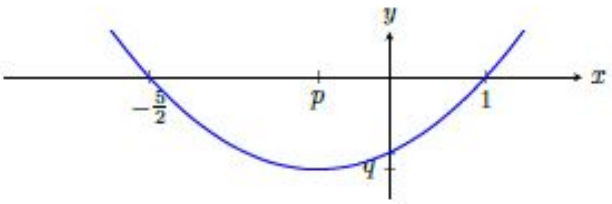
Solution

We first write $f(x)$ in the general form $f(x) = a(x-p)^2 + q$.

(a) $f(x) = 2x^2 + 3x - 5$.

$$\begin{aligned} 2x^2 + 3x - 5 &= 2\left(x^2 + \frac{3}{2}x - \frac{5}{2}\right). \quad \text{Completing the square in the bracket yields.} \\ &= 2\left[\left(x + \frac{3}{4}\right)^2 - \frac{5}{2} - \frac{9}{16}\right] = 2\left[\left(x + \frac{3}{4}\right)^2 - \frac{49}{16}\right] \\ &= 2\left(x + \frac{3}{4}\right)^2 - \frac{49}{8} \quad \Rightarrow \quad a = 2, p = -\frac{3}{4} \quad \text{and} \quad q = -\frac{49}{8}. \end{aligned}$$

- Since $a = 2 > 0$, then $(p, q) = \left(-\frac{3}{4}, -\frac{49}{8}\right)$ is a minimum point.
- The line of symmetry is $x = p = -\frac{3}{4}$.
- For the y-intercept, put $x = 0$. Thus, we have, $f(0) = 2(0)^2 + 3(0) - 5 = -5$. So, the curve $f(x) = 2x^2 + 3x - 5$ cuts the y-axis at $(0, -5)$.
- For the x-intercept, put $y = 0$. Thus, we have, $2x^2 + 3x - 5 = 0$. The roots are $x = 1$ or $x = -\frac{5}{2}$. Hence, the curve $f(x) = 2x^2 + 3x - 5$ cuts the x-axis at points $(-5/2, 0)$ and $(1, 0)$.



(b) $f(x) = 2 - 3x - 4x^2$.

$$\begin{aligned} -4x^2 - 3x + 2 &= -4\left(x^2 + \frac{3}{4}x - \frac{1}{2}\right). \quad \text{Completing the square in the bracket yields.} \\ &= -4\left[x^2 + \frac{3}{4}x + \left(\frac{3}{8}\right)^2 - \frac{1}{2} - \left(\frac{3}{8}\right)^2\right] = -4\left[\left(x + \frac{3}{8}\right)^2 - \frac{41}{64}\right] \\ &= -4\left(x + \frac{3}{8}\right)^2 + \frac{41}{16} \quad \Rightarrow \quad a = -4, p = -\frac{3}{8} \quad \text{and} \quad q = \frac{41}{16}. \end{aligned}$$

- Since $a = -4 < 0$, then $(p, q) = \left(-\frac{3}{8}, \frac{41}{16}\right)$ is a maximum point.
- The line of symmetry is $x = p = -\frac{3}{8}$.
- For the y-intercept, put $x = 0$. Thus, we have, $f(0) = 2$. So, the curve $f(x) = 2 - 3x - 4x^2$ cuts the y-axis at $(0, 2)$.
- For the x-intercept, put $y = 0$. Thus, we have, $-4x^2 - 3x + 2 = 0$. The roots are $x = 0.4$ or $x = -1.2$. Hence, the curve $f(x) = 2 - 3x - 4x^2$ cuts the x-axis at points $(-1.2, 0)$ and $(0.4, 0)$.

The sketch of the graph is an exercise.

Exercise:

1. Find the line of symmetry, maximum or minimum point, x and y-intercepts and then sketch the curve.
   (a) $f(x) = 2x^2 - 4x + 1$.
   (b) $f(x) = x^2 + x + \frac{1}{4}$.
   (c) $f(x) = -3x^2 + 3x - 6$.
   (d) $f(x) = 2 - 2x - x^2$.
2. Find the area and dimensions of the largest rectangular field that can be enclosed with 3000 meters of fence. From the dimensions, conclude about the shape of the field.

3.3 Quadratic Inequalities

If $b^2 - 4ac > 0$, the equation $ax^2 + bx + c = 0$ can be solved for real and distinct values. Let the roots be α and β (with $\beta > \alpha$). Then, $ax^2 + bx + c = a(x - \alpha)(x - \beta)$.

	$x - \alpha$	$x - \beta$	$(x - \alpha)(x - \beta)$	$a(x - \alpha)(x - \beta)$ for $a < 0$	$a(x - \alpha)(x - \beta)$ for $a > 0$
$x < \alpha$	-ve	-ve	+ve	-ve	+ve
$\alpha < x < \beta$	+ve	-ve	-ve	+ve	-ve
$\beta < x$	+ve	+ve	+ve	-ve	+ve

- If $a < 0$: $ax^2 + bx + c > 0$ when $\alpha < x < \beta$, and $ax^2 + bx + c < 0$ when $x < \alpha$ and $x > \beta$.
- If $a > 0$: $ax^2 + bx + c > 0$ when $x < \alpha$ and $x > \beta$, and $ax^2 + bx + c < 0$ when $\alpha < x < \beta$.

→ Note: sketches of quadratic functions can also be used to solve quadratic inequalities.

Examples:

1. Solve $2x^2 + 5x - 3 > 0$

Solution

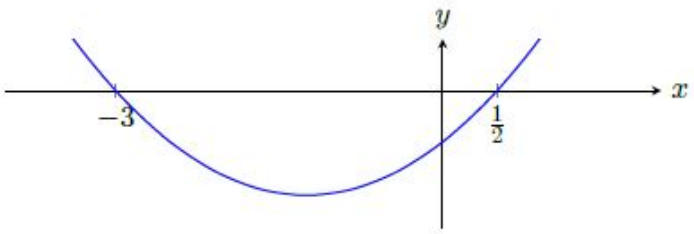
The roots of the equation $2x^2 + 5x - 3 = 0$ are $x = \frac{1}{2}$ and $x = -3$. Thus,

$$2x^2 + 5x - 3 = 2\left(x - \frac{1}{2}\right)(x + 3)$$

Here, $a = 2 > 0, \alpha = -3$, and $\beta = \frac{1}{2}$. Now,

	$x + 3$	$x - \frac{1}{2}$	$2(x + 3)\left(x - \frac{1}{2}\right)$
$x < -3$	-ve	-ve	+ve
$-3 < x < \frac{1}{2}$	+ve	-ve	-ve
$\frac{1}{2} < x$	+ve	+ve	+ve

Hence, $2x^2 + 5x - 3 > 0$ when $x < -3$ and when $x > \frac{1}{2}$. Graphically, we have



→ Note: $f(x) > 0$ if $x < -3$ and if $x > \frac{1}{2}$.

2. Find the set of values for which $1 + 2x - 3x^2 < 0$

Solution

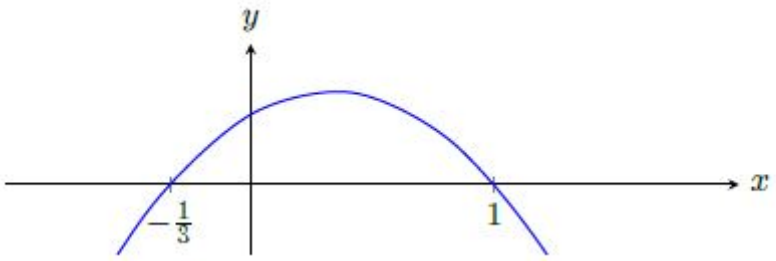
The roots of the equation $1 + 2x - 3x^2 = 0$ are $x = -\frac{1}{3}$ and $x = 1$. Thus,

$$1 + 2x - 3x^2 = -3\left(x + \frac{1}{3}\right)(x - 1)$$

Here, $a = -3 > 0, \alpha = -\frac{1}{3}$, and $\beta = 1$. Now,

	$x + \frac{1}{3}$	$x - 1$	$-3\left(x + \frac{1}{3}\right)(x - 1)$
$x < -\frac{1}{3}$	-ve	-ve	-ve
$-\frac{1}{3} < x < 1$	+ve	-ve	+ve
$1 < x$	+ve	+ve	-ve

Hence, $1 + 2x - 3x^2 < 0$ when $x < -\frac{1}{3}$ and when $x > 1$. Graphically, we have



Exercise:

1. Solve each of the inequalities.
   (a) $2x^2 + 5x - 3 \leq 0$
   (b) $1 + 2x - 3x^2 \geq 0$
   (c) $6 + x > 12x^2$
   (d) $(3 - 2x)(x + 5) > 0$
   (e) $(x - 1)^2 - 4x^2 > 0$
   (f) $x^2 - 8x + 15 < 0$
2. Sketch the graph of $y = -3x^2 + 5x + 2$ and hence use your graph to solve each of the following inequalities.
   (a) $-3x^2 + 5x + 2 > 0$
   (b) $-3x^2 + 5x + 2 \geq 0$
   (c) $-3x^2 + 5x + 2 < 0$
3. [***Assignment 1:***] Determine the values of k for which the equation $x^2 + 2(k+1)x + 2(k+5) = 0$ has real roots. [ans: $k \geq 3$ or $k \leq -3$]

Lecture 4

4 Factorizable polynomials

Let n be a non-negative integer. A polynomial of degree n in the variable x is an expression of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_3x^3 + a_2x^2 + a_1x + a_0,$$

where $a_0, a_1, \cdots, a_n$ are real numbers and $a_n \neq 0$. For example, $f(x) = 2x^3 - 4x^2 + x - 1$ is a polynomial of degree 3 in the variable x.

4.1 Long division of polynomials

When a polynomial $f(x)$ is divided by another polynomial $g(x)$, we obtain a quotient $Q(x)$ and a remainder $R(x)$. The degree of the remainder $R(x)$ is less than that of the divisor $g(x)$.

→ Note: generally, a polynomial is written as $f(x) = g(x) \cdot Q(x) + R(x)$.

Examples:

1. Divide

   (a) $f(x) = 3x^3 + 2x^2 + x - 1$ by $g(x) = x - 1$

   (b) $f(x) = 3x^3 + 2x^2 + x - 1$ by $g(x) = x^2 - 2$

   (c) $f(x) = 4x^3 - x + 2$ by $g(x) = 3x + 2$

Solution

(a) $f(x) = 3x^3 + 2x^2 + x - 1$ by $g(x) = x - 1$

$$\begin{array}{r|l} & 3x^2 + 5x + 6 \\ \hline x - 1 & 3x^3 + 2x^2 + x - 1 \\ & -3x^3 + 3x^2 \\ \hline & \quad 5x^2 + x \\ & \quad -5x^2 + 5x \\ \hline & \qquad 6x - 1 \\ & \qquad -6x + 6 \\ \hline & \qquad\quad 5 \end{array}$$

Thus, $Q(x) = 3x^2 + 5x + 6$ and $R(x) = 5$. Hence,

$$3x^3 + 2x^2 + x - 1 = (x - 1)(3x^2 + 5x + 6) + 5$$

(b) $f(x) = 3x^3 + 2x^2 + x - 1$ by $g(x) = x^2 - 2$

$$\begin{array}{r|l} & 3x + 2 \\ \hline x^2 - 2 & 3x^3 + 2x^2 + x - 1 \\ & -3x^3 \quad + 6x \\ \hline & \quad 2x^2 + 7x - 1 \\ & \quad -2x^2 \quad + 4 \\ \hline & \qquad 7x + 3 \end{array}$$

Thus, $Q(x) = 3x + 2$ and $R(x) = 7x + 3$. Hence,

$$3x^3 + 2x^2 + x - 1 = (x^2 - 2)(3x + 2) + (7x + 3)$$

(c) $f(x) = 4x^3 - x + 2$ by $g(x) = 3x + 2$

$$
\begin{array}{r|l}
 & \frac{4}{3}x^2 - \frac{8}{9}x + \frac{7}{27} \\
\hline
3x+2 & 4x^3 \qquad\quad - x + 2 \\
 & -4x^3 - \frac{8}{3}x^2 \\
\cline{2-2}
 & -\frac{8}{3}x^2 - x \\
 & \frac{8}{3}x^2 + \frac{16}{9}x \\
\cline{2-2}
 & \frac{7}{9}x + 2 \\
 & -\frac{7}{9}x - \frac{14}{27} \\
\cline{2-2}
 & \frac{40}{27}
\end{array}
$$

Exercise:

Divide:

(a) $f(x) = x^3 - 2x + 4$ by $g(x) = x - 1$

(b) $f(x) = 2x^3 - x^2 + 2$ by $g(x) = x - 3$

(c) $f(x) = x^4 - 3x^3 + 5x$ by $g(x) = 2x + 1$

(d) $f(x) = 9x^5 - 5x^2 + 2$ by $g(x) = 3x + 1$

(e) $f(x) = 5x^3 - 2x^2 + 1$ by $g(x) = x - 3$

4.2 Remainder theorem

Let $f(x)$ be a polynomial and a be a real number. If $f(x)$ is divided by $(x - a)$, then the remainder is $R = f(a)$.

Proof. The polynomial $f(x)$ can be written as

$$f(x) = (x - a)Q(x) + R$$

Putting $x = a$ yields $R = f(a)$ □

Corollary: If a polynomial $f(x)$ is divided by $(px + q)$, where $p \neq 0$, then the remainder is $R = f\left(-\frac{q}{p}\right)$.

Proof. Suppose R is the remainder when $f(x)$ is divided by $(px+q)$. Then, $f(x) = (px+q)\cdot Q(x)+R$. Putting $x = -\frac{q}{p}$, we get

$$f\left(-\frac{q}{p}\right) = \left[p\left(-\frac{q}{p}\right) + q\right] \cdot Q\left(-\frac{q}{p}\right) + R = 0 + R \quad \Rightarrow \quad R = f\left(-\frac{q}{p}\right)$$

Examples:

1. Find the remainder when $x^5 - 4x^3 + 2x + 3$ is divided by

   (a) $x - 1$

   (b) $x + 2$

   (c) $2x - 1$

Solution

Let $f(x) = x^5 - 4x^3 + 2x + 3$. Then,

(a) The remainder when $f(x)$ is divided by $(x - 1)$ is $R = f(1) = (1)^5 - 4(1)^3 + 2(1) + 3 = 2$.

(b) The remainder when $f(x)$ is divided by $(x+2)$ is $f(-2) = (-2)^5 - 4(-2)^3 + 2(-2) + 3 = -1$.

(c) The remainder when $f(x)$ is divided by $(2x-1)$ is $R = f(\frac{1}{2}) = (\frac{1}{2})^5 - 4(\frac{1}{2})^3 + 2(\frac{1}{2}) + 3 = \frac{113}{32}$.

Exercise:

1. Find the remainder when

   (a) $f(x) = x^3 - 2x^2 + 5x + 8$ is divided by $x - 2$. [ans: $R = f(2) =$]

   (b) $f(x) = x^3 + 3x^2 + 3x + 1$ is divided by $x + 2$. [ans: $R = f(-2) =$]

   (c) $f(x) = 4x^3 + 6x^2 + 3x + 2$ is divided by

   (i) $2x + 3$ [ans: $R = f(-\frac{3}{2}) =$]

   (ii) $-2x + 3$ [ans: $R = f(\frac{3}{2}) =$]

   (iii) $-2x - 3$ [ans: $R = f(-\frac{3}{2}) =$]

   (iv) x [ans: $R = f(0) = 2$]

4.3 Factor theorem

Let $f(x)$ be a polynomial and a be a real number. If $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$. That is, $(x - a)$ is a factor of $f(x)$ if and only if $f(x)$ leaves remainder zero when it is divided by $(x - a)$.

Examples:

1. Factorize $x^4 - 3x^3 + 4x^2 - 8$ completely.

Solution

We use trial and error method together with the factor theorem. Let $f(x) = x^4 - 3x^3 + 4x^2 - 8$. The factors of 8 are $\pm 1, \pm 2, \pm 4$ and ± 8. Now, by trial and error:

$$\text{Put } x = 1: \quad f(1) = -6 \neq 0 \quad \Rightarrow \quad (x-1) \quad \text{is not a factor of } f(x)$$

$$\text{Put } x = 2: \quad f(2) = (2)^4 - 3(2)^3 + 4(2)^2 - 8 = 0 \quad \Rightarrow \quad (x-2) \quad \text{is a factor of } f(x)$$

By long division, we have

$$\begin{array}{r|l} & x^3 - x^2 + 2x + 4 \\ \hline x-2 & x^4 - 3x^3 + 4x^2 - 8 \\ & -x^4 + 2x^3 \\ \hline & -x^3 + 4x^2 \\ & x^3 - 2x^2 \\ \hline & 2x^2 \\ & -2x^2 + 4x \\ \hline & 4x - 8 \\ & -4x + 8 \\ \hline & 0 \end{array}$$

$$\Rightarrow \quad x^4 - 3x^3 + 4x^2 - 8 = (x-2)(x^3 - x^2 + 2x + 4)$$

Next, we factorize $f_1(x) = x^3 - x^2 + 2x + 4$. The factors of 4 are $\pm 1, \pm 2$ and ± 4. Now, by trial and error:

$$\text{Put } x = 2: \quad f_1(2) = 12 \neq 0 \quad \Rightarrow \quad (x-2) \quad \text{is not a factor of } f_1(x)$$

$$\text{Put } x = -1: \quad f_1(-1) = (-1)^3 - (-1)^2 + 2(-1) + 4 = 0 \quad \Rightarrow \quad (x+1) \quad \text{is a factor of } f_1(x)$$

By long division, we have

$$\begin{array}{r|l} & x^2 - 2x + 4 \\ \hline x+1 & x^3 - x^2 + 2x + 4 \\ & -x^3 - x^2 \\ \hline & -2x^2 + 2x \\ & 2x^2 + 2x \\ \hline & 4x + 4 \\ & -4x - 4 \\ \hline & 0 \end{array}$$

$$\Rightarrow \quad x^3 - x^2 + 2x + 4 = (x+1)(x^2 - 2x + 4)$$

Since $(x^2 - 2x + 4)$ has no linear factors, therefore,

$$x^4 - 3x^3 + 4x^2 - 8 = (x-2)(x+1)(x^2 - 2x + 4)$$

2. Solve the cubic equation $x^3 - 7x - 6 = 0$.

Solution

Let $f(x) = x^3 - 7x - 6$. The factors of 6 are $\pm 1, \pm 2, \pm 3$ and ± 6. Clearly,

$$\text{Put } x = -1: \quad f(-1) = (-1)^3 - 7(-1) - 6 = 0 \quad \Rightarrow \quad (x+1) \quad \text{is a factor of } f(x)$$

$$\text{Put } x = -2: \quad f(-2) = (-2)^3 - 7(-2) - 6 = 0 \quad \Rightarrow \quad (x+2) \quad \text{is a factor of } f(x)$$

$$\text{Put } x = 3: \quad f(3) = (3)^3 - 7(3) - 6 = 0 \quad \Rightarrow \quad (x-3) \quad \text{is a factor of } f(x)$$

Therefore,

$$x^3 - 7x - 6 = (x+2)(x+1)(x-3) = 0. \quad \text{The roots are } x = -1, x = -2, x = 3$$

3. A polynomial $f(x)$ has remainder 9 when divided by $(x-3)$ and remainder -5 when divided by $(2x+1)$. Find the remainder when $f(x)$ is divided by $(x-3)(2x+1)$.

Solution

Since the divisor $(x-3)(2x+1)$ is of degree 2, the remainder R should be of degree 1 or 0. Let $R = ax+b$ ($a \neq 0$ if R is of degree 1 and $a = 0$ if R is of degree 0). Thus, the polynomial can be written as:

$$f(x) = (x-3)(2x+1) \cdot Q(x) + (ax+b)$$

- The remainder when $f(x)$ is divided by $(x-3)$ is $f(3) \equiv 3a+b = 9 \cdots (i)$.
- The remainder when $f(x)$ is divided by $(2x+1)$ is $f(-\frac{1}{2}) \equiv -\frac{1}{2}a+b = -5 \cdots (ii)$.

Solving equations (i) and (ii) simultaneously yields $a = 4$ and $b = -3$. Therefore, the remainder when $f(x)$ is divided by $(x-3)(2x+1)$ is $R = 4x-3$.

4. When a polynomial $f(x)$ is divided by $(x-1)$, the remainder is 3. When $f(x)$ is divided by $(x+1)$, the remainder is 5. When $f(x)$ is divided by $(x-2)$, the remainder is 20. Find the remainder when $f(x)$ is divided by $(x^2-1)(x-2)$.

Solution

Since the divisor $(x^2-1)(x-2) = (x-1)(x+1)(x-2)$ is of degree 3, the remainder R should be of degree 2 or 1 or 0. Let $R = ax^2+bx+c$. Thus, the polynomial can be written as:

$$f(x) = (x-1)(x+1)(x-2) \cdot Q(x) + (ax^2+bx+c)$$

- The remainder when $f(x)$ is divided by $(x-1)$ is $f(1) \equiv a+b+c = 3 \cdots (i)$.
- The remainder when $f(x)$ is divided by $(x+1)$ is $f(-1) \equiv a-b+c = 5 \cdots (ii)$.
- The remainder when $f(x)$ is divided by $(x-2)$ is $f(2) \equiv 4a+2b+c = 20 \cdots (iii)$.

Solving equations (i), (ii) and (iii) simultaneously yields $a = 6, b = -1$ and $c = -2$. Therefore, the remainder when $f(x)$ is divided by $(x^2-1)(x-2)$ is $R = 6x^2-x-2$.

Exercise:

1. Solve the equations
   (a) $x^4+5x^3+5x^2-5x-6 = 0$
   (b) $2x^3+11x^2+17x+6 = 0$
   (c) $3x^3+x^2-5x+2 = 0$
2. When the polynomial x^5+4x^2+ax+b is divided by (x^2-1), the remainder is $2x+3$. Find the values of a and b. [ans: $a = 1, b = -1$]
3. When the polynomial x^3+3x^2+ax+b is divided by (x^2-4), the remainder is $x+16$. Find the values of a and b. [ans: $a = -3, b = 4$]
4. A cubic polynomial ax^3+bx+6 is divisible by $(x+2)$. It leaves a remainder of -3 when divided by $(x-1)$. Determine the values of a and b and hence factorize the polynomial completely. [ans: $a = 4, b = -13$]
5. The expression ax^2+bx+c is divisible by $(x-1)$. It has remainder 2 when divided by $(x+1)$, and has remainder 8 when divided by $(x-2)$. Find the values of a, b and c. [ans: $a = 3, b = -1, c = -2$]
6. $(x-1)$ and $(x+1)$ are factors of the expression x^3+ax^2+bx+c. It leaves a remainder of 12 when divided by $(x-2)$. Find the values of a, b and c. [ans: $a = 2, b = -1, c = -2$]

8. The expression $ax^4 + bx^3 + 3x^2 - 2x + 3$ has a remainder of $x + 1$ when divided by $(x^2 - 3x + 2)$. Find the values of a and b. [ans: $a = 1, b = -3$]

9. What is the value of a if the polynomials $2x^2 - x - 6$, $3x^2 - 8x + 4$ and $ax^3 - 10x - 4$ have a common factor. [ans: $a = 3$ or $a = -250$]

10. Factorize the expression $4x^4 - 13x^2 + 9$ completely. [ans: $(2x - 3)(2x + 3)(x - 1)(x + 1)$]

Lecture 5

5 Permutations and Combinations

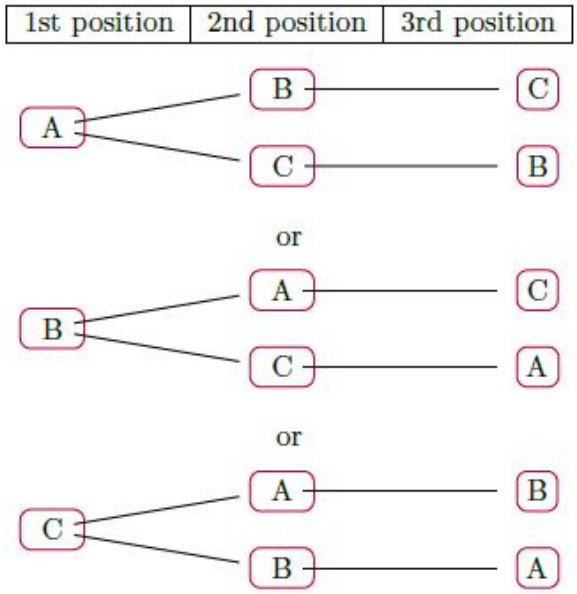
5.1 Permutations

A permutation is the number of possible ways of arranging a given number of objects in order. Now, if k independent choices are to be made where there are r_1 possibilities for the first choice, r_2 possibilities for the second choice, and so on, the the total number of choices is $r_1 \times r_2 \times \cdots \times r_k$.

Examples:

1. In how many ways can the letters A, B and C be arranged in three consecutive positions.

Solution



The 1st position can be filled in 3 different ways (we can have either A or B or C). Once the 1st position is filled, 2 letters remain and any of them can be used to fill the 2nd position. So, for each of the 3 ways of filling the 1st position we have 2 ways of filling the 2nd position. After filling the 1st and 2nd positions, the 3rd position can be filled in any 1 way as we are left with only 1 letter. Thus, the total number of distinct arrangements of the three letters is $3 \times 2 \times 1 = 6$ different ways. Each of these 6 arrangements is called a permutation of the letters A,B and C. The six permutations are ABC, ACB, BAC, BCA, CAB, and CBA.

Definition 1.6.1 (The factorial notation). *The number of ways of arranging n distinct objects is $n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$ which is denoted by $n!$ and read as "n factorial". For example,*

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120, \quad 4! = 4 \times 3 \times 2 \times 1 = 24, \quad 3! = 3 \times 2 \times 1 = 6, \quad 2! = 2 \times 1 = 2$$

By convention:

$1! = 1$ is the number of ways of arranging one object.

$0! = 1$ is the number of ways of arranging zero object.

Thus,

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1 \cdot 0!$$

→ Note: most calculators have a key for permutations.

Examples:

Evaluate:

(a) $\dfrac{9!}{2!7!}$

(b) $\dfrac{6!}{(3!)^2}$

(c) $\dfrac{6!2!}{8!}$

Solution

(a) $\dfrac{9!}{2!7!} = \dfrac{9 \times 8 \times \cancel{7!}}{2! \times \cancel{7!}} = \dfrac{9 \times 8}{2 \times 1} = 36$

(b) $\dfrac{6!}{(3!)^2} = \dfrac{6 \times 5 \times 4 \times \cancel{3!}}{3! \times \cancel{3!}} = \dfrac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$

(c) $\dfrac{6!2!}{8!} = \dfrac{\cancel{6!} \times 2!}{8 \times 7 \times \cancel{6!}} = \dfrac{2 \times 1}{8 \times 7} = \dfrac{1}{28}$

2. Write $40 \times 39 \times 38 \times 37$ in factorial notation.

Solution

$$40 \times 39 \times 38 \times 37 = \frac{40 \times 39 \times 38 \times 37 \times 36 \times 35 \times \cdots \times 3 \times 2 \times 1}{36 \times 35 \times \cdots \times 3 \times 2 \times 1} = \frac{40!}{36!}$$

3. How many arrangements are there for 2 objects chosen from 4 distinct objects?

Solution

There are 4 ways of filling the 1st position. Once the 1st position is filled, three objects remain. Thus, there are 3 ways of filling the 2nd position. Therefore, the number of arrangements of two objects chosen from four different objects is $4 \times 3 = 12$. But

$$4 \times 3 = \frac{4!}{2!} = \frac{4!}{(4-2)!}$$

Similarly, $5 \times 4 = \dfrac{5!}{3!} = \dfrac{5!}{(5-2)!} = 20$ is the number of arrangements of two objects from five distinct objects. Also, $5 \times 4 \times 3 = \dfrac{5!}{2!} = \dfrac{5!}{(5-3)!} = 60$ is the number of arrangements of three objects from five distinct objects.

Definition 1.6.2. *The number of permutations/arrangements of r objects chosen from n distinct objects is given by*

$$^nP_r = \frac{n!}{(n-r)!}$$

and is read as "n permutation r".

Examples:

1. (a) Suppose that 7 people enter a swim meet. Assuming that there are no ties, in how many ways could the gold, silver, and bronze medals be awarded?

Solution

Gold	Silver	Bronze
7	6	5

$$^{7}P_3 = \frac{7!}{(7-3)!} = \frac{7 \times 6 \times 5 \times \cancel{4!}}{\cancel{4!}} = 7 \times 6 \times 5 = 210 \text{ ways.}$$

(b) In a lottery, a total of 1000 tickets were sold. Determine the number of ways of obtaining winners of the 1st, 2nd and 3rd prizes if three tickets are drawn one after the other.

Solution

1st	2nd	3rd
1000	999	998

$$^{1000}P_3 = \frac{1000!}{(1000-3)!} = \frac{1000 \times 999 \times 998 \times \cancel{997!}}{\cancel{997!}} = 1000 \times 999 \times 998 = 997002000 \text{ ways.}$$

2. Five letter words are formed from the letters of the word INCOMPUTABLE. How many of these words have no repeated letters?

Solution

$$^{12}P_5 = \frac{12!}{(12-5)!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!}} = 12 \times 11 \times 10 \times 9 \times 8 = 95040 \text{ ways.}$$

3. (a) Find how many different arrangements of 12 letters can be obtained from the letters of the word SPEEDOMETERS

Solution

If the 12 letters are all different, there could be 12! different arrangements. In any of these arrangements, the two S's can be arranged in $2! = 2$ ways without altering the appearance of the letters in the given arrangement. Similarly, the four E's can be arranged in $4! = 24$ ways without altering the appearance of the letters in an arrangement. This in turn implies that in any arrangement, there can be $2! \times 4! = 48$ arrangements without altering the appearance of the letters of the given arrangement. Since the two S's and four E's are alike, these are

$$\frac{12!}{2!4!} = 9979200 \text{ arrangements}$$

In general, if we have n objects of which r_1 of the first kind are alike, r_2 of the second kind are alike,$\cdots$, r_k of the kth kind are alike, then the total number of distinct permutations is

$$\frac{n!}{r_1! \times r_2! \times \cdots \times r_k!}$$

(b) Determine the number of permutations of the letters of the words

(i) ASSIGNMENT
(ii) ASSASSINATION
(iii) MISSISSIPPI
(iv) FUNDAMENTALISM
(v) INTERCOMMUNICATION

(c) In how many ways can 5 blue beads, 4 green beads, 2 red beads and 1 white bead be arranged in a row if beads of the same color are indistinguishable. [ans: 83,160 ways]

4. How many different 7-digit telephone numbers are possible if the first digit cannot be zero and no digit may repeat?

Solution

Our number system consists of 10 digits, which are $\{0,1,2,3,4,5,6,7,8,$ and $9\}$. Since the first digit cannot be a zero, then there are 9 choices for the first digit, 9 choices for the second digit since a zero can be used and no digits may repeat, 8 choices for the third digit, 7 choices for the fourth digit, and so on. Thus, there are

9	9	8	7	6	5	4

$= 9\times9\times8\times7\times6\times5\times4=544{,}320$ possible telephone numbers.

5. How many even numbers, greater that 50000, can be formed using the digits 0,3,4,5,6,7

   (a) without replacement

   (b) if repetitions are allowed

Solution

The number can have either 5 or 6 digits and cannot begin with zero.

(a) If there is no repetition, the problem is split up into four cases:

i) Numbers with 5 digits, the ten thousand's (T Th) digit being even > 5

T Th	Th	H	T	O
1	4	3	2	2

$= 1 \times 4 \times 3 \times 2 \times 2 = 48$ possibilities

ii) Numbers with 5 digits, the T Th digit being odd ≥ 5

T Th	Th	H	T	O
2	4	3	2	3

$= 2 \times 4 \times 3 \times 2 \times 3 = 144$ possibilities

iii) Numbers with 6 digits, the H Th digit being even.

H Th	T Th	Th	H	T	O
2	4	3	2	1	2

$= 2 \times 4 \times 3 \times 2 \times 1 \times 2 = 96$ possibilities

iv) Numbers with 6 digits, the H Th digit being odd.

H Th	T Th	Th	H	T	O
3	4	3	2	1	3

$= 3 \times 4 \times 3 \times 2 \times 1 \times 3 = 216$ possibilities

Therefore, there are $48 + 144 + 96 + 216 = 504$ possible numbers.

(b) If repetition is allowed, we have two cases to consider:

i) Numbers with 5 digits, T Th digit ≥ 5

T Th	Th	H	T	O
3	6	6	6	3

$= 3 \times 6 \times 6 \times 6 \times 3 = 1944$ possibilities

ii) Numbers with 6 digits

H Th	T Th	Th	H	T	O
5	6	6	6	6	3

$= 5 \times 6 \times 6 \times 6 \times 6 \times 3 = 19440$ possibilities

Therefore, there are $1944 + 19440 = 21384$ possible numbers.

6. A man dines at the same hotel for three consecutive days and the menu each day is one of any 4 types of goat dish, or 2 types of chicken dish or 1 type of vegetarian dish. In how many ways can he arrange his lunches over the three days if he doesn't have a goat dish two days running nor repeat any dish.

Solution

We have four cases to consider

i) Goat dish on the first day:

Day 1 (Goat dish)	Day 2 (Other dish)	Day 3 (Any remaining dish)
4	3	5

$= 4 \times 3 \times 5 = 60$ ways

ii) Goat dish on the second day:

Day 1 (Other dish)	Day 2 (Goat)	Day 3 (Any of the two remaining other)
3	4	2

$= 3 \times 4 \times 2 = 24$ ways

iii) Goat dish on the third day:

Day 1 (Other dish)	Day 2 (Any of the two remaining other)	Day 3 (Goat)
3	2	4

$= 3 \times 2 \times 4 = 24$ ways

iv) No goat dish:

Day 1 (Other dish)	Day 2 (Any of the two remaining other dishes)	Day 3 (Single remaining other dish)
3	2	1

$= 3 \times 2 \times 1 = 6$ ways

Therefore, there are $60 + 24 + 24 + 6 = 114$ ways.

Exercise:

1. How many odd numbers greater than 500,000 can be made from the digits 2,3,4,5,6,7.
   (a) without repetition. [ans: 168 ways]
   (b) if repetition is allowed. [ans: 11,664 ways]
2. How many odd numbers greater than 70,000 can be formed using the digits 0,1,4,7,8,9.
   (a) without repetition. [ans: 456 ways]
   (b) if repetitions are allowed. [ans: 21,384 ways]
3. In how many ways can 8 people be scattered at a round table? [ans: $7! = 5040$]
4. How many arrangements of the letters of the word BEGIN are there which start with a vowel, without repetition? [ans: 48 ways]
5. In how many ways can 5 boys and 4 girls be arranged on a bench if boys and girls alternate? [ans: $5! \times 4! = 2880$ ways]
6. How many arrangements of the letters of the word REMAND are possible if
   (a) they begin with RE? [ans: $1! \times 4! = 24$ ways]
   (b) they don't begin with RE? [ans: $6! - 24 = 696$ ways]
7. How many numbers between 10 and 300 can be made from the digits 1,2,3 if
   (a) each digit may be used only once. [ans: 10 numbers]
   (b) each digit may be used more than once. [ans: 27 numbers]
8. Five letters from the word DRILLING are to be arranged in a row. Find the number of ways in which this can be done when the first letter is I and the last letter is L.
   (a) if no letter may be repeated. [ans: 24 ways]
   (b) if each letter may occur as many times as it does in DRILLING. [ans: 120 ways]
9. In how many ways can two mathematics books and 4 physics books be arranged in a shelf if
   (a) the mathematics books must be placed next to each other. [ans: $2! \times 5! = 240$ ways]
   (b) the mathematics books must NOT be placed next to each other. [ans: $6! - 240 = 480$ ways]

5.2 Combinations

In permutations, the order in which objects are chosen in important. However, in some cases, the order of selection is irrelevant. When the selection of objects is made randomly with no regard being paid to the order, it is referred to as a *combination*. Thus, ABC, ACB, BAC are different permutations but they are the same combination of letters.

Examples:

1. How many different committees of 3 people can be chosen to work on a special project from a group of 5 people? A,B,C,D and E?

   Solution

   There are 10 possible combinations of the committees of 3 people chosen from a group of 5 people: ABC, ABD, ABE, ACD, AEC, ADE, BCD, BCE, BDE, and CDE. Thus, there are 10 ways of selecting the project group.

   → Note: On the other hand, there are ${}^5P_3 = 60$ permutations of 3 objects from 5 distinct objects.

In general, the number of combinations/selections of r objects from n available objects can be found as follows:

- there are $\frac{n!}{(n-r)!}$ permutations of r objects chosen from n distinct objects.
- but each combination of r objects can be arranged in $r!$ ways, therefore,

$$\text{the number of permutations} = r! \times (\text{the number of combinations})$$

$$\text{i.e., } \frac{n!}{(n-r)!} = r! \times (\text{the number of combinations})$$

- so the total number of combinations/selections is $\frac{n!}{(n-r)!r!}$ written as

$${}^nC_r \text{ or } \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

→ Note: most calculators have a key for combinations.

1. Five policemen are to be selected for duty from a force of 20. In how many ways can this be done if

   (a) there is no restriction?

   (b) it has already been decided that a certain person must be on duty?

   Solution

   (a)
   $${}^{20}C_5 = \frac{20!}{(20-5)!5!} = \frac{20!}{15!5!} = 15,504$$

   (b) The problem has been reduced to selecting 4 more policemen from the remaining 19 policemen. Thus, we have ${}^{19}C_4 = \frac{19!}{(19-4)!4!} = \frac{19!}{15!4!} = 3876$. There are 3,876 ways of selecting the policemen for duty.

2. A mixed hokey team containing 5 men and 6 women is to be chosen from 7 men and 9 women. In how many ways can this be done?

Solution

Five men can be chosen from 7 men in ${}^7C_5$ ways and 6 women can be chosen from 9 women in ${}^9C_6$ ways. For each of the ${}^7C_5$ ways of selecting men, there are ${}^9C_6$ ways of selecting women. Therefore, the total number of selecting the team is

$$ {}^7C_5 \times {}^9C_6 = \frac{7!}{(7-5)!5!} \times \frac{9!}{(9-6)!6!} = \frac{7!}{2!5!} \times \frac{9!}{3!6!} = 21 \times 84 = 1764 \text{ ways} $$

3. A person wants to invite 8 friends but there is only room for 4 of them. In how many ways can the four to be invited be chosen if two of the eight are twins and must not be separated?

Solution

There are two cases to consider

i) The twins are invited: thus, we have to choose 2 out of the remaining 6 in ${}^6C_2 = 15$ ways

ii) The twins are left out: thus, we have to choose 4 out of the remaining 6 in ${}^6C_4 = 15$ ways

Therefore, the total number of ways is $15 + 15 = 30$.

4. Nine people are going to travel in two taxi. The larger has 5 seats and the smaller has 4 seats. In how many ways can the party be split?

Solution

Once the group of 5 has been selected, then the remaining 4 people will automatically comprise the other group. Thus, we have to select 5 from 9 in ${}^9C_5 = 126$ ways.

5. Five books are to be selected from 20 books of which 8 are paperback and 12 are hardback. How many selections are possible if at least one paperback book has to be included?

Solution

- If there is no restriction, we have ${}^{20}C_5 = 15,504$ ways of selecting the 5 books
- If only hardback books are selected, we have ${}^{12}C_5 = 792$ ways of selecting the 5 books

Therefore, the number of ways of selecting at least one paperback book is $15,504 - 792 = 14,712$.

Exercise:

1. A committee of six is to be formed from nine women and three men. In how many ways can the members be chosen so as to include at least one man? [ans: 840 ways]

2. A committee of ten is to be formed from nine men and six women. In how many ways can it be formed if at least four women are to be in the committee? [ans: 2142 ways]

3. In how many ways can a class of 20 children be split into two groups of 8 members and 12 members, respectively if there are two twins in the class who must not be separated. [ans: ${}^{18}C_6 + {}^{18}C_8 = 62,322$ ways]

4. Nine players are available to play for a table tennis team of 4 players. In how many ways can the team be selected if 2 of the players are brothers and must either both be included or both be excluded and if 2 other players have recently quarreled and should not both play in a team? [ans: 45 ways]

Lecture 6

6 Binomial expansion

A binomial is the sum or difference of two terms. For example,

$$\begin{aligned}(a+b)^0 &= 1\\ (a+b)^1 &= a+b\\ (a+b)^2 &= a^2+2ab+b^2\\ (a+b)^3 &= a^3+3a^2b+3ab^2+b^3\\ (a+b)^4 &= a^4+4a^3b+6a^2b^2+4ab^3+b^4\end{aligned}$$

If the coefficients are written alone, we have the following triangular array of numbers called the **Pascal's triangle**.

$n=0:$						1				
$n=1:$					1		1			
$n=2:$				1		2		1		
$n=3:$			1		3		3		1	
$n=4:$		1		4		6		4		1
$n=5:$	1		5		10		10		5		1
$n=6:$	1		6		15		20		15		6		1

Examples:

1. Expand $(a+b)^7$.

Solution

From the Pascal's triangle, the coefficients are 1,7,21,35,35,21,7,1. Therefore,

$$(a+b)^7 = a^7+7a^6b+21a^5b^2+35a^4b^3+35a^3b^4+21a^2b^5+7ab^6+b^7$$

2. Obtain the expansion of $\left(2x-\frac{1}{2}\right)^4$ in descending powers of x.

Solution

From the Pascal's triangle, the coefficients are 1,4,6,4,1. Therefore,

$$\begin{aligned}\left(2x-\frac{1}{2}\right)^4 &= (2x)^4+4(2x)^3\left(-\frac{1}{2}\right)+6(2x)^2\left(-\frac{1}{2}\right)^2+4(2x)\left(-\frac{1}{2}\right)^3+\left(-\frac{1}{2}\right)^4\\ &= 16x^4-16x^3+6x^2-x+\frac{1}{16}\end{aligned}$$

Exercise:

Use the Pascal's triangle to expand the following binomials

(a) $(1-x)^3$

(b) $(x+\frac{1}{x})^4$

(c) $(2x-1)^5$

(d) $(x+y)^7$

6.1 Binomial Theorem

If n is a positive integer, then

$$\begin{aligned}(a+b)^n &= \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + \binom{n}{n-1}an^{n-1} + \binom{n}{n}b^n \\ &= \sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r,\end{aligned}$$

where $\binom{n}{r} = {}^nC_r = \frac{n!}{(n-r)!r!}$. From the statement of the theorem, the term containing $a^{n-r}b^r$ is $\binom{n}{r}a^{n-r}b^r$.

Examples:

1. Find the coefficients of x^{10} in the expansion of $(2x-3)^{14}$.

Solution

The general term in the expansion is given by

$$\binom{14}{r}(2x)^{14-r}(-3)^r = \binom{14}{r}2^{14-r}(-3)^r \cdot x^{14-r}$$

The term in x^{10} is obtained when $14-r=10 \quad \Rightarrow \quad r=4$. Therefore, the required coefficient is $\binom{14}{4}2^{14-4}(-3)^4 = \frac{14!}{10!4!}2^{10}3^4$.

2. Find the coefficient of x in the expansion of $\left(2x^2-\frac{1}{x^2}\right)^{50}$.

Solution

The general term in the expansion is given by

$$\binom{50}{r}(2x^2)^{50-r}\left(-\frac{1}{x^2}\right)^r = \binom{50}{r}2^{50-r}(-1)^r \cdot \frac{x^{100-2r}}{x^{2r}} = \binom{50}{r}2^{50-r}(-1)^r x^{100-4r}$$

The term in x is obtained when $100-4r=1 \quad \Rightarrow \quad r=\frac{99}{4}$. But r is not a whole number. Thus, the expansion doesn't have a term in x.

3. Obtain the first four terms of the expansion $\left(1+\frac{1}{2}x\right)^{10}$ in ascending powers of x. Hence, find the value of $(1.005)^{10}$, correct to 4dp.

Solution

$$\begin{aligned}\left(1+\frac{1}{2}x\right)^{10} &= \binom{10}{0}(1)^{10}+\binom{10}{1}(1)^{9}\left(\frac{1}{2}x\right)+\binom{10}{2}(1)^{8}\left(\frac{1}{2}x\right)^{2}+\binom{10}{3}(1)^{7}\left(\frac{1}{2}x\right)^{3}+\cdots \\ &= 1+5x+\frac{45}{4}x^{2}+15x^{3}+\cdots\end{aligned}$$

Now, $(1.005)^{10}=(1+0.005)^{10}$ and comparing with $\left(1+\frac{1}{2}x\right)^{10}$, we get

$$\frac{1}{2}x=0.005 \quad \Rightarrow \quad x=0.01$$

Substituting this value of x in the above expansion, we obtain:

$$\begin{aligned}(1+0.005)^{10} &= 1+5(0.01)+\frac{45}{4}(0.01)^{2}+15(0.01)^{3}+\cdots \\ &= 1+0.05+0.001125+0.000015+\cdots \\ &= 1.0511 \quad \text{(to 4dp)}\end{aligned}$$

4. Obtain the expansion of $\left(1+x-2x^{2}\right)^{8}$ up to the term in x^{3}.

Solution

$$\begin{aligned}\left(1+x-2x^{2}\right)^{8} &= \left[1+(x-2x^{2})\right]^{8} \\ &= 1+\binom{8}{1}(x-2x^{2})+\binom{8}{2}(x-2x^{2})^{2}+\binom{8}{3}(x-2x^{2})^{3}+\cdots \\ &= 1+8(x-2x^{2})+28(x^{2}-4x^{3}+\cdots)+56(x^{3}+\cdots)+\cdots \\ &= 1+8x+12x^{2}-56x^{3}+\cdots\end{aligned}$$

Exercise:

1. Write down the coefficients of the terms indicated in the expansions of the following:
   (a) $(1+x)^{16}$, 3rd term.
   (b) $(3+2x)^{6}$, 4th term.
   (c) $(2-x)^{20}$, 18th term.
   (d) $(2+\frac{3}{2}x)^{8}$, 5th term.
2. In the expansion of $(1-2x+ax^2)^4$ as a series of powers of x, the coefficient of x^3 is zero. Show that $a=-\dfrac{4}{3}$ and find the coefficient of x^4. [ans: $=-37\frac{1}{3}$]
3. Use Binomial theorem to find the values of
   (a) $(1.01)^{10}$, correct to 3dp.
   (b) $(2.001)^{10}$, correct to 6 sf.
   (c) $(0.997)^{12}$, correct to 3dp.
   (d) $(1.998)^{8}$, correct to 2dp.
4. Expand the following as far as the term in x^3.
   (a) $(1+x+x^2)^3$
   (b) $(1+2x-x^2)^6$
   (c) $(2+x-2x^2)^7$
   (d) $(3-2x+x^2)^4$
5. Find the ratio of the term in x^7 to the term in x^8 in the expansion of $(3x+\frac{2}{3})^{17}$. [ans: $=\dfrac{8}{45x}$]
6. (a) Find, in factorial form, the coefficient of x in the expansion of $(2x^2-\frac{1}{x})^{50}$. [ans: $=-\dfrac{50!}{17!33!}2^{17}$]
   (b) Find the coefficient of y^8 in the expansion of $(2y+3)^{10}$. [ans: $=\dfrac{10!}{8!2!}2^8\cdot 3^2$]

6.2 Binomial theorem for any Index

For any rational number n, where n is not a positive integer, the binomial theorem is given by

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\cdots,$$

provided $-1<x<1$, i.e., $|x|<1$. To expand $(a+x)^n$ (where $a\neq 1$), we first need to express the binomial as $(a+x)^n=[a(1+\frac{x}{a})]^n=a^n(1+\frac{x}{a})^n$ and expand the expression in terms of $\dfrac{x}{a}$. Since the expansion holds only for certain values of x, this set of values must be stated.

Examples:

1. Expand the following in ascending powers of x as far as the term in x^3, stating the range of values of x for which the expansion is valid.

   (a) $(1+x)^{\frac{1}{3}}$

   (b) $(1-4x)^{-3}$

   (c) $(3-x)^{-2}$

   Solution

   (a)

$$\begin{aligned}(1+x)^{\frac{1}{3}} &= 1+\left(\frac{1}{3}\right)x+\frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}-1\right)}{2!}x^2+\frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}x^3+\cdots\\ &= 1+\frac{1}{3}x-\frac{1}{9}x^2+\frac{5}{81}x^3+\cdots \quad \text{(provided } -1<x<1 \text{ i.e., } |x|<1)\end{aligned}$$

   (b)

$$\begin{aligned}(1-4x)^{-3} &= 1+(-3)(-4x)+\frac{(-3)(-3-1)}{2!}(-4x)^2+\frac{(-3)(-3-1)(-3-2)}{3!}(-4x)^3+\cdots\\ &= 1+12x+96x^2+640x^3+\cdots \quad \text{(provided } -1<-4x<1 \text{ or } -\frac{1}{4}<x<\frac{1}{4})\end{aligned}$$

   (c) $(3-x)^{-2}=3^{-2}(1-\frac{x}{3})^{-2}$. Thus,

$$\begin{aligned}3^{-2}(1-\frac{x}{3})^{-2} &= \frac{1}{9}\left[1+(-2)\left(-\frac{x}{3}\right)+\frac{(-2)(-2-1)}{2!}\left(-\frac{x}{3}\right)^2+\frac{(-2)(-2-1)(-2-2)}{3!}\left(-\frac{x}{3}\right)^3+\cdots\right]\\ &= \frac{1}{9}\left[1+\frac{2}{3}x+\frac{1}{3}x^2+\frac{4}{27}x^3+\cdots\right]\\ &= \frac{1}{9}+\frac{2}{27}x+\frac{1}{27}x^2+\frac{4}{243}x^3+\cdots \quad \text{(provided } -1<-\frac{x}{3}<1 \text{ or } -3<x<3)\end{aligned}$$

2. Expand $\dfrac{4}{(1+4x)(1-2x)}$ as far as the term in x^5, stating the range of values of x for which the expansion is valid.

   Solution

$$\begin{aligned}\frac{4}{(1+4x)(1-2x)} &= \frac{4}{1+2x-8x^2}=4\left[1+(2x-8x^2)\right]^{-1}\\ &= 4\left[1+(-1)(2x-8x^2)+\frac{(-1)(-2)}{2!}(2x-8x^2)^2+\frac{(-1)(-2)(-3)}{3!}(2x-8x^2)^3\cdots\right]\\ &= 4\left[1-2x+8x^2+(4x^2-32x^3+\cdots)-(8x^3+\cdots)\cdots\right]\\ &= 4\left[1-2x+12x^2-40x^3+\cdots\right]\\ &= 4-8x+48x^2-160x^3+\cdots\end{aligned}$$

   Since we expanded $4(1+4x)^{-1}(1-2x)^{-1}$, the expansion is valid when $-1<4x<1$ and $-1<-2x<1$. That is, $-\frac{1}{4}<x<\frac{1}{4}$ and $-\frac{1}{2}<x<\frac{1}{2}$. This is so when $-\frac{1}{4}<x<\frac{1}{4}$.

3. Find the first four terms in the expansion of $\sqrt{1-8x}$ in ascending powers of x. Hence, substitute $x=0.01$ and obtain the value of $\sqrt{23}$ correct to 5 significant figures.

   Solution

$$\begin{aligned}\sqrt{1-8x} &= (1-8x)^{\frac{1}{2}}=1+\frac{1}{2}(-8x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(-8x)^2+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-8x)^3+\cdots\\ &= 1-4x-8x^2-32x^3+\cdots\end{aligned}$$

Putting $x = 0.01$ yields $\sqrt{1-8(0.01)} = \sqrt{0.92} = 1 - 4(0.01) - 8(0.01)^2 - 32(0.01)^3 \approx 0.959168$.
Now, expressing $\sqrt{23}$ in terms of $\sqrt{0.92}$ yields:

$$\sqrt{23} = \sqrt{\frac{92}{4}} = \sqrt{\frac{100 \times 0.92}{4}} = 5\sqrt{0.92} = 5 \times 0.959168 = 4.7958 \quad \text{(to 5 sf)}$$

4. Expand $\sqrt{\frac{1+x}{1-x}}$ in ascending powers of x as far as the term in x^3.

Solution

$\sqrt{\frac{1+x}{1-x}} = (1+x)^{\frac{1}{2}}(1-x)^{-\frac{1}{2}}$. Now,

$$\begin{aligned}
(1+x)^{\frac{1}{2}} &= 1 + \frac{1}{2}x + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}x^2 + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}x^3 + \cdots \\
&= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \cdots \\
(1-x)^{-\frac{1}{2}} &= 1 - \frac{1}{2}(-x) + \frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}(-x)^2 + \frac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-x)^3 + \cdots \\
&= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \cdots
\end{aligned}$$

Therefore,

$$\begin{aligned}
(1+x)^{\frac{1}{2}}(1-x)^{-\frac{1}{2}} &= \left(1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \cdots\right)\left(1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \cdots\right) \\
&= \left(1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3\right) + \frac{1}{2}x\left(1 + \frac{1}{2}x + \frac{3}{8}x^2\right) - \frac{1}{8}x^2\left(1 + \frac{1}{2}x\right) + \frac{1}{16}x^3(1) + \cdots \\
&= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \frac{1}{2}x + \frac{1}{4}x^2 + \frac{3}{16}x^3 - \frac{1}{8}x^2 - \cancel{\frac{1}{16}x^3} + \cancel{\frac{1}{16}x^5} + \cdots \\
&= 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots
\end{aligned}$$

Exercise:

1. Expand the following as a series of ascending powers of x upto and including the term in x^3.
   (a) $(1-3x)^{-\frac{1}{2}}$
   (b) $\frac{1}{\sqrt{2+x^2}}$
   (c) $\frac{x+2}{(1+x)^2}$
   (d) $\frac{\sqrt{1+2x}}{1-x}$
   (e) $\frac{1-x}{\sqrt{1+x}}$
2. Show that if x is small enough for its cube and higher powers to be neglected, then $\sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{x^2}{2}$. By putting $x = \frac{1}{8}$, show that $\sqrt{7} \approx 2\frac{83}{128}$.
3. Expand $\sqrt{\frac{1+2x}{1-2x}}$ as a series of ascending powers of x upto and including the term in x^2. By putting $x = 0.01$, find an approximation for $\sqrt{51}$ correct to 5 significant figures. [ans: $= 1 + 2x + 2x^2$, and $\sqrt{51} = 7.1414$]

Lecture 7

7 Sequences and Series

Definition 1.8.1 (Sequence). *A sequence is a list of numbers in a defined order with a rule for obtaining each of the numbers.*

The elements of a sequence are called *terms* of the sequence, where the nth term is denoted by u_n. For instance,

i) $5, 10, 15, 20, \cdots$ (multiples of 5; general term $5n,\ n \in \mathbb{N}$).

ii) $1, -3, 9, -27, \cdots$ (powers of -3; general term $(-3)^{n-1},\ n \in \mathbb{N}$).

iii) $27, 64, 125, 216, \cdots$ (cubes of consecutive integers in ascending order starting with 3^3; general term $(n+2)^3,\ n \in \mathbb{N}$).

Definition 1.8.2 (Finite and Infinite sequence). *A sequence that is made up of a finite number of terms is called a finite sequence; otherwise, it is called an infinite sequence.*

For example,

i) the sequence $10, 20, 30, \cdots, 100$ is finite while

ii) the sequence $10, 20, 30, \cdots$ is infinite

Definition 1.8.3 (Series). *A series is obtained by adding the terms of a sequence.*

A finite series is one that corresponds to a finite sequence while an infinite series is one that corresponds to an infinite sequence. For example,

i) $10 + 20 + 30 + \cdots + 100 = \sum_{n=1}^{10} 10n$ and $3^3 + 4^3 + \cdots + 10^3 = \sum_{n=3}^{10} n^3$ are finite series (each of 10 terms) while

ii) $1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^{n-1}$ is an in finite series (in this case, n has no upper limit).

7.1 Arithmetic Progression (A.P.)

An AP is a sequence in which any term differs from the previous by a constant number. In this case, the constant number is called the common difference. If an AP has its first term a and its common difference d, then its nth term is given by

$$\boxed{u_n = a + (n-1)d},$$

where n is the number of terms in the AP.

Examples:

1. Determine the first six terms of the AP whose first term is 18 and common difference is -5.

Solution

Given that $a = 18, d = -5$. The nth term is given by $u_n = a + (n-1)d = 18 - 5(n-1)$. Thus,

$$u_1 = 18,\ u_2 = 13,\ u_3 = 8,\ u_4 = 3,\ u_5 = -2, \text{ and } u_6 = -7$$

2. Determine the 4th and 12th terms of the AP whose first term is 19 and common difference is 6. [ans: $u_4 = 37,\ u_{12} = 85$]

Exercise:

(a) Write down the first five terms of the AP with first term 8 and common difference 7.

(b) Write down the first five terms of the AP with first term 2 and common difference -5.

(c) What is the common difference of the AP $11, -1, -13, -25, \ldots$?

(d) Find the 17th term of the arithmetic progression with first term 5 and common difference 2.

(e) Write down the 10th and 19th terms of the APs
   (i) $8, 11, 14, \ldots$,
   (ii) $8, 5, 2 \ldots$.

(f) An AP is given by $k, 2k/3, k/3, 0, \ldots$.
   (i) Find the sixth term.
   (ii) Find the n th term.
   (iii) If the 20th term is equal to 15, find k.

7.1.1 The sum of an AP

Consider an AP whose first term is a, whose common difference is d and with n terms. That is,

$$\text{AP:} \quad a, (a+d), (a+2d), \cdots, a+(n-3)d, a+(n-2)d, a+(n-1)d$$

The method of first principles (Euler's method) can be used to determine a formula for the sum of the first n terms of this AP as follows: Let L be the last term of the AP, i.e., $L = a+(n-1)d$. Adding the terms of the AP yields

$$S_n = a + (a+d) + (a+2d) + \cdots + (L-2d) + (L-d) + L \qquad (i)$$

Similarly, adding the terms in reverse order yields

$$S_n = L + (L-d) + (L-2d) + \cdots + (a+2d) + (a+d) + a \qquad (ii)$$

Adding the two series (i) and (ii), we have

$$2S_n = (a+L) + (a+L) + (a+L) + \cdots + (a+L) + (a+L) + (a+L)$$

$$\Rightarrow \quad 2S_n = n(a+L) \quad \Rightarrow \quad S_n = \frac{n}{2}(a+L)$$

Substituting $L = a+(n-1)d$, we get the formula for finding the sum of the first n terms of an AP as

$$\boxed{S_n = \frac{n}{2}\left[2a + (n-1)d\right]}$$

Examples:

1. The fourth term of an AP is 13 and the seventh term is 22. Determine:

   (a) the common difference

   (b) the value of n if the nth term is 100.

   (c) the sum of the first 20 terms

   (d) the value of m if the sum of the first m terms of the series 175.

   Solution

   (a) $u_4 = 13 \Rightarrow a + 3d = 13 - - - (i)$ and $u_7 = 22 \Rightarrow a + 6d = 22 - - - (ii)$. Solving equations (i) and (ii) simultaneously yields $a = 4, d = 3$. Thus, the common difference is 3.

   (b) $u_n = a + (n-1)d = 100 \Rightarrow 4 + 3(n-1) = 100 \Rightarrow n = 33.$

   (c) $S_{20} = \frac{20}{2}\Big[2(4) + 3(20-1)\Big] = 10[8 + 3 \times 19] = 650.$

   (d) $S_m = \frac{m}{2}\Big[2a + (m-1)d\Big] = 175 \Rightarrow m\Big[8 + 3(m-1)\Big] = 350 \Rightarrow 3m^2 + 5m - 350 = 0.$
   Solving this quadratic equation yields the roots $m = -\frac{35}{3}$ or $m = 10$. Since $m \in \mathbb{N}$, then $m = 10$.

2. The sum of the first n terms of a series is given by $S_n = 2n(n+6)$. Find the fifth term of the series and show that the terms are in AP.

   Solution

   (i) Fifth term: $u_5 = S_5 - S_4 = 2 \times 5(5+6) - 2 \times 4(4+6) = 110 - 80 = 30.$

(ii) We know that $u_n = S_n - S_{n-1}$. Now,

$$\begin{aligned} S_n &= u_1 + u_2 + u_3 + \cdots + u_{n-1} + u_n = 2n(n+6) \\ S_{n-1} &= u_1 + u_2 + u_3 + \cdots + u_{n-1} = 2(n-1)\left[(n-1)+6\right] = 2(n-1)(n+5) \end{aligned}$$

Therefore,

$$u_n = 2n(n+6) - 2(n-1)(n+5) = 2n^2 + 12n - 2n^2 - 8n + 10 = 4n + 10$$

$$u_{n-1} = 4(n-1) + 10 = 4n + 6$$

Now, $u_n - u_{n-1} = (4n+10) - (4n+6) = 4$. Since 4 is the common difference regardless of the value of n, then the terms of the series form an AP.

3. Determine the sum of all integers between 1 and 1000 (inclusive) which are multiples of 5 or 8 or both.

Solution

- Let S be the sum of a series with multiples of 5 i.e., $S = 5+10+15+\cdots+1000$. This is an AP whose $a = 5, d = 5$ and with nth term given by $[5+(n-1)\times 5] = 1000 \quad \Rightarrow \quad n = 200$. Thus,
$$S = \frac{200}{2}\Big[2(5) + (200-1)\times 5\Big] = 100\Big[10 + 199\times 5\Big] = 100500$$
- Let T be the sum of a series with multiples of 8 i.e., $T = 8+16+24+\cdots+1000$. This is an AP whose $a = 8, d = 8$ and with nth term given by $[8+(n-1)\times 8] = 1000 \quad \Rightarrow \quad n = 125$. Thus,
$$T = \frac{125}{2}\Big[2(8) + (125-1)\times 8\Big] = 62.5\Big[16 + 124\times 8\Big] = 63000$$
- Let U be the sum of a series with multiples of both 5 and 8 i.e., $U = 40+80+120+\cdots+1000$. This is an AP whose $a = 40, d = 40$ and with nth term given by $[40 + (n-1)\times 40] = 1000 \quad \Rightarrow \quad n = 25$. Thus,
$$U = \frac{25}{2}\Big[2(40) + (25-1)\times 40\Big] = 12.5\Big[80 + 24\times 40\Big] = 13000$$

Therefore, the sum of all integers between 1 and 1000 (inclusive) which are multiples of 5 or 8 or both is given by $S + T - U = 100500 + 63000 - 13000 = 150500$.

Find the sum of the first 50 terms of the sequence

1. $1, \quad 3, \quad 5, \quad 7, \quad 9, \quad \ldots .$

Solution

This is an arithmetic progression, and we can write down

$$a = 1\,, \qquad d = 2\,, \qquad n = 50\,.$$

We now use the formula, so that

$$\begin{aligned} S_n &= \tfrac{1}{2}n(2a + (n-1)d) \\ S_{50} &= \tfrac{1}{2} \times 50 \times (2 \times 1 + (50-1) \times 2) \\ &= 25 \times (2 + 49 \times 2) \\ &= 25 \times (2 + 98) \\ &= 2500\,. \end{aligned}$$

2. Find the sum of the series

$$1 + 3{\cdot}5 + 6 + 8{\cdot}5 + \ldots + 101\,.$$

Solution

This is an arithmetic series, because the difference between the terms is a constant value, $2{\cdot}5$. We also know that the first term is 1, and the last term is 101. But we do not know how many terms are in the series. So we will need to use the formula for the last term of an arithmetic progression,

$$\ell = a + (n-1)d$$

to give us

$$101 = 1 + (n-1) \times 2{\cdot}5\,.$$

Now this is just an equation for n, the number of terms in the series, and we can solve it. If we subtract 1 from each side we get

$$100 = (n-1) \times 2{\cdot}5$$

and then dividing both sides by $2{\cdot}5$ gives us

$$40 = n - 1$$

so that $n = 41$. Now we can use the formula for the sum of an arithmetic progression, in the version using ℓ, to give us

$$\begin{aligned} S_n &= \tfrac{1}{2}n(a + \ell) \\ S_{41} &= \tfrac{1}{2} \times 41 \times (1 + 101) \\ &= \tfrac{1}{2} \times 41 \times 102 \\ &= 41 \times 51 \\ &= 2091\,. \end{aligned}$$

3. An arithmetic progression has 3 as its first term. Also, the sum of the first 8 terms is twice the sum of the first 5 terms. Find the common difference.

Solution

We are given that $a = 3$. We are also given some information about the sums S_8 and S_5, and we want to find the common difference. So we shall use the formula

$$S_n = \tfrac{1}{2}n(2a + (n-1)d)$$

for the sum of the first n terms. This tells us that

$$S_8 = \tfrac{1}{2} \times 8 \times (6 + 7d)\,.$$

and that

$$S_5 = \tfrac{1}{2} \times 5 \times (6 + 4d)$$

So, using the given fact that $S_8 = 2S_5$, we see that

$$\begin{aligned} \tfrac{1}{2} \times 8 \times (6+7d) &= 2 \times \tfrac{1}{2} \times 5 \times (6+4d) \\ 4 \times (6+7d) &= 5 \times (6+4d) \\ 24 + 28d &= 30 + 20d \\ 8d &= 6 \\ d &= \tfrac{3}{4}\,. \end{aligned}$$

Exercise:

1. Find the sum of an AP of 10 terms whose first term is 7 and whose last term is 10.
2. Show that the terms of the series $\sum_{r=1}^{n} \log 5^r$ are in AP. Hence, find the sum of the first twenty terms of the series and also the least value of n for which the sum to n terms exceeds 400. [ans: $S_{20} = 146.78,\ \ n = 34$]
3. Find the difference between the sums of the first ten terms of the AP whose first terms are 12 and 8, and whose common differences are 2 and 3, respectively.
4. The first term of an AP is -12 and the last term is 40. If the sum of the progression is 196, find the number of terms and the common difference.
5. Find the sum of the odd numbers between 100 and 200.
6. Find the sum of the even numbers which are divisible by 3 and lie between 400 and 500.
7. The twenty-first term of an AP is $5\frac{1}{2}$, and the sum of the first twenty-one terms is $94\frac{1}{2}$. Find the first term, the common difference and the sum of the first 30 terms.
8. In an AP, the thirteenth term is 27, and the seventh term is three times the second term. Find the first term, the common difference and the sum of the first ten terms. [ans: $a = 3, d = 2$ and $S_{10} = 120$]
10. Evaluate $60 + 64 + 68 + 72 + \cdots + 120$ [ans: =1440]

(a) Find the sum of the first 23 terms of the AP $4, -3, -10, \ldots$.

(b) An arithmetic series has first term 4 and common difference $\frac{1}{2}$. Find
   (i) the sum of the first 20 terms,
   (ii) the sum of the first 100 terms.

(c) Find the sum of the arithmetic series with first term 1, common difference 3, and last term 100.

(d) The sum of the first 20 terms of an arithmetic series is identical to the sum of the first 22 terms. If the common difference is -2, find the first term.

7.1.2 Arithmetic mean

☐ **Arithmetic mean** - the arithmetic mean of two numbers P and Q is a number A such that the series $P + A + Q$ form an AP. Thus, the common difference is

$$A - P = Q - A \quad \Rightarrow \quad 2A = P + Q \quad \Rightarrow \quad \boxed{A = \frac{P+Q}{2}}$$

Examples:

1. Insert three arithmetic means between 8 and 18.

Solution

Let the arithmetic means be A, B and C. Thus, we have the series $8 + A + B + C + 18$. Now, $u_1 = 8 = a$ and $u_5 = 18$. Letting the common difference be d, we have $8+4d = 18 \quad \Rightarrow \quad d = 2.5$. Therefore,

$$A - 8 = 2.5 \quad \Rightarrow \quad A = 10.5$$
$$B - A = 2.5 \quad \Rightarrow \quad B = 13.0$$
$$C - B = 2.5 \quad \Rightarrow \quad C = 15.5$$

Exercise:

1. Find five arithmetic means between 12 and 21.6 [hint: $d = 1.6$]

7.2 Geometric Progression (G.P.)

A GP is a sequence in which the ration between two consecutive terms is the same. In this case, the constant multiplying factor is called the common ratio. In general, if a GP has its first term a and common ratio r, then the nth term is given by

$$\boxed{u_n = ar^{n-1}},$$

where n is the number of terms in the GP. Thus, a GP having n terms is given as

$$a, ar, ar^2, ar^3, \cdots, ar^{n-1}$$

For example, the sequence $36, 12, 4, \frac{4}{3}, \frac{4}{9}, \frac{4}{27}, \cdots$, is a GP with $a = 36$ and $r = \frac{1}{3}$.

Exercise:

(a) Write down the first five terms of the geometric progression which has first term 1 and common ratio $\frac{1}{2}$.

(b) Find the 10th and 20th terms of the GP with first term 3 and common ratio 2.

(c) Find the 7th term of the GP $2, -6, 18, \ldots$,

7.2.1 The sum of a GP

Consider a GP whose first term is a and common ratio is r, and with n terms. That is,

$$\text{GP:} \quad a, ar, ar^2, \cdots, ar^{n-3}, ar^{n-2}, ar^{n-1}$$

The method of first principles (Euler's method) can be used to determine a formula for the sum of the first n terms of a GP as follows: Adding the terms of the GP yields

$$S_n = a + ar + ar^2 + \cdots + ar^{n-3} + ar^{n-2} + ar^{n-1} \qquad (i)$$

Multiplying through by r, we get

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^{n-2} + ar^{n-1} + ar^n \qquad (ii)$$

Subtracting the two series (i) and (ii) yields

$$S_n - rS_n = a - ar^n \quad \Rightarrow \quad (1-r)S_n = a\,(1-r^n)$$

Making S_n the subject yields the formula for finding the sum of the first n terms of a GP as

$$\boxed{S_n = \frac{a\,(1-r^n)}{1-r}, \quad |r| \neq 1}$$

→ Note: if $|r| > 1$, we write $S_n = \dfrac{a\,(r^n-1)}{r-1}$.

Examples:

1. In a GP, the sum of the second and third terms is 6, and the sum of the third and fourth terms is -12. Find the first term, the common ratio and the sum of the first ten terms.

Solution

Let the first term be a and the common ration be r. Then, we have

$$ar + ar^2 = 6 \quad \Rightarrow \quad ar(1+r) = 6 \quad \text{and} \quad ar^2 + ar^3 = -12 \quad \Rightarrow \quad ar^2(1+r) = -12$$

Thus,

$$\frac{ar^2(1+r)}{ar(1+r)} = \frac{-12}{6} \quad \Rightarrow \quad r = -2$$

Substituting $r = -2$ in $ar(1+r) = 6$ yields $a = 3$. Since $|r| > 1$, we have

$$S_{10} = \frac{a\left(r^{10}-1\right)}{r-1} = \frac{3\left[(-2)^{10}-1\right]}{-2-1} = -\left[(-2)^{10}-1\right] = -1023$$

2. Determine the smallest number of terms of the GP $8+24+72+\cdots$ whose sum exceeds 6,000,000.

Solution

Here, $a = 8$ and $r = \frac{24}{8} = \frac{72}{24} = 3$. Since $|r| > 1$, we have

$$S_n = \frac{a\left(r^n-1\right)}{r-1} = \frac{8\left(3^n-1\right)}{3-1} = 4\left(3^n-1\right)$$

We require that $S_n > 6000000$. Thus, we have

$$4\left(3^n-1\right) > 6000000 \quad \Rightarrow \quad 3^n - 1 > 1500000 \quad \Rightarrow \quad 3^n > 1500001$$

From which we obtain

$$n > \frac{\log 1500001}{\log 3} = 12.94 \text{ to 4 s.f}$$

For the sum to exceed 6000000, then the smallest value of n is 13.

Example

Find the sum of the geometric series

$$8 - 4 + 2 - 1 + \ldots$$

where there are 5 terms in the series.

Solution

For this series, we have $a = 8$, $r = -\frac{1}{2}$ and $n = 5$. So

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} \\ S_5 &= \frac{8(1-(-\frac{1}{2})^5)}{1-(-\frac{1}{2})} \\ &= \frac{8(1-(-\frac{1}{32}))}{\frac{3}{2}} \\ &= \frac{2\times 8\times \frac{33}{32}}{3} \\ &= \frac{11}{2} \\ &= 5\frac{1}{2}\,. \end{aligned}$$

Example

Find the sum of the geometric series

$$2+6+18+54+..$$

where there are 6 terms in the series.

Solution

For this series, we have $a = 2$, $r = 3$ and $n = 6$. So

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} \\ S_6 &= \frac{2(1-3^6)}{1-3} \\ &= \frac{2(1-729)}{-2} \\ &= -(-728) \\ &= 728\,. \end{aligned}$$

Example

How many terms are there in the geometric progression

$$2,\ 4,\ 8,\ \ldots,\ 128\,?$$

Solution

In this sequence $a = 2$ and $r = 2$. We also know that the n-th term is 128. But the formula for the n-th term is ar^{n-1}. So

$$\begin{aligned} 128 &= 2 \times 2^{n-1} \\ 64 &= 2^{n-1} \\ 2^6 &= 2^{n-1} \\ 6 &= n-1 \\ n &= 7\,. \end{aligned}$$

So there are 7 terms in this geometric progression.

Example

How many terms in the geometric progression

$$1,\ 1{\cdot}1,\ 1{\cdot}21,\ 1{\cdot}331,\ \ldots$$

will be needed so that the sum of the first n terms is greater than 20?

Solution

The sequence is a geometric progression with $a = 1$ and $r = 1{\cdot}1$. We want to find the smallest value of n such that $S_n > 20$. Now

$$S_n = \frac{a(1-r^n)}{1-r}\,,$$

so

$$\begin{aligned} \frac{1 \times (1 - 1{\cdot}1^n)}{1 - 1{\cdot}1} &> 20 \\ \frac{1 - 1{\cdot}1^n}{-0{\cdot}1} &> 20 \\ (1{\cdot}1^n - 1) \times 10 &> 20 \\ 1{\cdot}1^n - 1 &> 2 \\ 1{\cdot}1^n &> 3\,. \end{aligned}$$

If we now take logarithms of both sides, we get

$$n \ln 1{\cdot}1 > \ln 3$$

and as $\ln 1{\cdot}1 > 0$ we obtain

$$n > \ln 3 / \ln 1{\cdot}1 = 11.5267\ldots$$

and therefore the smallest whole number value of n is 12.

Exercise:

1. The fourth term of a GP is -6 and the seventh term is 48. Find the first three terms of the progression.
2. Find the difference between the sums to ten terms of the AP and GP whose first two terms are -2 and 4.
3. In a GP, the sum of the second and third terms is 9. The seventh term is eight times the fourth. Find the first term, the common ratio, and the fifth term.
4. If S_n is the sum of the first n terms of a GP whose first term is a and whose common ration is r, show that $\dfrac{S_{3n} - S_{2n}}{S_n} = r^{2n}$.

(a) Find the sum of the first five terms of the GP with first term 3 and common ratio 2.

(b) Find the sum of the first 20 terms of the GP with first term 3 and common ratio 1.5.

(c) The sum of the first 3 terms of a geometric series is $\frac{37}{8}$. The sum of the first six terms is $\frac{3367}{512}$. Find the first term and common ratio.

(d) How many terms in the GP $4, 3.6, 3.24, \ldots$ are needed so that the sum exceeds 35?

7.2.2 Geometric mean

☐ **Geometric mean** - the geometric mean of two numbers P and Q is a number A such that the series $P + A + Q$ form a GP. Thus, the common ratio is

$$\frac{A}{P} = \frac{Q}{A} \quad \Rightarrow \quad A^2 = PQ \quad \Rightarrow \quad \boxed{A = \sqrt{PQ}}$$

Examples:

1. Insert four geometric means between 5 and 1215.

Solution

Let the geometric means be A, B, C and D. Thus, we have the series $5 + A + B + C + D + 1215$. Now, $u_1 = 5 = a$ and $u_6 = 1215$. Letting the common ratio be r, we have $ar^5 = 1215 \quad \Rightarrow r = \left(\dfrac{1215}{5}\right)^{1/5} = 3$. Therefore, $\dfrac{A}{5} = 3$, $\dfrac{B}{A} = 3$, $\dfrac{C}{B} = 3$, and $\dfrac{D}{C} = 3$. Hence, $A = 15, B = 45, C = 135$ and $D = 405$.

7.3 Applications of series

7.3.1 Compound Interest

Compound interest is when an initial investment is made and then interest is paid on the initial amount and on the interest already earned on a regular basis (usually every year).

Examples:

1. A customer makes a single deposit of Sh. 16000 in an account which pays compound interest at a rate of 6% p.a. Determine:

   (a) how much the investment is worth after 12 years

   (b) after how many years the investment will be worth three times its original value.

Solution

(a) Amount after

$$\text{1 year:} \quad 16000 + \frac{6}{100} \times 16000 = 16000(1.06)$$

$$\text{2 years:} \quad 16000(1.06) + \frac{6}{100} \times 16000(1.06) = 16000(1.06)^2$$

$$\text{3 years:} \quad 16000(1.06)^2 + \frac{6}{100} \times 16000(1.06)^2 = 16000(1.06)^3$$

This pattern shows that the investment after n years will be

$$A = 16000(1.06)^n = 16000(1 + \frac{6}{100})^n$$

So, the investment after 12 years will be $A = 16000(1.06)^{12} = \text{Sh. } 32,195$ (to the nearest shillings).

(b) We determine the value of n (in years) for which $16000(1.06)^n = 3 \times 16000$. Thus,

$$(1.06)^n = 3 \quad \Rightarrow \quad n = \frac{\log 3}{\log 1.06} = 18.85$$

Therefore, $n = 19$ complete years.

→ Note: in compound interest problems, the formula for the amount (value) A of a single investment (principal amount) P at a rate of $r\%$ per annum/unit time is

$$\boxed{A = P\left(1 + \frac{r}{100}\right)^n}$$

This formula will be used in all cases where the value of an item appreciates/depreciates at a constant rate per unit time. However, in the case of depreciation r will be negative.

2. Sh. 100,000 was invested on 1st January 1990. An additional Sh. 6,000 was added to the investment at the beginning of each subsequent year. The investment earns a compound interest of 8% p.a. Find the value of the investment on 31st December 2000.

Solution

There are a total of 11 years between 1st Jan 1990 and 31st Dec 2000. Now, amount on

31st December,

1990 (1 year): $100000 + \frac{8}{100} \times 100000 = 100000(1.08)$

1991 (2 years): $[100000(1.08) + 6000] + \frac{8}{100} \times [100000(1.08) + 6000] = 100000(1.08)^2 + 6000(1.08)$

1992 (3 years): $[100000(1.08)^2 + 6000(1.08) + 6000] \times 1.08 = 100000(1.08)^3 + 6000\left[1.08 + (1.08)^2\right]$

This pattern shows that the investment after n years will be

$$A = 100000(1.08)^n + 6000\sum_{k=1}^{n-1}(1.08)^k$$

So, the investment after 11 years will be

$$\begin{aligned} A &= 100000(1.08)^{11} + 6000\sum_{k=1}^{10}(1.08)^k \\ &= 100000(1.08)^{11} + 6000\left[1.08 + 1.08^2 + 1.08^3 + \cdots + 1.08^{10}\right] \\ &= 100000(1.08)^{11} + 6000\left\{\frac{1.08\left(1.08^{10} - 1\right)}{1.08 - 1}\right\} = 100000(1.08)^{11} + 81000\left(1.08^{10} - 1\right) \\ &= \text{Sh. } 327,037 \quad \text{(to the nearest shillings) after 11 years.} \end{aligned}$$

3. A ball is dropped from a height of 9m. It hits the ground and bounces to a height of 6m. It continues to bounce up and down. On each bounce, it rises $\frac{2}{3}$ of the height of the previous bounce. How far has the ball traveled (both up and down) when it hits the ground for the seventh time?

Solution

The ball originally drops 9m and when the ball hits the ground for the 7th time, it has completed 6 bounces. So, the total distance traveled until the ball hits the ground for the seventh time is given by the series

$$9 + \frac{2}{3}(9)(2) + \left(\frac{2}{3}\right)^2(9)(2) + \left(\frac{2}{3}\right)^3(9)(2) + \cdots + \left(\frac{2}{3}\right)^6(9)(2)$$

$$= 9 + \frac{a\left(1 - r^6\right)}{1 - r}, \quad \text{where} \quad a = \frac{2}{3}(9)(2) \quad \text{and} \quad r = \frac{2}{3}$$

$$= 9 + \frac{\frac{2}{3}(9)(2)\left[1 - \left(\frac{2}{3}\right)^6\right]}{1 - \frac{2}{3}} = 9 + 36\left[1 - \frac{64}{729}\right] = 41.84\text{m}$$

Worked example

A new virus is circulating on a remote island. On day one there were 10 people infected, with the number of new infections increasing at a rate of 40% per day.

a) Find the expected number of people newly infected on the 7th day.

Identify the geometric sequence:

$u_1 = 10$, $r = 1.4$ ← 40% increase so 140% of the day before

New infections : u_7

Formula for n^{th} term of a geometric series :

$$u_n = u_1 r^{n-1}$$

Sub in $u_1 = 10$, $r = 1.4$

$$u_7 = 10(1.4)^6 = 75.29...$$

Expected number of new infections = 75

b) Find the expected number of infected people after one week (7 days), assuming no one has recovered yet.

Total infections : S_7

Formula for the sum of a geometric series:

$$S_n = \frac{u_1(r^n - 1)}{r - 1}$$ ← $r > 1$ so this version is easier to use.

Sub in $u_1 = 10$, $r = 1.4$

$$S_7 = \frac{10(1.4^7 - 1)}{1.4 - 1} = 238.53...$$

Expected number of total infections = 239

Exercise:

1. A customer deposits Sh. 10,000 on first January every year for four years. How much is the investment worth at the end of the four years if it attracts a compound interest of 12% p.a.? [ans: Sh. 53,528]
2. A single deposit of Sh. 150,000 is invested for four years at compound interest. determine the rate at which the investment will be Sh. 182,326. [ans: $r = 5\%$]
3. A car has an initial value of Sh. 3,000,000. If it depreciates at a rate of 14% p.a., determine after how many complete years the car will be worth Sh. 1,000,000. [ans: $= 8$ years]
4. A man deposits Sh. 2000 in his savings account on 1st January each year for 20 years. If the account gives 3% compound interest p.a. What will be the total value of his savings at the end of 20 years? (Answer correct to the nearest shillings). [ans: = Sh. 55353]
5. To save for her daughter's college education, Ms. Mirada decides to put aside $ 50 every month in a credit union account paying 10% interest compounded monthly. She begins this savings program when her daughter is 3 years old. How much will she have saved by the time she makes the 180th deposit? How old is her daughter at this time?
6. Given that an experiment culture has an initial population of 50 bacteria and the population increased by 80% every 20 minutes. Determine the time it will take to have a population of 1.2 million bacteria. [ans:]

7.3.2 Exponential Growth and Decay

Exponential growth takes place when the amount of a substance increases at the same rate within equal time frames.
Examples include the multiplication of bacteria, a population of people etc. However, exponential decay happens when the amount of a substance reduces at the same rate over equal time intervals.
In applying the idea of exponential growth and decay, little changes will be made to the general formula:

$$y = ab^x$$

remember that **b** is the base that represents the **growth factor** or decay **factor.** this will be:

b = 1 + r for exponential growth

or

b = 1 - r for exponential decay

1 represents 100% and r represents the rate of increase in percentage.

Also, x will be changed to **t** which represents the **time interval** . Thus, the general equation changes to:

$y = a(1 + r)^t$ **for exponential growth**

or

$y = a(1 - r)^t$ **for exponential decay**

where y is the final amount, a is the starting value, r is the rate of increase and t is the time interval.

Note that the expression 1 + r or 1 - r is also referred to as the **multiplier** .

Exponential Growth	Exponential Decay
$f(x) = ab^x$	$f(x) = ab^{-x}$
$f(x) = a(1 + r)^t$	$f(x) = a(1 - r)^t$
$P = P_0e^{kt}$	$P = P_0e^{-kt}$

In the above formulas the 'a' or P_0 is the initial quantity of the substance.

Further for exponential growth $b = 1 + r = e^k$ and for exponential decay we have $b = 1 - r = e^{-k}$.

Example 1: What is the amount received from the investment fund after 2 years, if $.100,000 were invested at the compounding rate of 5% per every quarter?

Solution:

The invested principal is a = $100,000, the rate of compounding growth is r = 5% = 0.05 per quarter.

The time period is 2 years, and there are 4 quarters in a year, and we have t = 8.

Applying the concepts of exponential growth and decay we have the following expressions for exponential growth.

$f(x) = a(1 + r)^t$

$f(x) = 100,000(1 + 0.05)^8$

$f(x) = 1,00,000(1.05)^8 = 100,000 \times 1.47745544 = 147745.44$

Therefore an amount of $1,47, 746 is received after a period of 2 years.

Example 2: The radioactive material of thorium decays at the rate of 8% per minute. What part of 10 grams of thorium would be remaining after 5 minutes?

Solution:

The given initial quantity of thorium is a = 10grams, the rate of decay per minute is r = 8% = 8/100 = 0.08, and the time steps t = 5.

Here we can apply the concepts of exponential growth and decay, and the exponential decay formula for the decay of thorium is as follows.

$f(x) = a(1 - r)^t$

$^f(x) = 10(1 - 0.08)^5 = 10(0.92)^5 = 6.5908$

Therefore a quantity of 6.6 grams of thorium remains after 5 minutes.

Example 1:

James bought $200 worth of stock from an automobile company in Japan at the end of the year, and it is expected that the stock value is to increase by 25% annually.

a. Derive an exponential function that expresses James' stock value in t years

b. After 6 years, Mary purchased all of the same stock owned by James for $1000. How much has James gained or lost?

Solutions:

a. To decide what formula to apply, you need to know if there is an increase or decrease in the rate. In this case, there is a 25% increase.

So, we apply $y = a(1 + r)^t$

Let's define components of the formula

y is not given

a is $200

r is 25% = 0.25

t is given as t

Thus,

$y = 200(1 + 0.25)^t y = 200(1.25)^t$

so James 'stock value y in t years' time would be $200(1.25)^t$ dollars.

b. Since we have the equation expressing the exponential function of James' stock in t years, it would be easy to find his stock value in 6 years.

Thus, t = 6

Substitute the value of t as 6 in the equation:

$y = 200(1.25)^t y = 200(1.25)^6 y = 200 \times 3.8147 y = 762.9394 y = 763 to the nearest dollar$

Meanwhile, Mary purchases his stock of 6 years at the value of $1000 . This clearly shows that James has gained $1000-$763 = $237.

Example 2:

A certain parasite is said to double in the human body every six months. If the current population is 100 in an infected individual, what is expected to be the population of this parasite in the individual after two years if left untreated?

Solutions:

Watch out for the detail that this increase occurs every 6 months. This means per year, it increases twice.

Also, note that the rate of increase is double, which means that it is a 100% increase.

Remember that 1 in (1 + r) means 100%.

Since the parasite doubles, it means an additional whole increase, ie 100% + 100%.

b = 1 + r = 1 + 1 = 2

so, the growth factor is 2.

Thus, $\mathrm{y} = \mathrm{a}(1 + \mathrm{r})^{\mathrm{t}}$

$y = 100(1 + 1)^{2\times2}$

note that we are having a product of 2 and 2 because the increase occurs twice in two years

$y = 100(2)^4 y = 100 \times 16 y = 1600$

So, in 2 years, the parasite in the individual would have increased from 100 to 1600.

Therefore, when there is a doubling increase, b = 1 + r = 2; similarly, when the increase is triple, b = 3 and so on.

7.4 Convergence of a series

If S_n is the sum of the first n terms of any series and if $\lim_{n\to\infty} S_n$ exists and is finite, then the series S_n is said to converge. Here, the sum to infinity is given by

$$S = S_\infty = \lim_{n\to\infty} S_n$$

If the series doesn't converge i.e., $\lim_{n\to\infty} S_n = \pm\infty$, then it is said to be divergent. We consider the AP and GP.

i) **Convergence of an AP:** any AP where $d > 0$ will diverge i.e., $\lim_{n\to\infty} \frac{n}{2}[2a + (n-1)d] = \infty$ and if $d < 0$, $\lim_{n\to\infty} \frac{n}{2}[2a + (n-1)d] = -\infty$ meaning it will diverge too.

ii) **Convergence of a GP:** recall that for a general GP, the sum to n terms is given by $S_n = \frac{a(1-r^n)}{1-r}$.

- If $|r| < 1$, then $\lim_{n\to\infty} r^n = 0$. So, $\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r}$. Hence, a GP converges to the sum $\frac{a}{1-r}$, provided $|r| < 1$.
- If $|r| > 1$, then $\lim_{n\to\infty} r^n = \infty$ and the series diverges.

Example Find the sum to infinity of the geometric progression

$$1, \ \tfrac{1}{3}, \ \tfrac{1}{9}, \ \tfrac{1}{27}, \ \ldots .$$

Solution For this geometric progression we have $a = 1$ and $r = \frac{1}{3}$. As $-1 < r < 1$ we can use the formula, so that

$$S_\infty = \frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}} = \tfrac{3}{2}.$$

Key Point

The sum to infinity of a geometric progression with starting value a and common ratio r is given by

$$S_\infty = \frac{a}{1-r}$$

where $-1 < r < 1$.

Examples:

1. Consider the problem of the ball in example (3) above. Find the distance covered by the ball before coming to rest.

Solution

Theoretically, the ball bounces up and down indefinitely (practically, the ball comes to rest after a finite number of bounces). So, the distance traveled before coming to rest is

$$S_n = 9 + \frac{2}{3}(9)(2) + \left(\frac{2}{3}\right)^2 (9)(2) + \left(\frac{2}{3}\right)^3 (9)(2) + \cdots$$

$$S_\infty = 9 + \frac{a}{1-r} = 9 + \frac{\frac{2}{3}(9)(2)}{\left\{1 - \frac{2}{3}\right\}} = 9 + \frac{12}{\left\{1 - \frac{2}{3}\right\}} = 9 + 36 = 45\text{m}$$

2. The fifth term of a convergent GP is the arithmetic mean of the preceding two terms. Find the common ration given that the common ration and the first term are non-zero. If the first term is 12, find the sum to infinity.

Solution

i) Let the series be $a + ar + ar^2 + \cdots$. Then, $u_5 = ar^4 = \frac{1}{2}(ar^2 + ar^3)$.

$$\Rightarrow \quad 2ar^4 = ar^2(1+r) \quad \Rightarrow \quad 2r^2 - r - 1 = 0 \quad \Rightarrow \quad r = 1 \text{ or } r = -\frac{1}{2}$$

Since the series is convergent, then $|r| < 1$. Therefore, $r = -\frac{1}{2}$.

ii) When $a = 12$, the sum to infinity is given by $S_\infty = \frac{a}{1-r} = \frac{12}{1-\left(-\frac{1}{2}\right)} = 8$.

3. Express $0.\dot{4}\dot{5}$ as a fraction.

Solution

$$0.\dot{4}\dot{5} = 0.454545\cdots = \frac{45}{100} + \frac{45}{10000} + \frac{45}{1000000} + \cdots$$

This is a GP with $a = \frac{45}{100}$ and $r = \frac{1}{100}$. Therefore,

$$0.\dot{4}\dot{5} = S_\infty = \frac{45/100}{\left\{1 - \frac{1}{100}\right\}} = \frac{45}{100} \times \frac{100}{99} = \frac{45}{99} = \frac{5}{11}$$

Exercise:

(a) Find the sum to infinity of the GP with first term 3 and common ratio $\frac{1}{2}$.

(b) The sum to infinity of a GP is four times the first term. Find the common ratio.

(c) The sum to infinity of a GP is twice the sum of the first two terms. Find possible values of the common ratio.

1. Express the following recurring decimals as rational numbers

   (a) $0.0\dot{7}$ [ans: $= \frac{7}{90}$]

   (b) $1.00\dot{4}$ [ans: $= 1\frac{1}{225}$]

   (c) $2.9\dot{6}\dot{0}$ [ans: $= 2\frac{317}{330}$]

   (d) $0.2\dot{4}\dot{5}\dot{6}$ [ans: $= 2\frac{409}{1665}$]

2. Find the common ratio of a GP whose sum to infinity is three times the first term. [ans: $=\frac{2}{3}$]

3. The second term of a GP is 24 and its sum to infinity is 100. Find the two possible values of the common ratio and the corresponding first term.

7.4.1 Sequence and Series Formulas

*Here, a = first term, d = common difference, r = common ratio, n = position of term, l = last term.

	Arithmetic Progression	Geometric Progression
Sequence	a, a+d, a+2d,.....,a+(n-1)d,....	a, ar, ar^2,....,$ar^{(n-1)}$,...
Common Difference or Ratio	Successive term – Preceding term Common difference = d = $a_2 - a_1$	Successive term/Preceding term Common ratio = r = $ar^{(n-1)}/ar^{(n-2)}$
General Term (nth Term)	$a_n = a + (n-1)d$	$a_n = ar^{(n-1)}$
nth term from the last term	$a_n = l - (n-1)d$	$a_n = l/r^{(n-1)}$
Sum of first n terms	$s_n = n/2(2a + (n-1)d)$	$s_n = a(1 - r^n)/(1 - r)$ if $\|r\| < 1$ $s_n = a(r^n - 1)/(r - 1)$ if $\|r\| > 1$

7.4.2 Difference Between Sequences and Series

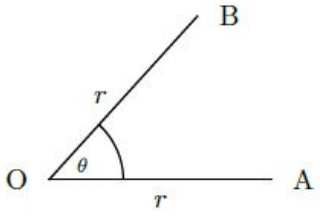
Sequences	Series
Set of elements that follow a pattern	Sum of elements of the sequence
Order of elements is important	Order of elements is not so important
Finite sequence: 1,2,3,4,5	Finite series: 1+2+3+4+5
Infinite sequence: 1,2,3,4,.....	Infinite Series: 1+2+3+4+.....

Lecture 8

8 Trigonometry

8.1 Circular functions

When a line OA is pivoted at O and rotates from its initial position OA to its new location OB. Then, the angle AOB is the measure of rotation of line OA.

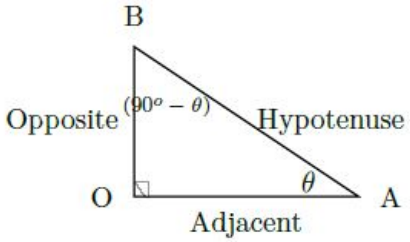


The angle θ is usually measured in either degree (^{o}) or radian (^{c}). One complete revolution $= 360^o$. One radian is the angle subtended by an arc whose length is equal to the radius of the circle. The number of radians in one complete revolution is given by the ratio $\frac{\text{circumference}}{\text{radius}} = \frac{2\pi r}{r} = 2\pi$ radians Thus, 2π radians $= 360^0$. Therefore, the degree and radian are related by:

$$\boxed{\pi^c = 180^o}$$

8.2 Trigonometric ratios

Consider the right angle triangle OAB. A simple rule to find the trigonometric ratios is SOHCAHTOA.



$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{\text{OB}}{\text{AB}}$$
$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{\text{OA}}{\text{AB}}$$
$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{\text{OB}}{\text{OA}}$$

Now, $\frac{\sin\theta}{\cos\theta} = \frac{\text{OB/AB}}{\text{OA/AB}} = \frac{\text{OB}}{\text{OA}} = \tan\theta$. Therefore,

$$\boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}}$$

Also, $\cos(90^o - \theta) = \frac{\text{OB}}{\text{AB}} = \sin\theta$. Therefore,

$$\boxed{\sin\theta = \cos(90^o - \theta)}$$

8.3 Inverse trigonometric ratios

The following inverse trigonometric ratios are useful

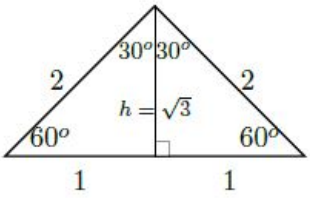
$$\text{cosec}\theta = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{1}{\tan\theta}$$

Exercise:

1. Use the formula $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$ to mark two sides of a right-angled triangle whose included angle is 2θ. Use Pythagoras theorem to find the length of the third side in terms of $\tan\theta$. Hence, write down a formula for $\sin 2\theta$ in terms of $\tan\theta$ and use it to find the smallest positive value of θ for which $2\tan\theta = \operatorname{cosec}\theta$.

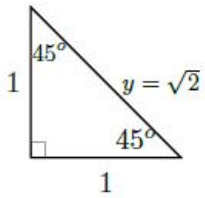
8.4 Trigonometric ratios of special angles ($30°, 45°, 60°$)

Consider an equilateral triangle



From Pythagoras theorem, we have $h^2 = 2^2 - 1^2 = 3 \quad \Rightarrow \quad h = \sqrt{3}$.
From the diagram, $\sin 30^o = \dfrac{1}{2}$, $\cos 30^o = \dfrac{\sqrt{3}}{2}$, and $\tan 30^o = \dfrac{1}{\sqrt{3}}$.
Similarly, $\sin 60^o = \dfrac{\sqrt{3}}{2}$, $\cos 60^o = \dfrac{1}{2}$, and $\tan 60^o = \sqrt{3}$.

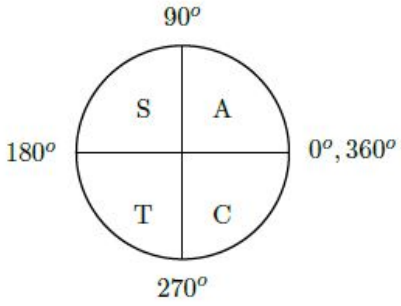
Consider the right angle triangle



From Pythagoras theorem, we have $y^2 = 1^2 + 1^2 = 2 \quad \Rightarrow \quad y = \sqrt{2}$.
From the diagram, $\sin 45^o = \dfrac{1}{\sqrt{2}}$, $\cos 45^o = \dfrac{1}{\sqrt{2}}$, and $\tan 45^o = 1$.

8.5 Trigonometric ratios of angles greater than $90°$

In this case, we use the unit circle. The unit circle consists of 4 quadrants.



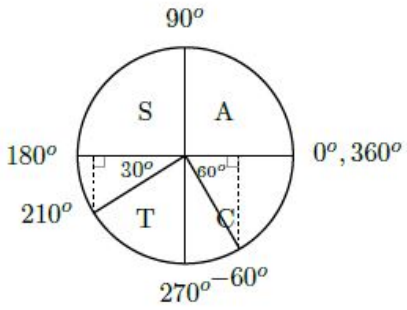
Trigonometric ratios of all (A) angles in the first quadrant are positive. Only the sine (S) of angles in the second quadrant is positive. Only the tangent (T) of angles in the third quadrant is positive. Only the cosine (C) of angles in the fourth quadrant is positive.

→ Note: positive angles are measured in anticlockwise direction while negative angles are measured in clockwise direction, from the principal axis.

Examples:

1. Evaluate $\sin 210^o$ and $\cos(-60^0)$ without using tables or calculator.

Solution



From the unit circle, $\sin 210^o = \dfrac{-0.5}{1} = -\dfrac{1}{2}$ and $\cos(-60^o) = \dfrac{0.5}{1} = \dfrac{1}{2}$.

2. Simplify the following without using mathematical tables or calculator, $\dfrac{\sec^2 150^o\,(1-\cos 300^o)}{(1+\tan^2 210^o)\,(1+\sin 330^o)}$. [ans: 1]

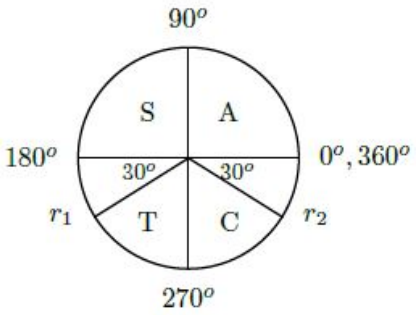
8.6 Trigonometric equations

Trigonometric equations have unlimited number of roots. For example, $\sin\theta = 0$ is satisfied by $\theta = 0^o, \pm 180^o, \pm 360^o, \cdots$. This can be written as $\theta = n\pi$, where $n = 0, \pm 1, \pm 2, \cdots$. So, there is a need to specify for what range of values the roots are required.

Examples:

1. Solve $\sin\theta = -\dfrac{1}{2}$ in the range $0^o \leq \theta \leq 360^o$.

Solution



From the unit circle, $\sin\theta$ is negative in the 3rd and 4th quadrants. Therefore, $180^o \leq \theta \leq 360^o$. Let the roots be r_1 and r_2. The included angle is given by $a = \sin^{-1}(\frac{1}{2}) = 30^o$. Hence, the roots are: $r_1 = (180^o + 30^o) = 210^o$ and $r_2 = (360^o - 30^o) = 330^o$.

Exercise:

1. Solve the following trigonometric equations

(a) $\cos 2\theta = 0.6428$ in the range $-180^o \leq \theta \leq 180^o$. [ans: $\theta = \pm 25^o, \pm 155^o$]

(b) $2\sin^2\theta = \sin\theta$ in the range $0^o \leq \theta \leq 360^o$. [ans: $\theta = 0^o, 30^o, 150^o, 180^o, 360^o$]

(c) $\tan\theta = 2\sin\theta$ in the range $0^o \leq \theta \leq 360^o$. [ans: $\theta = 60^o, 300^o$]

(d) $4\sin^2\theta = 3\cos^2\theta$ in the range $0^o \leq \theta \leq 360^o$. [ans: $\theta = 40.9^o, 220.9^o$]

8.7 Trigonometric identities

The following are the basic trigonometric identities

$$\begin{aligned}\sin(A+B) &= \sin A\cos B+\cos A\sin B\\ \sin(A-B) &= \sin A\cos B-\cos A\sin B\\ \cos(A+B) &= \cos A\cos B-\sin A\sin B\\ \cos(A-B) &= \cos A\cos B+\sin A\sin B\end{aligned}$$

Substituting $A=B=\theta$ in identity (6) yields $\cos 0=\cos^2\theta+\sin^2\theta$. But $\cos 0=1$, therefore,

$$\boxed{\cos^2\theta+\sin^2\theta=1}\quad---(i)$$

Dividing both sides of eq. (i) by $\cos^2\theta$, we get

$$1+\left(\frac{\sin\theta}{\cos\theta}\right)^2=\left(\frac{1}{\cos\theta}\right)^2\quad\Rightarrow\quad\boxed{1+\tan^2\theta=\sec^2\theta}\quad---(ii)$$

Dividing both sides of eq. (i) by $\sin^2\theta$, we get

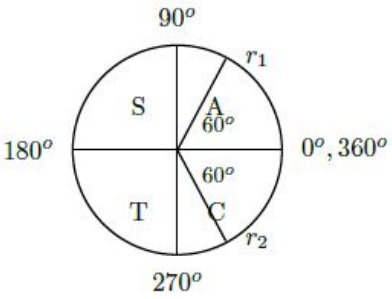
$$\left(\frac{\cos\theta}{\sin\theta}\right)^2+1=\left(\frac{1}{\sin\theta}\right)^2\quad\Rightarrow\quad\boxed{\cot^2\theta+1=\text{cosec}^2\theta}\quad---(iii)$$

Examples:

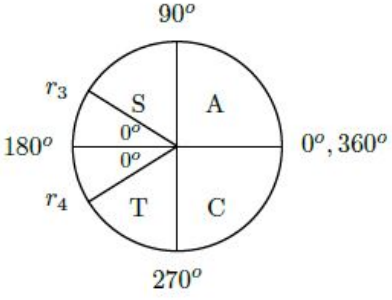
1. Solve $1+\cos\theta=2\sin^2\theta$ in the range $0^o\le\theta\le 360^o$.

Solution

From $\cos^2\theta+\sin^2\theta=1\quad\Rightarrow\quad\sin^2\theta=1-\cos^2\theta$. Substituting in the given equation yields $1+\cos\theta=2(1-\cos^2\theta)\quad\Rightarrow\quad 2\cos^2\theta+\cos\theta-1=0$. Let $\cos\theta=x$. Thus, we have $2x^2+x-1=0\quad\Rightarrow\quad(2x-1)(x+1)=0$. The roots are given by $x=\frac{1}{2}$ and $x=-1$. That is, $\cos\theta=\frac{1}{2}$ and $\cos\theta=-1$. From the unit circle, we have



Case 1: $\cos\theta=\frac{1}{2}$.
From the unit circle, cosine is +ve in the 1st and 4th quadrants.
Let the roots be r_1 and r_2.
The included angle is given by $a=\cos^{-1}\left(\frac{1}{2}\right)=60^o$.
Hence, the roots are: $r_1=60^o$ and $r_2=(360^o-60^o)=300^o$.



Case 2: $\cos\theta=-1$.
From the unit circle, cosine is -ve in the 2nd and 3rd quadrants.
Let the roots be r_3 and r_4.
The included angle is given by $a=\cos^{-1}(1)=0^o$.
Hence, the roots are: $r_3=r_4=180^o$.

Therefore, $\theta=60^o, 180^o$ and 300°.

2. Eliminate θ from $x = a\sin\theta$ and $y = b\tan\theta$.

Solution

From $x = a\sin\theta \quad \Rightarrow \quad \sin\theta = \frac{x}{a} \quad \Rightarrow \quad \text{cosec}\theta = \frac{a}{x}$ and $y = b\tan\theta \quad \Rightarrow \quad \tan\theta = \frac{y}{b} \quad \Rightarrow$ $\cot\theta = \frac{b}{y}$. From the relationship $1 + \cot^2\theta = \text{cosec}^2\theta$, we have

$$1 + \left(\frac{b}{y}\right)^2 = \left(\frac{a}{x}\right)^2 \quad \Rightarrow \quad 1 + \frac{b^2}{y^2} = \frac{a^2}{x^2}$$

3. Find without using tables or calculator the values of $\sin 165^o$, $\cos 75^o$, and $\cos 15^o$.

Solution

$$\begin{aligned}
\sin 165^o &= \sin(120^o + 45^o) = \sin 120^o\cos 45^o + \sin 45^o\cos 120^o = \sin 60^o\cos 45^o - \sin 45^o\cos 60^o \\
&= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4} \\
\cos 75^o &= \cos(120^o - 45^o) = \cos 120^o\cos 45^o + \sin 120^o\sin 45^o = -\cos 60^o\cos 45^o + \sin 60^o\sin 45^o \\
&= -\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4} \\
\cos 15^o &= \cos(60^o - 45^o) = \cos 60^o\cos 45^o + \sin 60^o\sin 45^o = \frac{1}{2}\cdot\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4}
\end{aligned}$$

4. Given that $\sin A = \frac{4}{5}$ and $\tan B = \frac{5}{12}$, where A is an obtuse angle and B is an acute angle. Find, $\cos(A + B)$ without using mathematics tables or calculators.

Solution

From $\sin A = \frac{4}{5}$ and A is obtuse means that A is in the 2nd quadrant. The base (adjacent) of a right angle triangle with opposite 4 units and hypotenuse 5 units is: adjacent $= \sqrt{5^2 - 4^2} = 3$ units. Thus, we have $\cos A = -\frac{3}{5}$ and $\tan A = -\frac{4}{3}$ (since A is obtuse). Similarly, from $\tan B = \frac{5}{12}$ and B is acute means that B is in the 1st quadrant. The hypotenuse of a right angle triangle with opposite 5 units and base (adjacent) 12 units is: hypotenuse $= \sqrt{5^2 + 12^2} = 13$ units. Thus, we have $\cos B = \frac{12}{13}$ and $\sin B = \frac{5}{13}$. Now,

$$\cos(A + B) = \cos A\cos B - \sin A\sin B = \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) - \left(\frac{4}{5}\right)\left(\frac{5}{13}\right) = -\frac{56}{65}$$

5. Find the formula for $\tan(A+B)$ and $\tan(A-B)$. Hence, if $\tan\theta = \frac{4}{3}$ and θ is acute. Find without tables or calculator the values of $\tan 2\theta$, $\tan 4\theta$, and $\tan(\theta/2)$.

Solution

$$\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)} = \frac{\sin A\cos B + \sin B\cos A}{\cos A\cos B - \sin A\sin B} = \frac{\frac{\sin A\cos B}{\cos A\cos B} + \frac{\sin B\cos A}{\cos A\cos B}}{\frac{\cos A\cos B}{\cos A\cos B} - \frac{\sin A\sin B}{\cos A\cos B}} = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

Similarly,

$$\tan(A-B) = \frac{\sin(A-B)}{\cos(A-B)} = \frac{\sin A\cos B - \sin B\cos A}{\cos A\cos B + \sin A\sin B} = \frac{\frac{\sin A\cos B}{\cos A\cos B} - \frac{\sin B\cos A}{\cos A\cos B}}{\frac{\cos A\cos B}{\cos A\cos B} + \frac{\sin A\sin B}{\cos A\cos B}} = \frac{\tan A - \tan B}{1 + \tan A\tan B}$$

Now,

$$\tan 2\theta = \tan(\theta+\theta) = \frac{\tan\theta + \tan\theta}{1 - \tan\theta\tan\theta} = \frac{\frac{4}{3} + \frac{4}{3}}{1 - \left(\frac{4}{3}\right)^2} = \frac{\frac{8}{3}}{1 - \frac{16}{9}} = -\frac{24}{7}$$

$$\tan 4\theta = \tan(2\theta+2\theta) = \frac{\tan 2\theta + \tan 2\theta}{1 - \tan 2\theta\tan 2\theta} = \frac{-\frac{24}{7} - \frac{24}{7}}{1 - \left(-\frac{24}{7}\right)^2} = \frac{-\frac{48}{7}}{1 - \frac{576}{49}} =$$

$$\tan\theta = \tan\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \frac{\tan(\theta/2) + \tan(\theta/2)}{1 - \tan(\theta/2)\tan(\theta/2)} = \frac{2\tan(\theta/2)}{1 - \tan^2(\theta/2)} \quad \Rightarrow \quad \frac{4}{3} = \frac{2\tan(\theta/2)}{1 - \tan^2(\theta/2)}$$

$$\Rightarrow \quad 2\tan^2(\theta/2) + 3\tan^2(\theta/2) - 2 = 0 \quad \Rightarrow \quad (2\tan(\theta/2) - 1)(\tan(\theta/2) + 2) = 0$$

Therefore, $\tan(\theta/2) = -2$ or $\tan(\theta/2) = \frac{1}{2}$.

6. Show that $\tan^{-1}2 + 2\tan^{-1}3 = \pi + \cot^{-1}2$.

Solution

Let $A = \tan^{-1}2$ and $B = \tan^{-1}3$, where A and B are acute. Thus, $\tan A = 2$ and $\tan B = 3$. Since $\tan A > 1 \quad \Rightarrow \quad 45^o \leq A \leq 90^o$. Also, $\tan B > 1 \quad \Rightarrow \quad 45^o \leq B \leq 90^o \quad \Rightarrow \quad 90^o \leq 2B \leq 180^o$. Hence, $2B$ is obtuse. Now,

$$\tan 2B = \tan(B+B) = \frac{\tan B + \tan B}{1 - \tan B\tan B} = \frac{3+3}{1-(3)(3)} = -\frac{3}{4}$$

Therefore,

$$\tan(A+2B) = \frac{\tan A + \tan 2B}{1 - \tan A\tan 2B} = \frac{2 - \frac{3}{4}}{1 - 2\left(-\frac{3}{4}\right)} = \frac{1}{2}$$

Since $\tan(A+2B)$ is positive, A is acute and $2B$ is obtuse, then $A+2B$ must be a reflex angle in the 3rd quadrant i.e., $180^o \leq A + 2B \leq 270^o$. From $\tan(A+2B) = \frac{1}{2} \quad \Rightarrow \quad \cot(A+2B) = 2$, the included angle is given by $a = \tan^{-1}\left(\frac{1}{2}\right) = \cot^{-1}2$. Therefore, $A + 2B = \pi + a$. Substituting yields $\tan^{-1}2 + 2\tan^{-1}3 = \pi + \cot^{-1}2$ as required.

7. Solve $\sin^{-1}\theta + \sin^{-1} 3\theta = \frac{\pi}{2}$.

Solution

Let $A = \sin^{-1}\theta$ and $B = \sin^{-1} 3\theta$. Thus, $\sin A = \theta$ and $\sin B = 3\theta$. Now,

$$\cos A = \sqrt{1-\sin^2 A} = \sqrt{1-\theta^2} \quad \text{and} \quad \cos B = \sqrt{1-\sin^2 B} = \sqrt{1-9\theta^2}$$

Therefore,

$$\cos(A+B) = \cos A\cos B - \sin A \sin B = \cos(\pi/2) \quad \Rightarrow \quad \sqrt{1-\theta^2}\sqrt{1-9\theta^2} - 3\theta^2 = 0$$

$$\Rightarrow \quad \sqrt{(1-\theta^2)(1-9\theta^2)} = 3\theta^2 \quad \Rightarrow \quad (1-\theta^2)(1-9\theta^2) = 9\theta^4 \quad \Rightarrow \quad \theta = \pm\frac{1}{\sqrt{10}}$$

Exercise:

1. Eliminate θ from the equations

   (a) $x = 1 - \sin\theta$ and $y = 1 + \cos\theta$. [ans: $x^2 + y^2 - 2x - 2y = -1$]

   (b) $x = a\text{cosec}\theta$ and $y = b\sec\theta$. [ans: $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 1$]

2. Find to the nearest integer the values of θ in the interval $-180^o \le \theta \le 180^o$, for which $3\sin^2 3\theta - 7\cos 3\theta - 5 = 0$. [ans: $\theta = \pm 37^o, \pm 84^o, \pm 157^o$]

3. Solve the equations, giving the general solutions.

   (a) $2\tan^2\theta + \sec\theta = 1$. [ans: $\theta = 0^o, 90^o, 131.8^o, 228.2^o, 270^o$]

   (b) $\sec\theta + 3 = \cos\theta + (2 + \sin\theta)\tan\theta$. [ans: $\theta = 56.3 + n\pi$, where $n \in \mathbb{N}$]

   (c) $3\sin^2\theta - \sin\theta\cos\theta - 4\cos^2\theta = 0$. [ans: $\theta = 45^o, 53.1^o, 135^o$]

   (d) $2\sin\theta = \cos(30^o - \theta)$. [ans: $\theta = \frac{\pi}{6} + n\pi, \ n \in \mathbb{Z}$]

4. Prove the following identities

   (a) $\sec\theta\cot^2\theta + \sec\theta = \sec\theta\text{cosec}^2\theta$.

   (b) $\dfrac{1-\cos\theta}{\sin\theta} = \dfrac{1}{\cos\theta + \cot\theta}$.

   (c) $\dfrac{\tan\theta + \cot\theta}{\sec\theta + \text{cosec}\theta} = \dfrac{1}{\sin\theta + \cos\theta}$.

   (d) $\dfrac{1+\sin\theta}{1-\sin\theta} = (\sec\theta + \tan\theta)^2$.

   (e) $\sec\theta - \sin\theta = \dfrac{\tan^2\theta + \cos^2\theta}{\sec\theta + \sin\theta}$.

   (f) $\sec^2\theta = \dfrac{\cos\theta}{\text{cosec}\theta - \sin\theta}$.

   (g) $\dfrac{\tan\theta - \cot\theta}{\tan\theta + \cot\theta} = (\sin\theta - \cos\theta)(\sin\theta + \cos\theta)$.

   (h) $\dfrac{1}{(\text{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)} = \tan\theta + \cot\theta$.

5. (a) If $\sin A = \frac{4}{5}$ and $\cos B = \frac{12}{13}$, where A is obtuse and B is acute. Find the values of
   (i) $\sin(A-B)$. [ans: $\frac{63}{65}$]
   (ii) $\cos(A+B)$. [ans: $-\frac{56}{65}$]
   (iii) $\sin(2A+B)$. [ans: $-\frac{323}{325}$]
   (iv) $\tan(A+B)$. [ans: $-\frac{13}{56}$]

   (b) If $\sin A = \frac{3}{5}$ and $\cos B = \frac{15}{17}$, where A is obtuse and B is acute. Find the exact values of
   (i) $\sin(A+B)$. [ans: $\frac{13}{85}$]
   (ii) $\cos(A-B)$. [ans: $-\frac{36}{85}$]
   (iii) $\tan(2A+B)$. [ans: $-\frac{304}{297}$]

6. Simplify the following without using tables and calculator, $\dfrac{\sin 18^o \cos 12^o + \cos 162^o \cos 102^o}{\sin 22^o \cos 8^o + \cos 158^o \cos 98^o}$. [ans: 1]

8.7.1 Double angle formulae

(1) Substituting $A = B = \theta$ in identity (5) yields $\cos 2\theta = \cos^2\theta - \sin^2\theta - - - (*)$. Putting $\sin^2\theta = 1 - \cos^2\theta$ in $(*)$, we get

$$\boxed{\cos 2\theta = 2\cos^2\theta - 1}$$

Similarly, putting $\cos^2\theta = 1 - \sin^2\theta$ in $(*)$, we get

$$\boxed{\cos 2\theta = 1 - 2\sin^2\theta}$$

(2) Substituting $A = B = \theta$ in identity (3) yields

$$\boxed{\sin 2\theta = 2\sin\theta\cos\theta}$$

Examples:

1. Find the values of $\sin 2\theta$ and $\cos 2\theta$ if (a) $\sin\theta = \frac{3}{5}$ and (b) $\cos\theta = \frac{12}{13}$, given that theta is acute.

Solution

(a) From $\sin\theta = \frac{3}{5}$, the base (adjacent) of a right angle triangle with opposite 3 units and hypotenuse 5 units is: adjacent $= \sqrt{5^2 - 3^2} = 4$ units. Thus, we have $\cos\theta = \frac{4}{5}$ and $\tan\theta = \frac{3}{4}$. Therefore,

$$\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = \left(\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \frac{7}{25}$$

(b) From $\cos\theta = \frac{12}{13}$, the height (opposite) of a right angle triangle with base (adjacent) 12 units and hypotenuse 13 units is: opposite $= \sqrt{13^2 - 12^2} = 5$ units. Thus, we have $\sin\theta = \frac{5}{13}$ and $\tan\theta = \frac{5}{12}$. Therefore,

$$\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{5}{13} \cdot \frac{12}{13} = \frac{120}{169}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = \left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2 = \frac{119}{169}$$

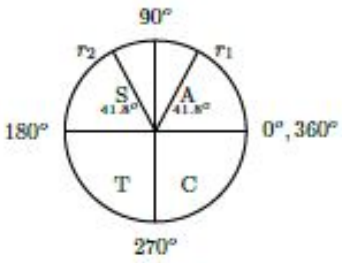
2. Solve the equation $3\cos 2\theta + \sin\theta = 1$ in the range $0^o \leq \theta \leq 360^o$.

Solution

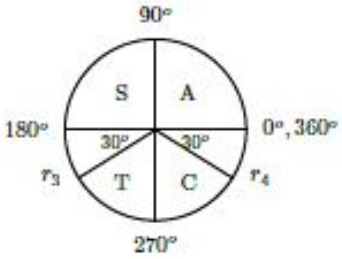
Substituting $\cos 2\theta = 1 - 2\sin^2\theta$ in the given equation, we have

$$3\left(1 - 2\sin^2\theta\right) + \sin\theta = 1 \quad \Rightarrow \quad 6\sin^2\theta - \sin\theta - 2 = 0 \quad \Rightarrow \quad (3\sin\theta - 2)(2\sin\theta + 1) = 0$$

Solving yields $\sin\theta = \frac{2}{3}$ or $\sin\theta = -\frac{1}{2}$.



Case 1: $\sin\theta = \frac{2}{3}$.
From the unit circle, sine is +ve in the 1st and 2nd quadrants.
Let the roots be r_1 and r_2.
The included angle is given by $a = \sin^{-1}\left(\frac{2}{3}\right) = 41.8^o$.
Hence, the roots are: $r_1 = 41.8^o$ and $r_2 = (180^o - 41.8^o) = 138.2^o$.



Case 2: $\sin\theta = -\frac{1}{2}$.
From the unit circle, sine is -ve in the 3rd and 4th quadrants.
Let the roots be r_3 and r_4.
The included angle is given by $a = \sin^{-1}\left(\frac{1}{2}\right) = 30^o$.
Hence, the roots are: $r_3 = (180^o + 30^o) = 210^o$
and $r_4 = (360^o - 30^o) = 330^o$.

Therefore, $\theta = 41.8^o, 138.2^o, 210^o$ and 330°.

3. Find x in the range $0 \leq x \leq 2\pi$ given $12\cos x - 5\sin x + 3 = 0$.

Solution

The given equation can be written as $12\cos x - 5\sin x = -3$. We need to express $12\cos x - 5\sin x$ in the form $R\cos(x + \alpha)$, where R and α are constants to be determined. Now,

$$\begin{aligned} R\cos(x+\alpha) &= 12\cos x - 5\sin x \\ \Rightarrow \quad R\cos x\cos\alpha - R\sin x\sin\alpha &= 12\cos x - 5\sin x \end{aligned}$$

Equating coefficients, we have

$$R\cos\alpha = 12 \quad \text{and} \quad R\sin\alpha = 5$$

Now,

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{5}{12} \quad \Rightarrow \quad \tan\alpha = \frac{5}{12} \quad \Rightarrow \quad \alpha = \tan^{-1}(5/12) = 0.394$$

and

$$(R\cos\alpha)^2 + (R\sin\alpha)^2 = (12)^2 + (5)^2 \quad \Rightarrow \quad R^2 = 169 \quad \Rightarrow \quad R = 13$$

Therefore, the given equation can be written as

$$13\cos(x + 0.394) = -3 \quad \Rightarrow \quad \cos(x + 0.394) = -\frac{3}{13}$$

Using a unit circle, we have $x + 0.394 = \pi \pm 1.3379$. Thus, $x = 1.4097$ and $x = 4.0855$.

4. (a) Solve for θ given that $\sin 4\theta - \sin 2\theta + 2\cos 3\theta = 0$ in the range $0^o \leq \theta \leq 360^o$.

Solution

$$\begin{aligned} \sin 4\theta &= \sin(3\theta + \theta) = \sin 3\theta\cos\theta + \sin\theta\cos 3\theta \\ \sin 2\theta &= \sin(3\theta - \theta) = \sin 3\theta\cos\theta - \sin\theta\cos 3\theta \end{aligned}$$

Subtracting yields $\sin 4\theta - \sin 2\theta = 2\sin\theta\cos 3\theta$. Therefore, the given equation becomes

$$\sin\theta\cos 3\theta + 2\cos 3\theta = 0 \quad \Rightarrow \quad \cos 3\theta(\sin\theta + 1) = 0$$

Solving yields $\cos 3\theta = 0$ or $(\sin\theta + 1) = 0 \quad \Rightarrow \quad \sin\theta = -1$. From $\cos 3\theta = 0$, we have $3\theta = 90^o, 270^o \quad \Rightarrow \quad \theta = 30^o, 90^o$. From $\sin\theta = -1$, we have $\theta = 270^o$. Therefore, $\theta = 30^o, 90^o$ and 270°.

(b) Find the values of θ, in the range $0 \leq \theta \leq \pi$, that satisfy the equation $\sin 5\theta + \sin 3\theta = 0$. [hint: $\sin 5\theta = \sin(4\theta + \theta), \sin 3\theta = \sin(4\theta - \theta)$, ans: $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$]

Exercise:

1. Solve for θ in the interval $0^o \leq \theta \leq 360^o$.

(a) $\cos 2\theta + \cos\theta + 1 = 0$. [ans: $\theta = 90^o, 120^o, 240^o, 270^o$]

(b) $\sin 2\theta\cos\theta + \sin^2\theta = 1$. [ans: $\theta = 30^o, 90^o, 150^o, 270^o$]

(c) $4\tan\theta\tan 2\theta = 1$. [ans: $\theta = 18.4^o, 161.6^o$]

(d) $3\cot 2\theta + \cot\theta = 1$. [ans: $\theta = 45^o, 59.03^o, 120.97^o, 135^o$]

(e) $\sin\theta = 6\sin 2\theta$. [ans: $\theta = 85.2^o, 274.8^o$]

(f) $12\cos^2\theta + \sin\theta = 11$. [ans: $\theta = 19.5^o, 160.5^o, 194.5^o, 345.5^o$]

2. Show that

(a) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

(b) $\cos 3\theta = 4\cos^3\theta - 3\sin\theta$

(c) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

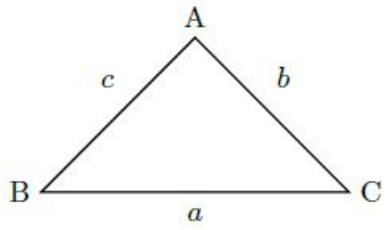
3. Find the general solution of the equations

(a) $12\sec^2\theta - 13\tan\theta - 9 = 0$ [ans: $\theta = 18.4 + n\pi, 36.9 + n\pi$, where $n \in \mathbb{Z}$]

(b) $\cos 2\theta - 3\cos\theta + 2 = 0$ [ans: $\theta = \pm\frac{\pi}{3} + 2n\pi, 2n\pi$, where $n \in \mathbb{Z}$]

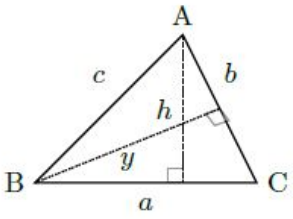
8.8 Solution of triangles

Consider triangle ABC.



The sides are usually labeled a, b, c so that a is opposite to vertex A, b is opposite to vertex B and c is opposite to vertex C. The angles of the triangle are defined by capital letters corresponding to each vertex.

8.8.1 The sine rule/formula



$\sin B = \frac{h}{c} \quad \Rightarrow \quad h = c\sin B - - - (i)$

$\sin C = \frac{h}{b} \quad \Rightarrow \quad h = b\sin C - - - (ii)$

Equating (i) and (ii), we get $c\sin B = b\sin C$

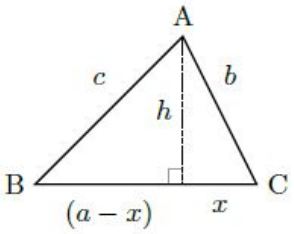
Dividing through by $\sin B \sin C$ yields $\frac{c}{\sin C} = \frac{b}{\sin B} - - - (1)$

Similarly, $\sin A = \frac{y}{c} \quad \Rightarrow \quad y = c\sin A - - - (iii)$ and $\sin C = \frac{y}{a} \quad \Rightarrow \quad y = a\sin C - - - (iv)$. Equating (iii) and (iv), we get $c\sin A = a\sin C$. Dividing through by $\sin A \sin C$ yields $\frac{c}{\sin C} = \frac{a}{\sin A} - - - (2)$. From equations (1) and (2), we have.

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}} - - - (3)$$

Equation (3) is called the sine rule. The rule is applicable when two angles are given and only one side is known.

8.8.2 The cosine rule/formula



$\cos C = \frac{x}{b} \quad \Rightarrow \quad x = b\cos C - - - (i)$

Using Pythagoras theorem, we have $x^2 + h^2 = b^2 - - - (ii)$

Also, $(a-x)^2 + h^2 = c^2 \quad \Rightarrow \quad a^2 - 2ax + x^2 + h^2 = c^2 - - - (iii)$

Substituting (ii) into (iii) yields $a^2 - 2ax + b^2 = c^2 - - - (iv)$

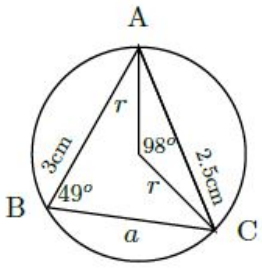
Substituting (i) into (iv) yields $\boxed{c^2 = a^2 + b^2 - 2ab\cos C} - - - (4)$

Similarly, $\boxed{a^2 = b^2 + c^2 - 2bc\cos A} - - - (5)$ and $\boxed{b^2 = a^2 + c^2 - 2ac\cos B} - - - (6)$. Equations (4), (5), and (6) are the cosine rule. The rule is applicable when only one included angle and two sides are given.

Examples:

1. Find the radius of a circle which passes through the vertices of triangle ABC with sides AB=3cm, AC=2.5cm and $\langle ABC = 49^o$.

Solution



From properties of a circle, angle subtended by an arc at the center is twice angle subtended by the same arc on the circumference.
Using cosine rule, we have $(2.5)^2 = r^2 + r^2 - 2 \cdot r \cdot r \cos 98^o$.
Therefore, $6.25 = 2r^2 - 2r^2 \cos 98^o \quad \Rightarrow \quad r^2 = \dfrac{6.25}{2 - 2\cos 98^o}$

$$\Rightarrow \quad r = \sqrt{\frac{6.25}{2 - 2(-0.1392)}} = 1.6563 \text{ cm}.$$

8.9 Area of a triangle

The following formulae are useful in finding the area of triangle ABC. Using either the formula

$$(1) \text{ Area } = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B$$

or using the Hero's formula

$$(2) \text{ Area } = \sqrt{s(s-a)(s-b)(s-c)}, \quad \text{where} \quad s = \frac{1}{2}(a+b+c)$$

Exercise:

1. In the cosine rule, substitute $\cos A = 1 - 2\sin^2(A/2)$ and hence prove that $\sin\left(\frac{A}{2}\right) = \sqrt{\dfrac{(s-b)(s-c)}{bc}}$.

2. In the cosine rule, substitute $\cos A = 2\cos^2(A/2) - 1$ and hence prove that $\cos\left(\frac{A}{2}\right) = \sqrt{\dfrac{s(s-a)}{bc}}$.

3. Use the result of equations (1) and (2) to prove Hero's formula for the area of a triangle given by Area $= \sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{1}{2}(a+b+c)$.

4. Prove Hero's formula by eliminating A from the formula $a^2 = b^2 + c^2 - 2bc\cos A$ and using the area of the triangle given by Area $= \frac{1}{2}bc\sin A$. [hint: use $\cos^2 A + \sin^2 A = 1$]

Lecture 9

9 Statistics

Statistics is a branch of applied mathematics that deals with collection, presentation, analysis, and interpretation of data.

- Data → raw facts collected from the field
- Information → typically formatted facts
- Sample → a portion of interest
- Population → the entire group in a statistical investigation

Data can be summarized using concise numerical values. A numerical summary for a set of data is referred to as a statistic if the data set is a sample and a parameter if the data set is a population. The numerical summaries are categorized as

1. Measures of location
   - Measures of central tendency
   - Measures of relative positioning (N-tiles/quantiles)
2. Measures of dispersion (spread)

9.1 Measures of central tendency

They include arithmetic mean, median, and mode.

9.1.1 Arithmetic mean

The mean of individual observations is the sum (or total) of all the observations divided by the total number of observations. Let $x_1, x_2, \cdots, x_n$ be n observations. Their mean is defined as:

$$\text{Mean, } \bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

For grouped distribution, the arithmetic mean is given by the formula: Mean, $\bar{x} = \frac{\sum fx}{\sum f}$.

9.1.2 Median

This is the value above which and below which half of the observations fall (if ranked in order of size). The median is the $\left(\frac{n+1}{2}\right)^{\text{th}}$ value in a sorted data set. Thus, if n is even we average the middle two terms. For grouped data median is estimated using the formula:

$$\text{Median } = \text{L.C.B } + \left(\frac{\frac{n+1}{2} - Cf_a}{f}\right) \times i,$$

where L.C.B is the lower class boundary of the median class, Cf_a is the cumulative frequency of the class above the median class, f is the frequency of the median class, and i is the class interval of the median class.

9.1.3 Mode

This is the value occurring most frequently in a set of observations or statistical data. For grouped data, mode is estimated using the formula:

$$\text{Mode} = \text{L.C.B} + \left(\frac{f - f_a}{2f - f_a - f_b}\right) \times i,$$

where L.C.B is the lower class boundary of the modal class, f_a is the frequency of the class above the modal class, f_b is the frequency of the class below the modal class, f is the frequency of the modal class, and i is the class interval of the modal class.

9.2 Measures of relative positioning

The commonly used N-tiles are quartiles, deciles, and percentiles. They divide a sorted data set into 4, 10, and 100 equal parts, respectively. Thus, there are three quartiles (i.e., Q_1, Q_2, Q_3), nine deciles (i.e., $D_1, D_2, \cdots, D_9$), and ninety nine percentiles (i.e., $P_1, P_2, \cdots, P_{99}$). The value corresponding to Q_2, D_5 and P_{50} is the median of the sorted data set.

9.2.1 Quartiles

The kth quartile is the $\frac{k}{4}(n+1)^{\text{th}}$ value, and is given by the formula:

$$Q_k = \text{LCB} + \left(\frac{\frac{k(n+1)}{4} - Cf_a}{f}\right) \times i,$$

where LCB is the lower class boundary of the kth quartile class, Cf_a is the cumulative frequency of the class above the kth quartile class, f is the frequency of the kth quartile class, and i is the class interval of the kth quartile class.

9.3 Measures of dispersion

Dispersion is the degree of variation of the data about the central value. Examples of measures of dispersion include:

- Range $= x_{\max} - x_{\min}$
- Interquatile range (IQR) $= Q_3 - Q_1$
- Semi-interquatile range (SIQR) $= \frac{1}{2}(Q_3 - Q_1)$
- Mean absolute deviation (MAD) $= \dfrac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$
- Variance and standard deviation

9.3.1 Variance

The variance refers to the mean of the squared deviations from the arithmetic mean. It is denoted by S^2. Thus,

$$\text{Variance, } S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n} = \frac{\sum_{i=1}^{n} x_i^2}{n} - (\bar{x})^2$$

For grouped distribution, the variance is given by: $S^2 = \frac{\sum f(x-\bar{x})^2}{\sum f} = \frac{\sum fx^2}{\sum f} - (\bar{x})^2$.

9.3.2 Standard deviation

Standard deviation refers to the root mean square deviations from the arithmetic mean. Thus, standard deviation, $S = \sqrt{\text{variance}}$.

Examples:

1. The masses, in grams of, 50 small fruits are as shown in the table below.

Mass (g)	50-59	60-69	70-79	80-89	90-99	100-109	110-119
No. of fruits	4	5	9	14	7	6	5

(a) State the modal class and calculate the mode.

(b) Calculate the mean, variance and standard deviation.

(c) Calculate the median and semi-interquartile range.

Solution

Classes	Mid point (x)	f	fx	fx^2	c.f
50-59	54.5	4	218	11881	4
60-69	64.5	5	322.5	20801.25	9
70-79	74.5	9	670.5	49952.25	18
80-89	84.5	14	1183	99963.5	32
90-99	94.5	7	661.5	62511.75	39
100-109	104.5	6	627	65521.5	45
110-119	114.5	5	572.5	65551.25	50
		$\sum f = 50$	$\sum fx = 4255$	$\sum fx^2 = 376182.5$	

(a) Modal class is 80-89 since it has the highest frequency. Therefore,

$$\text{Mode} = \text{LCB} + \left(\frac{f - f_a}{2f - f_a - f_b}\right) \times i = 79.5 + \left(\frac{14 - 9}{2(14) - 9 - 7}\right) \times 10 = 83.67 \text{ grams.}$$

(b) (i) Mean, $\bar{x} = \dfrac{\sum fx}{\sum f} = \dfrac{4255}{50} = 85.1$ grams.

(ii) Variance, $S^2 = \dfrac{\sum fx^2}{\sum f} - (\bar{x})^2 = \dfrac{65551.25}{50} - (85.1)^2 = 281.64$ grams.

(iii) Standard deviation, $S = \sqrt{281.64} = 16.7821$ grams.

(c) (i) Median is the $\left(\dfrac{n+1}{2}\right)^{\text{th}}$ value i.e., $\dfrac{50+1}{2} = 25.5^{\text{th}}$ value. Thus, the median class is 80-89. Therefore,

$$\text{Median, } Q_2 = \text{LCB} + \left(\frac{\frac{n+1}{2} - Cf_a}{f}\right) \times i = 79.5 + \left(\frac{25.5 - 18}{14}\right) \times 10 = 84.8571 \text{ grams.}$$

(ii) Lower quartile (Q_1) is the $\left(\dfrac{n+1}{4}\right)^{\text{th}}$ value i.e., $\dfrac{50+1}{4} = 12.75^{\text{th}}$ value. Thus, Q_1 falls in the class 70-79. Therefore,

$$Q_1 = \text{LCB} + \left(\frac{\frac{n+1}{4} - Cf_a}{f}\right) \times i = 69.5 + \left(\frac{12.75 - 9}{9}\right) \times 10 = 73.6667 \text{ grams.}$$

Upper quartile (Q_3) is the $\left(\dfrac{3(n+1)}{4}\right)^{\text{th}}$ value i.e., $\dfrac{50+1}{4} = 38.25^{\text{th}}$ value. Thus, Q_3 falls in the class 90-99. Therefore,

$$Q_3 = \text{LCB} + \left(\frac{\frac{3(n+1)}{4} - Cf_a}{f}\right) \times i = 89.5 + \left(\frac{38.25 - 32}{7}\right) \times 10 = 98.4286 \text{ grams.}$$

Semi-interquartile range (SIQR) $= \dfrac{1}{2}(Q_3 - Q_1) = \dfrac{1}{2}(98.4286 - 73.6667) = 12.3810$ grams.

Exercise:

1. The weights, to the nearest kilogram, of heifers in a farm are as shown below.

Weight (Kg)	170-179	180-189	190-199	200-209	210-219	220-229
No. of heifers	3	8	13	9	7	5

   (a) State the modal class.

   (b) Calculate the variance.

   (c) Calculate the median and semi-interquartile range.

Lecture 10

10 Probability

Probability refers to the numerical value that measures the likelihood of an event happening. We assign a value between 0 and 1 to an event such that the magnitude gives the likelihood (degree of certainty) that the event will occur. A value close to 0 means that the event is unlikely to occur while a value close to 1 means that the event is very likely to occur.

10.1 Definition of terms

Definition 4.1.1 (Experiment). *This is the process by which an observation or measurement is obtained. E.g., tossing a coin, throwing a die, etc.*

Definition 4.1.2 (Outcome). *This is the possible result of an experiment. E.g., a head when a coin is tossed once.*

Definition 4.1.3 (Sample space). *This is the set of all possible outcomes of a random experiment. For instance, when a coin is tossed once the sample space is given by* $S = \{H, T\}$.

The sample space when a die is rolled once is given by $S = \{1, 2, 3, 4, 5, 6\}$. The number of outcomes in the sample space S is denoted by $n(S)$.

Definition 4.1.4 (Event). *This is a subset of the sample space. For instance, the possibility of observing two heads if a fair coin is tossed twice. It is usually denoted by uppercase letters, say* A, B, C, *etc. The number of outcomes in an event* A *is denoted by* $n(A)$.

10.2 Classical definition of probability

In a classic sense, it means that every statistical experiment will contain elements that are equally likely to happen (equal chances of occurrence of something). Suppose an experiment with a sample space S consists of $n(S)$ equally likely outcomes. The probability of an event A happening is the number of times the event can happen, divided by the number of possible outcomes. That is,

$$P(A) = \frac{n(A)}{n(S)}$$

Examples:

1. Let A be the event of observing two heads if a fair coin is tossed twice. Write down the sample space and find the probability of getting exactly two heads.

   Solution

   The sample space is given by $S = \{HH, HT, TH, TT\}$. The event A is $A = \{HH\}$. Now,

   $$P(A) = \frac{n(A)}{n(S)} = \frac{1}{4}$$

2. A fair coin is tossed 3 times. Find the sample space and the probability of observing exactly two heads.

   Solution

   The sample space is given by $S = \{HHH, HTH, THH, TTH, HHT, HTT, THT, TTT\}$. The event A is $A = \{HHT, HTH, THH\}$. Now,

   $$P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$$

10.3 Rules of probability

(1) The probability of any event A lies between 0 and 1. That is, $0 \leq P(A) \leq 1$.

Proof. Since $A \subseteq S \quad \Rightarrow \quad 0 \leq n(A) \leq n(S)$. Dividing through by $n(S)$ yields

$$0 \leq \frac{n(A)}{n(S)} \leq \frac{n(S)}{n(S)} \quad \Rightarrow \quad 0 \leq P(A) \leq 1$$

□

(2) Addition rule: let A and B be two events of the same experiment. The probability of event A or B or both happening is given by $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ or in symbol notation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Examples:

1. A number is randomly selected from integers between 20 and 24 (inclusive). Find the probability that

   (a) Its 2nd digit is greater than its 1st digit.

   (b) It is divisible by 4.

   (c) Its 2nd digit is greater than its 1st digit and it is divisible by 4.

   Solution

   The sample space is $S = \{20, 21, 22, 23, 24\}$.

   (a) Let event A be: $A = \{\text{2nd digit} > \text{1st digit}\} = \{23, 24\}$. Then, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{2}{5}$

   (b) Let event B be: $B = \{x : x \text{ is divisible by } 4\} = \{20, 24\}$. Then, $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{2}{5}$

   (c) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$. But $A \cup B = \{20, 23, 24\}$.
   So, $P(A \cup B) = \dfrac{n(A \cup B)}{n(S)} = \dfrac{3}{5}$. Therefore, $P(A \cap B) = \dfrac{2}{5} + \dfrac{2}{5} - \dfrac{3}{5} = \dfrac{1}{5}$

2. Two fair dice labeled 1 to 6 bare rolled. Let A be the event that the product of the two numbers showing up is greater than 21 and let B be the event that the product is divisible by 6. Find $P(A), P(B), P(A \cap B)$, and $P(A \cup B)$.

   Solution

   We use the following multiplication table

	1	**2**	**3**	**4**	**5**	**6**
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

   The number of elements in the sample space is $n(S) = 36$. Let A and B be the events defined by

   A: product > 21.

   B: product is divisible by 6.

   Thus, $n(A) = 6$ (all the circled elements), $n(B) = 15$ (all red) and $n(A \cap B) = 5$ (all red and circled elements).

   Therefore, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{6}{36} = \dfrac{1}{6}$, $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{15}{36} = \dfrac{5}{12}$, $P(A \cap B) = \dfrac{n(A \cap B)}{n(S)} = \dfrac{5}{36}$,
   and $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \dfrac{1}{6} + \dfrac{5}{12} - \dfrac{5}{36} = \dfrac{16}{36} = \dfrac{4}{9}$.

10.4 Conditional probability

This is the probability of an event A happening given that event B has already happened. It is denoted by $P(A/B)$ and is defined as

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Examples:

1. A deck of 52 cards has 13 spades. If two cards are drawn from the deck at random, what is the probability that both are spades?

Solution

Here, $n(S) = 52$. Let A be the event that the 1st card drawn is a spade. Let B be the event that the 2nd card drawn is a spade. Now, $P(A) = \frac{13}{52}$ and $P(B/A) = \frac{12}{51}$. Hence,

$$P(A \cap B) = P(A) \times P(B/A) = \frac{13}{52} \times \frac{12}{51} = \frac{1}{17}$$

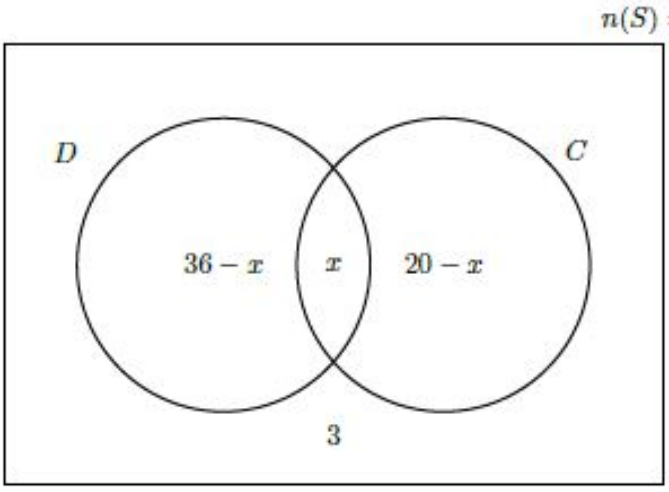
Alternatively, we can use a tree diagram.

2. In a survey of 50 students at JKUAT-IT centre, it was found that 36 are in diploma program, 20 have personal computers and only 3 are neither diploma students nor have personal computers. With the help of Venn diagram or otherwise, find the probability that a randomly selected student

   (a) Has a computer but not in diploma program.

   (b) Has a computer if he/she is in diploma program.

Solution



Here, $n(D) = 36, n(C) = 20$.
Therefore, $P(D) = \frac{n(D)}{n(S)} = \frac{36}{50}$
and $P(C) = \frac{n(C)}{n(S)} = \frac{20}{50}$
Let $n(D \cap C) = x$. Thus,
$(36 - x) + x + (20 - x) + 3 = 50$
$\Rightarrow \quad x = 9$.

(a) $n(C - D) = 20 - x = 20 - 9 = 11$. Therefore, $P(C - D) = \frac{n(C - D)}{n(S)} = \frac{11}{50}$

(b) $P(C/D) = \frac{P(C \cap D)}{P(D)} = \frac{n(C \cap D)/n(S)}{n(D)/n(S)} = \frac{n(C \cap D)}{n(D)} = \frac{9}{36} = \frac{1}{4}$

Exercise:

1. In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats, and 20 purchased an alarm system and bucket seats. If a car buyer chosen at random bought an alarm system, what is the probability they also bought bucket seats? [ans: $P(B/A) = \frac{1}{2}$]

2. For two events A and B, $P(\sim A) = \frac{1}{3}, P(\sim B) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{4}$. Find $P(A/B)$. [ans: $\frac{5}{8}$]

3. Two dice, coloured red and green, are thrown in a game. The player has to score a total of 9 or more in order to stay in the game. Find the probability that:

   (a) The player scores a total of 9 or more. [ans: $\frac{5}{18}$]

   (b) The player scores a total of 9 or more given that the score on the green die is 3. [ans: $\frac{1}{6}$]

   (c) The player scores a total of 9 or more given that the score on the green die is 6. [ans: $\frac{2}{3}$]

4. Two fair six-faced dice are thrown and the events A, B are defined as follows.

   A: the score on the topmost face of the first die is 5.

   B: the sum of the scores on the topmost faces on both dice is at least 9.

   Determine each of the following probabilities:

   (a) $P(A)$ [ans: $= \frac{1}{6}$]

   (b) $P(B)$ [ans: $= \frac{5}{18}$]

   (c) $P(A \cap B)$ [ans: $= \frac{1}{12}$]

   (d) $P(A \cup B)$ [ans: $= \frac{13}{36}$]

   (e) $P(A/B)$ [ans: $= \frac{3}{10}$]

   (f) $P(B/A)$ [ans: $= \frac{1}{2}$]

   (g) $P(B/\sim A)$ [ans: $= \frac{7}{30}$]

10.5 Independent events

Two events A and B are said to be independent if the occurrence of one doesn't affect the occurrence of the other. For example, consider the experiment of tossing a fair coin and rolling a die. The occurrence of a head (H) and a 5 or a 6 in the two experiments are independent. Thus,

$$P(A \cap B) = P(A) \times P(B)$$

Examples:

1. Let $P(A) = 0.8, P(B) = 0.7$. Find $P(A \cap B)$ and $P(A \cup B)$ if A and B are

   (a) Independent events

   (b) Mutually exclusive events

   Solution

   (a) $P(A \cap B) = P(A) \times P(B) = 0.8 \times 0.7 = 0.56$. Therefore,

   $$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.7 - 0.56 = 0.94$$

   (b) $P(A \cap B) = 0$. Therefore, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.7 - 0 = 1.5 > 1$???. In conclusion, two events A and B cannot be independent and then at the same time mutually exclusive and vice versa.

10.6 Binomial formula

It is used to solve problems emanating from the use of tree diagrams. The binomial formula is used when

i) The experiment is to be performed n independent times.

ii) The outcomes of the experiment can be classified into two categories i.e., success or failure with probabilities p and q, respectively, where $q = 1 - p$.

iii) When the interest is on the number of successes.

The probability of observing x successes out of n trials is given by

$$\boxed{P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}}, \quad x = 0, 1, 2, \cdots, n$$

→ Note: $p^x(1-p)^{n-x}$ is the result for only one branch and $\binom{n}{x}$ gives the number of favoured branches.

Examples:

1. A fair coin is tossed 10 times. What is the probability that one observes exactly 8 heads.

Solution

Here, $n = 10, x = 8, p = 0.5, q = 1 - p = 1 - 0.5 = 0.5$. Therefore,

$$P(X = 8) = \binom{10}{8} (0.5)^8 (1-0.5)^{10-8} = \binom{10}{8} (0.5)^{10} = \frac{10!}{2!8!} (0.5)^{10} = 0.044$$

2. The probability that it rains in any day of the coming week is 0.73. Find the probability that:
   (a) It rains in exactly 5 days of the coming week.
   (b) There will be no rainy day in the coming week.
   (c) It rains for at least 5 days of the coming week.
   (d) It rains between the 2 to 4 days of the coming week (inclusive).

Solution

Here, $n = 7, p = 0.73, q = 1 - p = 1 - 0.73 = 0.27$. Therefore,

(a) $P(X = 5) = \binom{7}{5}(0.73)^5(1-0.73)^{7-5} = \frac{7!}{2!5!}(0.73)^5(0.27)^2 = 0.3174$

(b) $P(X = 0) = \binom{7}{0}(0.73)^0(1-0.73)^{7-0} = (0.27)^7 = 0.0001046035$

(c)

$$\begin{aligned} P(X \geq 5) &= P(X=5) + P(X=6) + P(X=7) \\ &= \binom{7}{5}(0.73)^5(0.27)^2 + \binom{7}{6}(0.73)^6(0.27)^1 + \binom{7}{7}(0.73)^7(0.27)^0 \\ &= \frac{7!}{2!5!}(0.73)^5(0.27)^2 + 7(0.73)^6(0.27) + (0.73)^7 = \end{aligned}$$

(d)

$$\begin{aligned} P(2 \leq X \leq 4) &= P(X=2) + P(X=3) + P(X=4) \\ &= \binom{7}{2}(0.73)^2(0.27)^5 + \binom{7}{3}(0.73)^3(0.27)^4 + \binom{7}{4}(0.73)^4(0.27)^3 \\ &= \frac{7!}{5!2!}(0.73)^2(0.27)^5 + \frac{7!}{4!3!}(0.73)^3(0.27)^4 + \frac{7!}{3!4!}(0.73)^4(0.27)^3 = \end{aligned}$$

PHA 2116: BASIC MATHEMATICS **CAT 1** **TIME: 1 HOUR**

(a) i) Differentiate between a surd and an irrational number. [2 Marks]

ii) Express $\frac{3\sqrt{2}+5\sqrt{6}}{3\sqrt{2}-5\sqrt{6}}$ in the form $a+b\sqrt{c}$, where a, b and c are rational numbers. [3 Marks]

(b) Solve for x given that $\log_3 x - 2\log_x 3 = 1$. [4 Marks]

(c) The roots of the equation $5x^2 - 3x + 2 = 0$ are α and β. Find the equation with integral coefficients whose roots are $\frac{1}{\alpha^3}$ and $\frac{1}{\beta^3}$. [5 Marks]

(d) Find the values of p and q given that $x^2 + 3x - 5 \equiv (x-p)^2 + q$. [2 Marks]

(e) A cubic polynomial $ax^3 + bx + 6$ is divisible by $(x+2)$. It leaves a remainder of -3 when divided by $(x-1)$. Determine the values of the constants a and b and hence factorize the polynomial completely. [6 Marks]

(f) How many odd numbers greater than 70,000 can be formed using the digits 0,1,4,7,8, and 9, if each digit should be used only once. [4 Marks]

(g) A committee of ten is to be formed from nine men and seven women. In how many ways can it be formed if at least two men and three women are to be in the committee? [4 Marks]

PHA 2116:BASIC MATHEMATICS **CAT 2** **TIME: 1 HOUR**

(a) In an AP, the thirteenth term is 27, and the seventh term is three times the second term. Find the first term, the common difference and the sum of the first ten terms. [4 Marks]

(b) A customer deposits Sh. 2000 in his savings account on 1st January each year for 20 years. If the account gives compound interest at the rate of 3% p.a. What will be the total value of his savings at the end of 20 years? (Answer correct to the nearest shillings). [6 Marks]

(c) Given that $\sin A = \frac{4}{5}$ and $\tan B = \frac{5}{12}$, where A is an obtuse angle and B is an acute angle. Find, $\cos(A+B)$ without using mathematics tables or calculators. [4 Marks]

(d) Solve the equation $2\cos^2(2\theta) - \sin(2\theta) - 1 = 0$ in the range $0^\circ \leq \theta \leq 360^\circ$. [3 Marks]

(f) The weights, to the nearest kilogram, of heifers in a farm are as shown below.

Weight (Kg)	170-179	180-189	190-199	200-209	210-219	220-229
No. of heifers	3	8	13	9	7	5

(i) State the modal class. [1 Mark]

(ii) Calculate the variance. [4 Marks]

(iii) Calculate the median and semi-interquartile range. [4 Marks]

(g) In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats, and 20 purchased an alarm system and bucket seats. If a car buyer chosen at random bought an alarm system, what is the probability they also bought bucket seats? [3 Marks]

PHA 2116: BASIC MATHEMATICS **CAT 2** **TIME: 1 HOUR**

(a) Expand $(x+5)^5$ and simplify your answer. Use the expansion to estimate $(3.02)^5$ correct to four decimal places. [4 Marks]

(b) In an AP, the thirteenth term is 27, and the seventh term is three times the second term. Find the first term, the common difference and the sum of the first ten terms. [3 Marks]

(c) A customer deposits Sh. 2000 in his savings account on 1st January each year for 20 years. If the account gives compound interest at the rate of 3% p.a. What will be the total value of his savings at the end of 20 years? (Answer correct to the nearest shillings). [5 Marks]

(d) Given $\sin A = \frac{4}{5}$ and $\tan B = \frac{5}{12}$, where A is an obtuse angle and B is an acute angle. Find $\cos(A+B)$ without using SMP table or calculator. [4 Marks]

(e) Solve for θ given $2\cos^2(2\theta) - \sin(2\theta) - 1 = 0$ in the range $-180^o \leq \theta \leq 180^o$. [4 Marks]

(f) The table below shows the number of people in certain age groups. Compute the mean, mode, median, variance and standard deviation. [8 Marks]

Age group	19-27	28-36	37-45	46-54	55-63	64-72	73-81	82-90	91-99
No. of people	4	6	9	15	19	16	7	5	2

(g) A fair die is rolled 20 times. What is the probability that the face labeled 5 will show up exactly 3 times. [2 Marks]

THE END